

COMPUTING WITH FOURIER EXPANSIONS OF HILBERT MODULAR FORMS

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ABSTRACT. We describe algorithms for efficient computations with q -expansions of Hilbert modular forms, including Eisenstein series. Applications include computing graded rings of Hilbert modular forms, partial weight 1 forms and plurigenera of certain Hilbert modular surfaces.

CONTENTS

TODO	2
1. Introduction	2
1.1. Motivation	2
1.2. Results	3
1.3. Previous work	5
1.4. Contents	5
1.5. Acknowledgements	6
2. Preliminaries	6
2.1. Basic notation	6
2.2. Hilbert modular varieties	6
2.3. Hilbert modular forms	7
2.4. Fourier expansions	9
2.5. Hecke characters	11
2.6. Nebentypus characters	13
2.7. Unit characters	14
3. Arithmetic of q -expansions	16
3.1. Encoding in bits	16
3.2. Computing the shadow.	19
3.3. Arithmetic operations	24
3.4. Hecke operators	26
3.5. Inclusion and degeneracy operators	26
3.6. Actions by field automorphisms	27
3.7. Restriction to modular curves	28
4. Sources of Hilbert modular forms	29
4.1. Eisenstein series	29
4.2. Quaternionic methods	30
4.3. Theta series associated with characters of quadratic extensions	30
4.4. Pullbacks of Siegel modular forms	32

5. Computing graded rings	34
5.1. Preparatory lemmas and functions	34
5.2. An algorithm for computing graded rings	37
5.3. Examples	38
Hilbert modular surface for $\mathbb{Q}(\sqrt{13})$	40
6. Hecke stability	43
6.1. The method	44
6.2. A non-dihedral form of weight $(1, 1)$	45
6.3. Non-CM partial weight one forms	48
7. Computing plurigena	49
7.1. Pluricanonical forms and cusp forms	49
7.2. The cusp test at infinity	51
7.3. The other cusps via Atkin–Lehner involutions	51
7.4. Plurigenus computation	53
7.5. Example: $X_0(\mathfrak{p}_{31})$ over $\mathbb{Q}(\sqrt{5})$	54
Appendix A. Eisenstein Series	57

TODO

- (1) Verify the Threefold examples (Edgar, Jean (§7))
- (2) Compute a model for a symmetric Hilbert modular variety (pick an example from van der Geer).
- (3) contact Adam Logan to see which ones he was able to figure out using defects. (Eran)
- (4) cite Adam’s work
- (5) Add table about what models can we compute (Edgar)
- (6) Compare them over finite fields.
- (7) Expand tests in Magma (Edgar)
- (8) Add base change to section 4 (Abhijit)
- (9) When computing shadows, use the L1/L2 ball and prune using range queries instead of what we currently do. JK: [Made the change in the text.]

1. INTRODUCTION

JK: [Do we want to add a couple words on graded rings in the abstract? We’re able to reproduce the content of half a dozen papers on this topic, I find it quite impressive.] Edgar: [YES! we need to do that!] JK: [done, feel free to change]

1.1. Motivation. Modular forms remain a central object of study in computational number theory. A key tool for working with classical modular forms is provided by their q -expansions. Indeed, representing modular forms by their Fourier expansions at ∞ not only makes them accessible, but the coefficients of these power series also contain essential arithmetic, analytic, and combinatorial information.

Here we consider the generalization to Hilbert modular forms, enlarging the base field from the rational numbers to a totally real number field. We present versatile algorithms for manipulating q -expansions of Hilbert modular forms, we analyze their efficiency, and we describe an implementation in the computer algebra system Magma [**Magma**], available

online [**ourcode**]. A key idea we present, improving upon naive expansion, is to diagonalize the space for the action of the unit diamond operators, decomposing according to what we call *unit characters* (see Section 2.7); this allows us to restrict to Fourier coefficients up to the action of totally positive units in significant generality.

Our main goal is to contribute a practical tool to facilitate the deeper investigation of Hilbert modular varieties that need not take advantage of special features; the arithmetic of q -expansions is formulated for arbitrary narrow class group, level, and finite-order Hecke nebentypus, once the relevant input spaces are available. In particular, we include Eisenstein series (see Section 4.1), which remain enduringly useful in applications (for example, see Dasgupta–Kakde [**DasguptaKakde**] for their use in the Brumer–Stark conjecture). The q -expansions for Eisenstein series are explicitly given [**Dasgupta-Darmon-Pollack**], and we take care of the details and arithmetic content in the computation of their coefficients.

JK: [Maybe reorder this to respect the order of sections or conversely.]

We anticipate many applications, and we present three such instances here. First, we compute with partial weight 1 Hilbert modular forms, including nonparallel weights. Previously, the only (non-CM) example computed is due to Moy–Specter [**MoySpecter**]; we follow and extend their approach, which generalizes the Hecke stability method of Schaeffer [**Schaeffer**]. Second, we consider an application to computing numerical invariants of Hilbert modular surfaces, building on our previous work [**HMSDB**]. Under some additional hypotheses, we describe a method to compute the plurigenera of (smooth) Hilbert modular surfaces, which further provides the Kodaira–Enriques type in certain cases. Our method analyzes explicitly the q -expansion at the cusps to determine when a cusp form extends to a holomorphic pluricanonical form.

Finally, we apply the q -expansions machinery exhibited here to compute presentations for graded rings of Hilbert modular forms, giving canonical equations for Hilbert modular varieties. It would also be interesting to use q -expansions for evaluation and integration of Hilbert modular forms, with applications to periods and Heegner and Darmon points [**DarLog**].

1.2. Results. Our first result is an efficient algorithm to compute products of q -expansions of Hilbert modular forms. Let F be a totally real field with ring of integers $\mathbb{Z}_F$.

Proposition 1.2.1 (Eq. (3.3.3)). *Let B be a positive integer, and let f and g be Hilbert modular forms represented as in Section 3.1 to precision B . Assuming unit-cost arithmetic in a fixed coefficient field, the product $h = fg$ can be computed to precision B using $O_F(B \log B)$ arithmetic operations.*

JV: [The duplicate labels for the theorems is a bit annoying; I prefer to say “This is proven in Eq. (3.3.3), where we also blah blah.” Or just “See blah”.]

Using Eisenstein series (Eq. (4.1.3)) and quaternionic methods (Eq. (4.2.1)), we construct q -expansions for bases in the ranges covered by those theorems; theta series (Eq. (4.3.5)) provide additional explicit forms. The above proposition allows us to multiply these efficiently, giving rise to several applications.

Our first main result is as follows. For $\mathfrak{N} \subseteq \mathbb{Z}_F$ be a nonzero ideal, let

$$M(\Gamma_0(\mathfrak{N})) := \bigoplus_{k \in \mathbb{Z}_{\geq 0}} M_{(k, \dots, k)}(\Gamma_0(\mathfrak{N}), \mathbb{Q})$$

be the graded ring of parallel-weight Hilbert modular forms whose Fourier expansions have coefficients in $\mathbb{Q}$. Taking Proj of this graded ring gives the spin log-canonical model (taking those of even weight gives the log-canonical model) whose complexification defines the Baily–Borel compactification. Edgar: [we should be able to do $\mathbb{Q}$ instead of $\mathbb{C}$] JK: [I agree. We know how to compute Galois actions, so it should suffice to find QQ-bases for each graded piece.] JV: [We talked about in fact doing this over $\mathbb{Z}$.] JV: [My preference is to work with the log canonical model, so even weight forms, with a forward reference that we can also compute spin log canonical, including odd weight forms. With trivial character, we're not missing much in odd weight; this is more familiar to the modular curve case; it's also more geometrically interesting, in light of the birational classification of surfaces.]

Theorem 1.2.2 (Eq. (5.2.5)). *There exists an algorithm (Algorithm 5.2.2) that on input a nonzero ideal $\mathfrak{N} \subseteq \mathbb{Z}_F$ returns as output a finite presentation for the $\mathbb{Z}_{\geq 0}$ -graded $\mathbb{Q}$ -algebra $M(\mathfrak{N})$.*

JV: [Changed this from $S(\mathfrak{N})$, which looks like the graded module of cusp forms, which is not a ring; also not sure about losing Γ_1 etc. so prefer $M(\Gamma_0(\mathfrak{N}))$ if e.g. we're going to write $M_k(\Gamma_0(\mathfrak{N}))$]

Eran: [Add a paragraph about specific interesting examples, maybe a threefold if we are able to compute it.]

As another application of the arithmetic of q -expansions, we are able to apply the Hecke stability method. This allows us to produce a new example of a parallel weight $(1, 1)$ Hilbert modular form which is not dihedral, and in fact compute that its projective Galois image is A_4 .

Theorem 1.2.3 (Eq. (6.2.3)). *There exists a Hilbert modular newform f over $F = \mathbb{Q}(\sqrt{5})$ of weight $k = (1, 1)$, level $\mathfrak{N} = (18\sqrt{5})$, and order 6 nebentypus of conductor $\mathfrak{N}$, such that the projective Galois representation associated to f has image isomorphic to A_4 . Moreover, f is not the base change to F of a classical modular form.*

The Hecke stability method further allows us to find a new example of a non-CM Hilbert modular eigenform of partial weight 1.

Theorem 1.2.4 (Eq. (6.3.1)). *There exists a non-CM Hilbert modular newform f over $F = \mathbb{Q}(\sqrt{5})$ of weight $(1, 3)$, level $\mathfrak{N} = 18\mathbb{Z}_F$, quadratic Hecke nebentypus χ with ray conductor $3\mathbb{Z}_F \cdot \infty_1 \infty_2$, and with coefficients in $E := \mathbb{Q}[\alpha]/(\alpha^4 + 3\alpha^2 + 1) \simeq F[\alpha]/(\alpha^2 - (\sqrt{5} - 3)/2)$.*

As a final application, we can compute the spaces of modular forms exhibiting certain vanishing conditions along the cuspidal and elliptic divisors, which allows us to compute certain geometric invariants of the surface, such as plurigenera. Edgar: [we could also go for Equation (7.4.7)] Edgar: [we have not yet defined $X_0(\mathfrak{N})$]. JK: [I agree with Edgar that 7.4.7 is more impressive. We don't have to define the toroidal compactification because plurigenera are birational invariants. We have forward references to notation in the previous theorems too.]

Theorem 1.2.5 (Eq. (7.3.5)). *Let F be a real quadratic field with $h^+(F) = 1$ and ring of integers $\mathbb{Z}_F$. Assume $\mathfrak{N} \subseteq \mathbb{Z}_F$ is a squarefree ideal. Let $X_0(\mathfrak{N}) \rightarrow \overline{Y}_0(\mathfrak{N})$ be a toroidal compactification. For every $m \geq 1$, there exists a subspace $V_m \subseteq S_{2m}(\Gamma_0(\mathfrak{N}))$ determined by vanishing of a finite number of q -expansion coefficients under the action of the Atkin–Lehner*

operators, such that there is an isomorphism

$$(1.2.6) \quad H^0(\overline{Y}_0(\mathfrak{N}), K_{\overline{Y}_0(\mathfrak{N})}^{\otimes m}) \simeq V_m.$$

In particular,

$$(1.2.7) \quad P_m(\overline{Y}_0(\mathfrak{N})) = \dim_{\mathbb{C}} V_m.$$

This can be used to complete the classification work of Hilbert modular surfaces begun in [HMSDB], and as an example we provide a proof of the following statement, classifying $X_0(\mathfrak{p}_{31})$ as an elliptic surface, where $\mathfrak{p}_{31}$ is a prime above 31 in $\mathbb{Q}(\sqrt{5})$, see Section 7.5.

Theorem 1.2.8 (Eq. (7.5.1), Eq. (7.5.3)). *The Hilbert modular surface $X_0(\mathfrak{p}_{31})$ is an elliptic surface with an elliptic fibration $\varphi_2 : X_0(\mathfrak{p}) \rightarrow \mathbb{P}^1$, having a unique multiple fiber, of multiplicity 2, and the reduced fiber has arithmetic genus 1. The other singular fibers are*

$$I_6 + 2I_4 + I_2 + 8I_1 \quad (\text{twelve fibers}).$$

JV: [Do we want to mention the provenance, that it may at first look like it should be a K3 surface but turns out to be an honestly elliptic surface and somehow that motivates what we're doing?]

Eran: [Maybe add a paragraph about possible use for threefolds once we have a proper plurigenus theorem.]

1.3. Previous work. The graded ring $M(\Gamma)$ for real quadratic fields and small level has been computed in the literature via several approaches. For example, for $F = \mathbb{Q}(\sqrt{5})$ and $\Gamma = \mathrm{SL}_2(\mathbb{Z}_F)$, Gundlach uses Eisenstein series to find generators in the case of parallel weight [Gundlach]. Hirzebruch determines relations among these generators and describes the ring completely, using the resolution of singularities at the cusps [Hirzebruch77]. For other approaches in the case of $F = \mathbb{Q}(\sqrt{5})$, see [Resnikoff]; for the same field with the principal congruence subgroup $\Gamma(2)$, see [Nagaoka].

When $F = \mathbb{Q}(\sqrt{8})$, Müller [Muller8] describes the graded ring for full level building on the techniques introduced in [Gundlach]. Hammond uses modular embeddings to describe generators for the ring of symmetric Hilbert modular forms of even weight [Hammond]. For the mixed-weight case, the graded rings for $F = \mathbb{Q}(\sqrt{2})$ or $F = \mathbb{Q}(\sqrt{5})$, and full level $\mathrm{SL}_2(\mathbb{Z}_F)$, are described in [AokiTakemori].

For higher discriminant, and in the case of full level, van der Geer studies the case $F = \mathbb{Q}(\sqrt{6})$ [Geer88]; van der Geer and Zagier carry out this computation for $F = \mathbb{Q}(\sqrt{13})$ using a birational equivalence [GeerZagier]; Mayer uses Borcherds products for $F = \mathbb{Q}(\sqrt{17})$ [Mayer]; and Williams uses Borcherds products and their restrictions to Hirzebruch–Zagier curves when $F = \mathbb{Q}(\sqrt{29})$ and $F = \mathbb{Q}(\sqrt{37})$ [Williams]. For surveys on this topic, see [Geer88; Williams].

The first equations for Hilbert modular surfaces were given by [Wilson] for $F = \mathbb{Q}(\sqrt{5})$ and $\Gamma = \mathrm{SL}_2(\mathbb{Z}_F)$. In [ElkiesKumar], Elkies and Kumar find equations for Hilbert modular surfaces for $F = \mathbb{Q}(\sqrt{D})$ with $1 < D < 100$ and D a fundamental discriminant by connecting the surfaces to moduli spaces of elliptic K3 surfaces. The same method was used for square discriminants $1 < D \leq 121$ [Kumar]. For partial weight one forms, Moy and Specter computed a non-CM example using a Hilbert modular analogue of Schaeffer’s Hecke stability method [MoySpecter; Schaeffer].

Edgar: [Add [LoganThreefolds]]

1.4. **Contents.** In section 2, we quickly review notation and the relevant background. We present the nuts and bolts of our algorithms for manipulating q -expansions in section 3. To instantiate these methods, we survey in section 4 the various possible sources of Hilbert modular forms; a complete description of Eisenstein series is given in appendix A. In section 5 we present an algorithm that computes the graded ring of a Hilbert modular variety, and provide examples. Then, in section 6, we explain the method of Hecke stability to compute (partial) weight 1 forms and give examples. In section 7, we apply the methods of the previous sections to computing plurigenera and deducing geometric invariants of Hilbert modular surfaces.

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2. PRELIMINARIES

For further background on Hilbert modular forms we recommend [Freitag; Geer88; DembeleVoight].

2.1. **Basic notation.** We adopt the same notation as in our previous work [HMSDB]. Let F be a totally real field of degree $n := [F : \mathbb{Q}]$ with discriminant d_F , ring of integers $\mathbb{Z}_F$, and class group $\text{Cl } \mathbb{Z}_F$ with cardinality $h(\mathbb{Z}_F) := \# \text{Cl } \mathbb{Z}_F$. We number the real places $v_i : F \hookrightarrow \mathbb{R}$ with $i = 1, \dots, n$ (e.g., ordering the roots of the `polredabs` [`pari'gp`] defining polynomial for F). For $a \in F$, we write $a_i := v_i(a) \in \mathbb{R}$, extended entrywise to $\alpha \in \text{M}_2(F)$ so that $\alpha_i \in \text{M}_2(\mathbb{R})$. Let

$$(2.1.1) \quad F_{>0}^\times := \{a \in F^\times : a_v > 0 \text{ for all real places } v\}$$

be the group of totally positive elements in F . For a fractional $\mathbb{Z}_F$ -ideal $\mathfrak{a}$, let $\mathfrak{a}_{>0}^\times := \mathfrak{a} \cap F_{>0}^\times$, and $\mathbb{Z}_{F,>0}^\times := \mathbb{Z}_F^\times \cap F_{>0}^\times$. Let $\text{Cl}^+ \mathbb{Z}_F$ be the narrow class group of $\mathbb{Z}_F$ and $h^+(\mathbb{Z}_F) := \# \text{Cl}^+ \mathbb{Z}_F$ its cardinality. We write $\text{Tr} := \text{Tr}_{F/\mathbb{Q}} : F \rightarrow \mathbb{Q}$ for the absolute trace. To avoid confusion, we write tr for matrix trace.

2.2. **Hilbert modular varieties.** Let $\mathfrak{H}_1$ denote the complex upper half-plane. The group

$$(2.2.1) \quad \text{GL}_2^+(F) := \{\gamma \in \text{GL}_2(F) : \det \gamma \in F_{>0}^\times\}$$

acts by orientation-preserving isometries on $\mathcal{H} := \mathfrak{H}_1^n$ via coordinate-wise linear fractional transformations.

For a nonzero ideal $\mathfrak{N} \subseteq \mathbb{Z}_F$ and a (nonzero) fractional $\mathbb{Z}_F$ -ideal $\mathfrak{b} \subset F$, we define the standard congruence subgroups of level $\mathfrak{N}$ by

$$\begin{aligned}
(2.2.2) \quad \Gamma_0(\mathfrak{N})_{\mathfrak{b}} &:= \begin{pmatrix} \mathbb{Z}_F & \mathfrak{b}^{-1} \\ \mathfrak{N}\mathfrak{b} & \mathbb{Z}_F \end{pmatrix} \cap \det^{-1}(\mathbb{Z}_{F,>0}^{\times}) \\
&= \left\{ \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2^+(F) : a, d \in \mathbb{Z}_F, \ b \in \mathfrak{b}^{-1}, \ c \in \mathfrak{N}\mathfrak{b}, \ \det \gamma \in \mathbb{Z}_{F,>0}^{\times} \right\}, \\
\Gamma_1(\mathfrak{N})_{\mathfrak{b}} &:= \begin{pmatrix} 1 + \mathfrak{N} & \mathfrak{b}^{-1} \\ \mathfrak{N}\mathfrak{b} & \mathbb{Z}_F \end{pmatrix} \cap \det^{-1}(\mathbb{Z}_{F,>0}^{\times})
\end{aligned}$$

so that $\Gamma_1(\mathfrak{N})_{\mathfrak{b}} \trianglelefteq \Gamma_0(\mathfrak{N})_{\mathfrak{b}} < \mathrm{GL}_2^+(F)$. Conjugation by $\mathrm{diag}(\alpha^{-1}, 1)$ for $\alpha \in F_{>0}^{\times}$ defines an isomorphism $\Gamma_i(\mathfrak{N})_{\mathfrak{b}} \simeq \Gamma_i(\mathfrak{N})_{\alpha\mathfrak{b}}$ for $i = 0, 1$, so without loss of generality we take the fractional ideals $\mathfrak{b}$ among a choice of representatives of $\mathrm{Cl}^+ \mathbb{Z}_F$. Finally, we define

$$(2.2.3) \quad \Gamma_i^1(\mathfrak{N})_{\mathfrak{b}} := \ker(\det|_{\Gamma_i(\mathfrak{N})_{\mathfrak{b}}}) = \Gamma_i(\mathfrak{N})_{\mathfrak{b}} \cap \mathrm{SL}_2(F)$$

for $i = 0, 1$, giving an exact sequence

$$(2.2.4) \quad 1 \rightarrow \Gamma_i^1(\mathfrak{N})_{\mathfrak{b}} \rightarrow \Gamma_i(\mathfrak{N})_{\mathfrak{b}} \rightarrow \mathbb{Z}_{F,>0}^{\times} \rightarrow 1.$$

Let $\Gamma = \Gamma_i(\mathfrak{N})_{\mathfrak{b}} < \mathrm{GL}_2^+(F)$ be a standard congruence subgroup. Then, the action of Γ on $\mathcal{H}$ is by a discrete group acting properly. The quotient $Y(\Gamma)(\mathbb{C}) := \Gamma \backslash \mathcal{H}$ is a (connected) complex orbifold of dimension n , an n -dimensional complex analytic space which is the analytification of a quasi-projective variety $Y(\Gamma)$ called a **Hilbert modular variety**. Hilbert modular varieties are coarse moduli spaces for polarized abelian varieties with real multiplication and level structure.

The Baily–Borel compactification of $Y(\Gamma)$ is obtained by adding finitely many points as follows. Let $\mathcal{H}^* := \mathcal{H} \cup \mathbb{P}^1(F)$ be the completed product of upper half-planes with its natural topology and the extended action of Γ . The **cusps** of Γ are the orbits of $\mathbb{P}^1(F)$ under Γ . The quotient $\overline{Y}(\Gamma)(\mathbb{C}) := \Gamma \backslash \mathcal{H}^*$ obtained from adding cusps is compact; moreover, the variety $\overline{Y}(\Gamma)$ is proper but singular at the cusps when $F \neq \mathbb{Q}$.

To obtain a variety with a canonical model defined over a number field admitting the full action of Hecke operators, we gather components (in a union over idèlically conjugate groups) and define

$$(2.2.5) \quad \overline{Y}_i(\mathfrak{N}) := \bigsqcup_{[\mathfrak{b}] \in \mathrm{Cl}^+ \mathbb{Z}_F} \overline{Y}(\Gamma_i(\mathfrak{N})_{\mathfrak{b}})$$

for $i = 0, 1$, and similarly the variant $\overline{Y}_i^1(\mathfrak{N})$. The variety $\overline{Y}_i(\mathfrak{N})$ arises naturally from an idèlic description [Voight], better for theoretical purposes; the equivalent description (2.2.5) is more accessible for our algorithmic purposes.

Note that it may happen that different standard congruence subgroups give rise to the same Hilbert modular variety. For instance, in the special case where $\mathrm{Cl}^+ \mathbb{Z}_F = \mathrm{Cl} \mathbb{Z}_F$, we have $\mathbb{Z}_{F,>0}^{\times} = \mathbb{Z}_F^{\times 2}$ so the images of $\Gamma_0^1(\mathfrak{N})_{\mathfrak{b}}$ and $\Gamma_0(\mathfrak{N})_{\mathfrak{b}}$ agree in $\mathrm{PGL}_2^+(F)$ for every $\mathfrak{b}$. The associated Hilbert modular varieties $\overline{Y}_0(\mathfrak{N})$ and $\overline{Y}_0^1(\mathfrak{N})$ are then identical because the action of $\mathrm{GL}_2^+(F)$ on $\mathcal{H}$ factors through $\mathrm{PGL}_2^+(F)$.

ub:hmf-def}

2.3. Hilbert modular forms. We now define the central objects of study in this paper.

Definition 2.3.1. A **weight** is a vector $(k, t) \in \mathbb{Z}^n \times \mathbb{R}^n$. When all k_i are equal, we say the weight k is **parallel**; when all k_i have the same parity (congruent modulo 2), we say k is **paritious**.

For $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2^+(F)$ and $z \in \mathcal{H}$, we define the **automorphy factor** for weight (k, t) as

$$(2.3.2) \quad j_{k,t}(\alpha; z) := \prod_i \frac{(c_i z_i + d_i)^{k_i}}{\det(\alpha_i)^{k_i + t_i - 1}},$$

where we take the positive real value of $\det(\alpha_i)^{k_i + t_i - 1}$ in the denominator. Given a holomorphic function $f: \mathcal{H} \rightarrow \mathbb{C}$ and a matrix $\alpha \in \mathrm{GL}_2^+(F)$, we define

$$(f|_{k,t}\alpha)(z) := j_{k,t}^{-1}(\alpha; z)f(\alpha z).$$

{def:HMF}

Definition 2.3.3. Let Γ be a standard congruence subgroup, as in Eq. (2.2.2). A **Hilbert modular form** over F of weight (k, t) and level Γ is a holomorphic function $f: \mathcal{H} \rightarrow \mathbb{C}$ such that

$$(2.3.4) \quad (f|_{k,t}\gamma) = f$$

for all $\gamma \in \Gamma$, together with the additional condition that f remains bounded in vertical strips if $F = \mathbb{Q}$.

We write $M_{k,t}(\Gamma)$ for the $\mathbb{C}$ -vector space of Hilbert modular forms for Γ of weight (k, t) .

Remark 2.3.5. The boundedness condition in Equation (2.3.3) is automatic when $F \neq \mathbb{Q}$ (equivalently, $n \geq 2$) by the Koecher principle.

We have $\dim_{\mathbb{C}} M_{k,t}(\Gamma) < \infty$. Different choices of t are convenient for different applications. In this paper, we take $t = \tilde{t}(k) := 1 - \frac{k}{2}$.

Remark 2.3.6. This choice is convenient from an analytic/representation theoretic point of view as the automorphic forms associated to elements of $M_{k,\tilde{t}(k)}(\Gamma)$ have central characters with trivial infinity type. Furthermore, this choice means that the product of forms of weight k and k' is a form of weight $k + k'$.

Proposition 2.3.7. *The space $M_{k,t}(\Gamma)$ is $\{0\}$ unless both of the following conditions hold:*

- (1) $k_i + 2t_i$ is independent of i ;
- (2) Either $k_i > 0$ for all i , or $k = 0$ for all i .

Proof. The standard congruence subgroups always contain $\begin{pmatrix} \varepsilon & 0 \\ 0 & \varepsilon \end{pmatrix}$ for ε in some finite-index subgroup of $\mathbb{Z}_{F,>0}^\times$. Plugging this into Eq. (2.3.3), we see by Dirichlet's unit theorem that $M_{k,t}(\Gamma) = \{0\}$ unless $k_i + 2t_i$ is independent of i . This proves the first claim.

By Remark 2.3 of [DembeleLoefflerPacetti], if we replace t by $t' := t + h \cdot \mathbf{1}$ for any $h \in \mathbb{R}$, the resulting space $M_{k,t'}(\Gamma)$ is isomorphic to $M_{k,t}(\Gamma)$. Taking $t = \tilde{t}(k)$, we can then apply Proposition 1.1 of [Shimura] to prove the second claim. $\square$

Let

$$(2.3.8) \quad j_k(\alpha; z) := j_{k, \tilde{t}(k)}(\alpha, z) := \prod_i \frac{(c_i z_i + d_i)^{k_i}}{\det(\alpha_i)^{k_i/2}},$$

where we take positive square roots in the denominator, and similarly

$$M_k(\Gamma) := M_{k, \tilde{t}(k)}(\Gamma).$$

Example 2.3.9. If $\alpha = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$ is a diagonal matrix, then since we take positive square roots in Eq. (2.3.8),

$$(2.3.10) \quad j_k(\alpha; z) = \prod_i \frac{d_i^{k_i}}{(a_i d_i)^{k_i/2}} = \prod_i \operatorname{sgn}(d_i)^{k_i} \left(\frac{d_i}{a_i} \right)^{k_i/2}$$

using that $x = \operatorname{sgn}(x)(x^{1/2})^2$ for $x \in \mathbb{R}^\times$. In particular, when $\alpha = a \cdot 1_2$ where $a \in F^\times$, we get $j_k(\alpha; z) = \prod_i \operatorname{sgn}(a_i)^{k_i}$; and if further each k_i is even, then $j_k(\alpha; z) = 1$.

For $(f_1, f_2) \in M_{k_1}(\Gamma) \times M_{k_2}(\Gamma)$ we have $f_1 \cdot f_2 \in M_{k_1+k_2}(\Gamma)$, which leads to the definition of the graded ring of Hilbert modular forms of level Γ

$$(2.3.11) \quad M(\Gamma) := \bigoplus_k M_k(\Gamma).$$

Definition 2.3.12. A Hilbert modular form $f \in M_k(\Gamma)$ is a **cusp form** if $f(z) \rightarrow 0$ as $z \rightarrow c$ for c ranging over a set of representatives of the cusps of Γ .

We write $S_k(\Gamma) \subseteq M_k(\Gamma)$ for the subspace of cusp forms. There is a natural integration pairing, called the **Petersson inner product**, between $M_k(\Gamma)$ and $S_k(\Gamma)$ which provides an orthogonal decomposition

$$(2.3.13) \quad M_k(\Gamma) = \mathcal{E}_k(\Gamma) \oplus S_k(\Gamma);$$

we call $\mathcal{E}_k(\Gamma)$ the **Eisenstein subspace**. See Section 4.1 for further discussion of Eisenstein series.

Finally, we gather components as in (2.2.5).

Definition 2.3.14. A Hilbert modular form of weight k on $\overline{Y}_i(\mathfrak{N})$ is a tuple $(f_{\mathfrak{b}})_{[\mathfrak{b}] \in \mathrm{Cl}^+(\mathbb{Z}_F)}$ where each $f_{\mathfrak{b}}$ is a Hilbert modular form of weight k on $\Gamma_i(\mathfrak{N})_{\mathfrak{b}}$.

In light of Eq. (2.3.3), this is a slight abuse of notation, but it will be clear in context whether or not we are referring to the form on a single component or on the entire modular variety. We write

$$(2.3.15) \quad M_k(\Gamma_0(\mathfrak{N})) := \bigoplus_{[\mathfrak{b}] \in \mathrm{Cl}^+ R} M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}})$$

for the $\mathbb{C}$ -vector space of Hilbert modular forms of weight k on $\overline{Y}_0(\mathfrak{N})$. We define similarly all the variants $S_k(\Gamma_0(\mathfrak{N}))$, $M_k(\Gamma_1^1(\mathfrak{N}))$, etc.

2.4. Fourier expansions. In this paper, we work explicitly with Hilbert modular forms via their Fourier expansions at the cusp ∞ , which we now introduce.

The stabilizer of the cusp $\infty \in \mathbb{P}^1(F)$ under the action of a standard congruence subgroup $\Gamma_{\mathfrak{b}}$ is the subgroup of elements of the form

$$\begin{pmatrix} \mathbb{Z}_F^\times & \mathfrak{b}^{-1} \\ 0 & \mathbb{Z}_F^\times \end{pmatrix} \cap \Gamma_{\mathfrak{b}}$$

acting by $z \mapsto (a_i z_i + b_i)_i$ with $a \in \mathbb{Z}_{F, > 0}^\times$ and $b \in \mathfrak{b}^{-1}$. In particular, in each case this stabilizer group contains the subgroup of translations by elements in $\mathfrak{b}^{-1}$; since $j_k(\gamma; z) = 1$ for such a translation, a Hilbert modular form for $\Gamma_{\mathfrak{b}}$ is invariant under this subgroup and therefore admits a Fourier expansion indexed by the dual lattice to $\mathfrak{b}^{-1}$. More precisely, for any $\mathbb{Z}$ -submodule $I \subset F$ let

$$(2.4.1) \quad I^\sharp := \{x \in F : \text{Tr}(xI) \subseteq \mathbb{Z}\}$$

be the dual of I under the trace pairing. If $I = \mathfrak{a}$ is a fractional $\mathbb{Z}_F$ -ideal, then so is $\mathfrak{a}^\sharp$. The fractional $\mathbb{Z}_F$ -ideal $\mathbb{Z}_F^\sharp \supseteq \mathbb{Z}_F$ is the **codifferent** of $\mathbb{Z}_F$, and the **different** $\mathfrak{d}_F := (\mathbb{Z}_F^\sharp)^{-1}$ has norm $\text{Nm}((\mathbb{Z}_F^\sharp)^{-1}) = d_F \mathbb{Z}$, generated by the discriminant of F .

Now let $f \in M_k(\Gamma_{\mathfrak{b}})$. Since f is holomorphic, the Fourier expansion of f at ∞ is a series indexed by 0 and totally positive elements of the fractional ideal

$$(2.4.2) \quad \mathfrak{b}' := (\mathfrak{b}^{-1})^\sharp = \mathfrak{b} \mathbb{Z}_F^\sharp = \mathfrak{b} \mathfrak{d}_F^{-1}.$$

written

$$(2.4.3) \quad f(z) = a_0 + \sum_{\nu \in (\mathfrak{b} \mathbb{Z}_F^\sharp)_{> 0}} a_\nu \exp(2\pi\sqrt{-1}(\nu_1 z_1 + \cdots + \nu_n z_n))$$

with $a_\nu = a_\nu(f) \in \mathbb{C}$ (and with $z = (z_i)_i \in \mathcal{H}$). We will use the rather abusive—but nevertheless convenient—abbreviation using multi-index notation

$$(2.4.4) \quad q^\nu := \exp(2\pi\sqrt{-1}(\nu_1 z_1 + \cdots + \nu_n z_n)).$$

With this, we can rewrite Eq. (2.4.3) as

$$(2.4.5) \quad f(z) = a_0 + \sum_{\nu \in (\mathfrak{b} \mathbb{Z}_F^\sharp)_{> 0}} a_\nu q^\nu.$$

Remark 2.4.6. To work with a general congruence subgroup, the subgroup of translations (whose totally nonnegative elements index Fourier expansions) need not be a fractional $\mathbb{Z}_F$ -ideal; but we could still describe it as a module for an order $R \subseteq \mathbb{Z}_F$ and work with a fractional R -ideal, or instead just describe it as a $\mathbb{Z}$ -submodule of F with a basis.

The action of the unit group implies further relations among these coefficients for forms in $M_k(\Gamma_1(\mathfrak{N}))$. More precisely, for any $\varepsilon \in \mathbb{Z}_{F, > 0}^\times$, let $\gamma := \begin{pmatrix} 1 & \\ & \varepsilon \end{pmatrix} \in \Gamma_1(\mathfrak{N})$. Then for $f \in M_k(\Gamma_{\mathfrak{b}})$ we obtain

$$(2.4.7) \quad f(\gamma z) = f(\varepsilon^{-1} z) = j_k(\gamma; z) f(z)$$

where by Eq. (2.3.10),

$$(2.4.8) \quad j_k(\gamma; z) = \text{sgn}(a)^k \varepsilon^{k/2}.$$

If $M_k(\Gamma_{\mathfrak{b}}) \neq \{0\}$, substituting (2.4.7) into the Fourier expansion (2.4.5), by the uniqueness of the q -expansion we have

$$(2.4.9) \quad a_{\nu\varepsilon}(f) = \varepsilon^{k/2} a_{\nu}(f)$$

for all ν .

Continue assuming that $f \in M_k(\Gamma_1(\mathfrak{N}))$. Following [Shimura], if $\mathfrak{n}$ is the integral ideal $\nu\mathfrak{b}'^{-1}$, then we define

$$(2.4.10) \quad a_{\mathfrak{n}} := a_{\nu}\nu^{-k/2} \text{Nm}(\mathfrak{n})^{k_0/2}.$$

By Eq. (2.4.9), this is independent of the choice of ν .

Remark 2.4.11. If f is a normalized Hecke eigenform in $S_k(\Gamma_0(\mathfrak{N}), \chi)$ (see Eq. (2.6.5)), the local factor of the (appropriately normalized) L -function associated to f at a prime ideal $\mathfrak{p} \subset \mathbb{Z}_F$ is

$$(2.4.12) \quad L_{\mathfrak{p}}(s) := (1 - a_{\mathfrak{p}} \text{Nm}(\mathfrak{p})^{-s} + \chi(\mathfrak{p}) \text{Nm}(\mathfrak{p})^{k_0-1-2s})^{-1}$$

and the local factor at an archimedean place v is (up to constants) $\Gamma\left(s - \frac{k_0 - k_v}{2}\right)$, where k_v is the component of the weight associated to v and where $k_0 := \max_v k_v$. This explains the strange-looking normalization in Eq. (2.4.10) – it lets us relate the Fourier coefficients of f to the Satake parameters of the associated (appropriately normalized) automorphic representation. Many formulas in the literature for producing Hilbert modular forms from other sources are given by specifying the $a_{\mathfrak{n}}$ for integral ideals $\mathfrak{n}$, and we can obtain the Fourier coefficients (for a given choice of ideal class representative) using Eq. (2.4.10).

Remark 2.4.13. Looking at Eq. (2.4.9), it may seem that the field of definition of coefficients needs to include square roots of totally positive units when k is odd. However, because ε is totally positive, $\varepsilon^{k_0/2} = 1$ and we have

$$\varepsilon^{k/2} = \varepsilon^{(k-k_0)/2} \varepsilon^{k_0/2} = \varepsilon^{(k-k_0)/2}.$$

As long as k is paritious, this means that we do not need to worry about square roots of units when transforming coefficients as in Eq. (2.4.9). Even when k is nonparitious, if $h^+(F) = 1$, every totally positive unit is a square and so we again can avoid computing square roots of units.

Remark 2.4.14. By the same argument we can compare Fourier expansions corresponding to different choices of representative ideals for the narrow class group. For all $\alpha \in F_{>0}^{\times}$, as noted after (2.2.2), a form $f(z)_{\mathfrak{b}}$ on $\Gamma_{\mathfrak{b}}$ gives rise to a form $f(\alpha z)_{\alpha\mathfrak{b}}$ on $\Gamma_{\alpha\mathfrak{b}}$ with Fourier coefficients

$$a_{\alpha\nu}(f(\alpha z)_{\alpha\mathfrak{b}}) = a_{\nu}(f(z)_{\mathfrak{b}}).$$

2.5. Hecke characters. In this subsection, we review the theory of Hecke characters. This will be necessary both to define Hilbert modular forms with nebentypus in Section 2.3 and when discussing automorphic induction in Section 4.3.

Let K be a number field. Let $\mathbb{A}_K^{\times}$ be the idèle group of K .

Definition 2.5.1. A Hecke character χ over the number field K is a continuous homomorphism (not necessarily finite-order) $\chi: K^{\times} \backslash \mathbb{A}_K^{\times} \rightarrow \mathbb{C}^{\times}$.

The notation $\mathfrak{p}$ will always denote a (finite) prime of $\mathbb{Z}_K$ and we write $K_{\mathfrak{p}}$ and $\mathbb{Z}_{\mathfrak{p}}$ for the completions at $\mathfrak{p}$ of K and $\mathbb{Z}_K$, respectively. Let $\widehat{K}^{\times}$ be the restricted product $\prod'_{\mathfrak{p}}(K_{\mathfrak{p}}^{\times}; \mathbb{Z}_{\mathfrak{p}}^{\times})$. Because $\mathbb{C}^{\times}$ has no small subgroups¹, the kernel of $\chi|_{\widehat{K}^{\times}}$ is an open subgroup of $\widehat{K}^{\times}$. At each finite prime $\mathfrak{p}$, let $\varpi_{\mathfrak{p}}$ be a uniformizer of $\mathbb{Z}_{\mathfrak{p}}$, and let $v_{\mathfrak{p}}$ be the smallest integer such that $1 + \varpi_{\mathfrak{p}}^{v_{\mathfrak{p}}} \mathbb{Z}_{\mathfrak{p}} \subset \ker \chi|_{\widehat{K}^{\times}}$. We define the **conductor** of χ , denoted by $\text{cond}(\chi)$, to be the ideal $\prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}}$. The conductor does not depend on the choices of uniformizers.

Again choosing uniformizers at every finite place, we decompose $K_{\mathfrak{p}}^{\times} \cong \mathbb{Z}_{\mathfrak{p}}^{\times} \times \varpi_{\mathfrak{p}}^{\mathbb{Z}}$. This will be used to send fractional ideals to finite idèles, and we will show later that the choices of uniformizers do not matter. Let r and s be the number of real and complex places of K , respectively. Let $I_K^{\mathfrak{N}}$ and $I_{K,\mathfrak{N}}$ be the groups of fractional ideals of $\mathbb{Z}_K$ coprime to $\mathfrak{N}$ and supported only on the primes dividing $\mathfrak{N}$, respectively. We can decompose the idèle group as

$$(2.5.2) \quad \mathbb{A}_K^{\times} \simeq ((\mathbb{R}^{\times})^r \times (\mathbb{C}^{\times})^s) \times I_K^{\mathfrak{N}} \times I_{K,\mathfrak{N}} \times \left(\prod_{\mathfrak{p} \subset \mathbb{Z}_K} \mathbb{Z}_{\mathfrak{p}}^{\times} \right).$$

As such, we can decompose any Hecke character χ as the product $\chi_{\infty} \cdot \chi_I^{\mathfrak{N}} \cdot \chi_{I,\mathfrak{N}} \cdot \chi_0$, where the four factors are the restriction of χ to the four factors in (2.5.2) respectively. Because χ is trivial on $K^{\times}$, weak approximation tells us that $\chi_{I,\mathfrak{N}}$ is determined by χ_{∞} , $\chi_I := \chi_I^{\mathfrak{N}}$, and χ_0 . In practice we specify a Hecke character by giving these three characters. These three factors are also not independent. First, we must have $\chi_0|_{\mathbb{Z}_K^{\times}} = \chi_{\infty}^{-1}|_{\mathbb{Z}_K^{\times}}$, since χ_I is trivial on $(\varepsilon) = (1)$ for $\varepsilon \in \mathbb{Z}_K^{\times}$. Second, for any $\alpha \in K^{\times}$ coprime to $\mathfrak{N}$, we must have $\chi_I((\alpha)) = (\chi_{\infty}\chi_0)^{-1}(\alpha)$. By the previous sentence, $\chi_{\infty}\chi_0$ is unchanged by replacing α with $\varepsilon\alpha$ for $\varepsilon \in \mathbb{Z}_K^{\times}$, so this condition is well-defined.

We call χ_{∞} the **infinity type** of χ , $\chi_I := \chi_I^{\mathfrak{N}}$ the **ideal character** corresponding to χ , and χ_0 the **Dirichlet character** corresponding to χ . The conductor of χ_0 as a Dirichlet character is the same as $\text{cond}(\chi)$.

Remark 2.5.3. This is a useful way to construct Hecke characters in practice. First pick an infinity type χ_{∞} . Then, choose a Dirichlet character χ_0 such that $\chi_{\infty}\chi_0$ is trivial on units. Lastly, choose a character χ_I of $I_K^{\mathfrak{N}}$ which agrees with $(\chi_{\infty}\chi_0)^{-1}$ on principal ideals coprime to $\mathfrak{N}$.

Recall that the continuous $\mathbb{C}^{\times}$ -valued characters of $\mathbb{R}^{\times}$ are of the form $x \mapsto \text{sgn}(x)^{\varepsilon} |x|^s$ for $\varepsilon \in \{0, 1\}$ and $s \in \mathbb{C}$, and the continuous $\mathbb{C}^{\times}$ -valued characters of $\mathbb{C}^{\times}$ are $z \mapsto \left(\frac{z}{|z|}\right)^n |z|^s = z^{s+\frac{n}{2}} \bar{z}^{s-\frac{n}{2}}$ for $n \in \mathbb{Z}$ and $s \in \mathbb{C}$. Therefore, we may write $\chi_{\infty} = (\chi_v)_{v|\infty}$

$$\begin{cases} \chi_v(x) = \text{sgn}(x)^{\varepsilon_v} |x|^{n_v} & v \text{ real.} \\ \chi_v(z) = z^{a_v} \bar{z}^{b_v} & v \text{ complex} \end{cases}$$

for $\varepsilon_v \in \{0, 1\}$ and $a_v, b_v, n_v \in \mathbb{C}$ such that $a_v - b_v \in \mathbb{Z}$. Because Hecke characters must be trivial on units, one can check using the Dirichlet unit theorem that there exists a real number $c \in \mathbb{R}$ such that for every infinite place v of K ,

$$(2.5.4) \quad c = \begin{cases} 2n_v & v \text{ real} \\ a_v + b_v & v \text{ complex.} \end{cases}$$

¹https://en.wikipedia.org/wiki/No_small_subgroup

It follows that n_v is the same for all real places. A Hecke character is finite order if and only if all the a_v, b_v , and n_v are 0. In particular, if K is totally real, then $\chi| \cdot |^{-c}$ is finite order.

Remark 2.5.5. This remark relates the idèlic notion of Hecke characters that we have introduced in Eq. (2.5.1) to the classical notion of Hecke characters and the notion used in Magma. By finiteness of the image of χ_0 and of the ideal class group, χ has finite order if and only if χ_∞ is trivial. Equivalently, χ is finite order if $a_v = b_v = n_v = 0$ in Section 2.5. Such a χ is entirely determined by its associated ideal character. By strong approximation, it is even determined by its ideal character χ_I restricted to the ideals coprime to $\mathfrak{N}$. Since $\chi_I \chi_0 = 1$ on principal ideals, χ_I should be 1 on any (α) , $\alpha \in K^\times$, for which $\chi_0(\alpha) = 1$. In particular, if $\tilde{\mathfrak{N}} = \mathfrak{N}\mathfrak{N}_\infty$ is the conductor of χ , χ_I should be trivial on

$$\text{Prin}_K^{\mathfrak{N},1} := \{(\alpha) : v_{\mathfrak{p}}(\alpha - 1) \geq v_{\mathfrak{p}}(\mathfrak{N}) \text{ for all } \mathfrak{p} \mid \mathfrak{N} \text{ and } v(\alpha) > 0 \text{ for all } v \in \mathfrak{N}_\infty\},$$

where $v_{\mathfrak{p}}$ is the discrete valuation associated to the prime ideal $\mathfrak{p}$. Therefore, we can view finite order Hecke characters as characters of the ray class group $\text{Cl}_K^{\mathfrak{N}} := I_K^{\mathfrak{N}} / \text{Prin}_K^{\mathfrak{N},1}$ with conductor $\mathfrak{N}$.

If $\tilde{\mathfrak{N}} = \mathfrak{N}\mathfrak{N}_\infty$ and $\tilde{\mathfrak{M}} = \mathfrak{M}\mathfrak{M}_\infty$, we write $\tilde{\mathfrak{N}} \mid \tilde{\mathfrak{M}}$ if $\mathfrak{N} \mid \mathfrak{M}$ and $\mathfrak{N}_\infty \subseteq \mathfrak{M}_\infty$. There are quotient maps $\text{Cl}_K^{\tilde{\mathfrak{M}}} \rightarrow \text{Cl}_K^{\tilde{\mathfrak{N}}}$ whenever $\tilde{\mathfrak{N}} \mid \tilde{\mathfrak{M}}$. Any ray class character χ of $\text{Cl}_K^{\tilde{\mathfrak{M}}}$ is of **modulus** $\mathfrak{M}$, and its **conductor** is the largest ideal $\mathfrak{N}$ such that χ factors through $\text{Cl}_K^{\tilde{\mathfrak{N}}}$. A ray class character is **primitive** if its modulus is equal to its conductor. Given a ray class character χ of conductor $\mathfrak{N}$, we write χ_0 for the character of modulus $\mathfrak{N}$ through which χ factors. The ray class characters which arise as the ideal characters of Hecke characters will always be primitive.

What Magma calls a Hecke character (i.e. an object of type `GrpHeckeElt`) is a (not necessarily primitive) ray class character. What Magma calls the "Dirichlet restriction" of a Hecke character is then the restriction of this ray class character to the principal ideals (α) which are coprime to the finite modulus.

nebensubtypus}

2.6. Nebentypus characters. In the classical theory of modular forms, the group $\Gamma_0(N)$ acts on the space $M_k(\Gamma_1(N))$ through the quotient $\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$, so we have a decomposition

$$(2.6.1) \quad M_k(\Gamma_1(N)) \cong \bigoplus_{\psi: (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times} M_k(\Gamma_0(N), \psi).$$

Similarly, in the Hilbert modular setting there is an isomorphism of groups

$$(2.6.2) \quad \Gamma_0(\mathfrak{N})_{\mathfrak{b}} / \Gamma_1(\mathfrak{N})_{\mathfrak{b}} \rightarrow (\mathbb{Z}_F / \mathfrak{N})^\times$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto a$$

giving an action of $(\mathbb{Z}_F / \mathfrak{N})^\times$ on $M_k(\Gamma_1(\mathfrak{N}))$, denoted $\langle a \rangle$ and called **diamond operators**. Decomposing according to eigenspaces for this action, we obtain the following analogue of Eq. (2.6.1).

$$(2.6.3) \quad M_k(\Gamma_1(\mathfrak{N})_{\mathfrak{b}}) \cong \bigoplus_{\psi: (\mathbb{Z}_F / \mathfrak{N})^\times \rightarrow \mathbb{C}^\times} M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}}, \psi).$$

Definition 2.6.4. We define $M_k(\mathfrak{N}, \psi)_{\mathfrak{b}} := M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}}, \psi)$ to be the subspace of $M_k(\Gamma_1(\mathfrak{N})_{\mathfrak{b}})$ such that for any $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(\mathfrak{N})_{\mathfrak{b}}$,

$$f(\gamma z) = \psi(a)j_k(\gamma; z)f(z).$$

Before we proceed, we introduce some notation. We say that a character $\psi: (\mathbb{Z}_F/\mathfrak{M})^\times \rightarrow \mathbb{C}^\times$ is **primitive** if it does not factor through some $(\mathbb{Z}_F/\mathfrak{N})^\times$ for $\mathfrak{N}$ strictly dividing $\mathfrak{M}$. Otherwise, we write ψ_{prim} for the primitive character through which ψ factors.

With this notation in hand, we can refine the decomposition in Eq. (2.6.3) further for each ψ by using (finite order) Hecke characters χ such that $\chi_0 = \psi_{\text{prim}}$. By Proposition 2.1 of [Shimura], for each ψ in Eq. (2.6.3),

$$(2.6.5) \quad M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}}, \psi) \cong \bigoplus_{\chi: \chi_0 = \psi_{\text{prim}}} M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}}, \chi).$$

When the class number of F is greater than one, this is a nontrivial refinement of the decomposition in Eq. (2.6.3). If $f \in M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}}, \chi)$, we say that χ is the **nebentypus** of f . In this paper, we will always take our nebentypen χ to be Hecke characters, and write χ_0 whenever we want to work with the associated Dirichlet characters. Given a Hecke character χ , we write

$$M_k(\mathfrak{N}, \chi) := \bigoplus_{\mathfrak{b}} M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}}, \chi).$$

Let

$$(2.6.6) \quad T := \begin{pmatrix} \mathbb{Z}_F^\times & 0 \\ 0 & \mathbb{Z}_F^\times \end{pmatrix} \subset \text{GL}_2^+(F).$$

Definition 2.6.7. We say that a Hecke character χ is **unit compatible** with a weight k if for all $\varepsilon \in T \cap \Gamma_{\mathfrak{b}}$ we have

$$(2.6.8) \quad \chi_0(\varepsilon) = \text{sgn}(\varepsilon)^k := \prod_{i=1}^n \text{sgn}(\varepsilon_i)^{k_i},$$

and we say χ has **compatible parity** with weight k if $\chi(-1) = (-1)^{\sum_{i=1}^n k_i}$.

In particular, if χ is unit compatible, then $\chi_0(\varepsilon) = 1$ for all $\varepsilon \in \mathbb{Z}_{F, >0}^\times \cap \Gamma_{\mathfrak{b}}$.

Lemma 2.6.9. *We have $M_k(\Gamma_0(\mathfrak{N}), \chi) = \{0\}$ if χ is not unit compatible with k , and $M_k(\Gamma_0^1(\mathfrak{N}), \chi) = \{0\}$ if χ does not have compatible parity with k .*

Proof. Let $f \in M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}}, \chi)$. If $\alpha = a \cdot 1_2 \in \Gamma_0(\mathfrak{N})_{\mathfrak{b}}$, where $a \in \mathbb{Z}_F^\times$, then α acts trivially on $\mathcal{H}$. By Equation (2.3.9),

$$f(z) = f(\alpha z) = \chi(a)j_k(\alpha; z)f(z) = \chi(a) \text{sgn}(a)^k f(z)$$

for all $z \in \mathcal{H}$, where $\text{sgn}(a)^k = \prod_i \text{sgn}(a_i)^{k_i}$. Therefore, by Eq. (2.6.8), $f = 0$ unless χ is compatible with k .

A similar argument with the scalar matrix -1 applies for $\Gamma_0^1(\mathfrak{N})_{\mathfrak{b}}$. □

In particular, we may restrict the sums (2.6.3) and (2.6.5) to the characters χ which are compatible with k .

c:unitchar}

2.7. Unit characters. In this subsection, we give a decomposition (Eq. (2.7.6)) of $M_k(\Gamma_i^1(\mathfrak{N}))$ into spaces indexed by characters of a finite abelian group $U_i(\mathfrak{N})$ (defined in Eq. (2.7.3)), which we call “unit characters”. Readers interested only in the case of level $\Gamma_0(\mathfrak{N})$ and $\Gamma_1(\mathfrak{N})$ may safely skip this section.

Let T be as in (2.6.6) and consider the group homomorphism

$$(2.7.1) \quad \mu: \Gamma \cap T \rightarrow \mathbb{Z}_{F,>0}^\times$$

$$\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \mapsto a/d$$

which we call the **unit ratio** map. The unit ratio map evidently factors through the quotient by the unit scalar subgroup $D := \mathbb{Z}_F^\times \cap \Gamma$. Define

$$\mathbb{Z}_{F,\mathfrak{N}}^1 := \{\varepsilon \in \mathbb{Z}_{F,>0}^\times : \varepsilon \equiv 1 \pmod{\mathfrak{N}}\}$$

and let $(\mathbb{Z}_{F,\mathfrak{N}}^1)^2 \leq \mathbb{Z}_{F,\mathfrak{N}}^1$ be the subgroup of squares.

unitratio}

Lemma 2.7.2. *The unit ratio map has image*

$$\mu(\Gamma \cap T) = \begin{cases} \mathbb{Z}_{F,>0}^\times, & \text{if } \Gamma = \Gamma_0(\mathfrak{N})_{\mathfrak{b}} \text{ or } \Gamma = \Gamma_1(\mathfrak{N})_{\mathfrak{b}}; \\ (\mathbb{Z}_F^\times)^2, & \text{if } \Gamma = \Gamma_0^1(\mathfrak{N})_{\mathfrak{b}}; \\ (\mathbb{Z}_{F,\mathfrak{N}}^1)^2, & \text{if } \Gamma = \Gamma_1^1(\mathfrak{N})_{\mathfrak{b}}. \end{cases}$$

Proof. Direct calculation. □

The map μ induces an isomorphism

$$(2.7.3) \quad (\Gamma_i(\mathfrak{N})_{\mathfrak{b}} \cap T)/D \cdot (\Gamma_i^1(\mathfrak{N})_{\mathfrak{b}} \cap T) \xrightarrow{\sim} U_i(\mathfrak{N}) := \mathbb{Z}_{F,>0}^\times / \mu(\Gamma_i^1(\mathfrak{N})_{\mathfrak{b}} \cap T),$$

where the codomain $U_i(\mathfrak{N})$ is a finite abelian group. The point of all this is that by considering the action of $\Gamma_i(\mathfrak{N})_{\mathfrak{b}} \cap T$ on $M_k(\Gamma_i^1(\mathfrak{N}))$, we are able to obtain a decomposition of $M_k(\Gamma_i^1(\mathfrak{N}))$ indexed by characters of $U_i(\mathfrak{N})$.

Because $\Gamma_i^1(\mathfrak{N})_{\mathfrak{b}} \cap T$ is a normal subgroup of $\Gamma_i(\mathfrak{N})_{\mathfrak{b}} \cap T$, there is an action of $\Gamma_i(\mathfrak{N})_{\mathfrak{b}} \cap T$ on $M_k(\Gamma_i^1(\mathfrak{N}))$. This furnishes us with a decomposition

$$(2.7.4) \quad M_k(\Gamma_i^1(\mathfrak{N})_{\mathfrak{b}}) \cong \bigoplus_{\omega} M_k(\Gamma_i(\mathfrak{N})_{\mathfrak{b}}, \omega)$$

for characters $\omega: (\Gamma_i(\mathfrak{N})_{\mathfrak{b}} \cap T) \rightarrow \mathbb{C}^\times$, where $M_k(\Gamma_i(\mathfrak{N})_{\mathfrak{b}}, \omega)$ denotes the ω -eigenspace of the action of $\Gamma_i(\mathfrak{N})_{\mathfrak{b}} \cap T$.

The characters ω in (2.7.4) are highly constrained. For $\gamma = \begin{pmatrix} a & \\ & d \end{pmatrix} \in \Gamma_i^1(\mathfrak{N})_{\mathfrak{b}} \cap T$, we have

$$(2.7.5) \quad j_k(\gamma; z) = \text{sgn}(d)^k \left(\frac{a}{d}\right)^{\frac{k}{2}} = \text{sgn}(d)^k \mu(\gamma)^{\frac{k}{2}}.$$

Define $\omega_0: (\Gamma_i^1(\mathfrak{N})_{\mathfrak{b}} \cap T) \rightarrow \mathbb{C}^\times$ to be the character sending γ to $\text{sgn}(d)^k \mu(\gamma)^{\frac{k}{2}}$ – we think of this as the “default” action of $\Gamma_1(\mathfrak{N})_{\mathfrak{b}} \cap T$. ?? implies that $f|_k \gamma = \omega_0(\gamma)f(z)$ for $\gamma \in \Gamma_i^1(\mathfrak{N})_{\mathfrak{b}} \cap T$, so ω must coincide with ω_0 on $\Gamma_i^1(\mathfrak{N})_{\mathfrak{b}} \cap T$. Thus, any character ω appearing in (2.7.4) has the form $\omega'\omega_0$, where ω' is a character of $U_i(\mathfrak{N})$, and we can refine (2.7.4) as follows:

$$(2.7.6) \quad M_k(\Gamma_i^1(\mathfrak{N})_{\mathfrak{b}}) \cong \bigoplus_{\omega': U_i(\mathfrak{N}) \rightarrow \mathbb{C}^\times} M_k(\Gamma_i(\mathfrak{N})_{\mathfrak{b}}, \omega'\omega_0)$$

We call any ω appearing in (2.7.6) a **unit character** for $\Gamma_i(\mathfrak{N})_{\mathfrak{b}}$. In terms of Fourier coefficients, for any Hilbert modular form $f \in M_k(\Gamma_i(\mathfrak{N})_{\mathfrak{b}}, \omega)$, any $\nu \in (\mathfrak{b}\mathbb{Z}_F^{\sharp})_{>0}$ and any $\varepsilon \in \mathbb{Z}_{F,>0}^{\times}$, we have

$$(2.7.7) \quad a_{\nu\varepsilon}(f) = \omega(\varepsilon)a_{\nu}(f).$$

Remark 2.7.8. The presence of the $\text{sgn}(d)^k$ factor in the definition of ω_0 is related to the compatible parity condition of Eq. (2.6.9). Note that if $i = 1$, $\text{sgn}(d) = 1$, so this factor is only relevant when $i = 0$.

Example 2.7.9. In ??, we consider forms on the group $\Gamma_0^1(\mathfrak{N})_{\mathfrak{b}}$ (this corresponds to considering automorphic forms for SL_2 instead of for GL_2). The unit characters are indexed by $\mathbb{Z}_{F,>0}^{\times}/(\mathbb{Z}_F^{\times})^2$ by Eq. (2.7.2). By (2.7.6), the ω_0 is the only possible unit character exactly when this group is trivial, or equivalently, when $\text{Cl}^+ \mathbb{Z}_F = \text{Cl} \mathbb{Z}_F$ (confirming (2.2.4)).

3. ARITHMETIC OF q -EXPANSIONS

In this section, we present the essential algorithms for working with q -expansions. We first discuss how to represent q -expansions efficiently and present algorithms for the basic arithmetic operations. Then we exhibit methods to compute Hecke operators, inclusion and degeneracy operators, actions by field automorphisms, and restrictions to modular curves.

3.1. Encoding in bits. We begin by describing how we encode Hilbert modular forms, using finitely many coefficients of a Fourier expansion.

We encode spaces of Hilbert modular forms by their weight k , the type of standard congruence subgroup (Γ_0 , Γ_0^1 , Γ_1 , or Γ_1^1), and the level $\mathfrak{N}$. When applicable, we further refine (sub)spaces by additional attributes such as the nebentypus (Section 2.6) or unit character (Section 2.7).

We encode Hilbert modular forms $f = (f_{\mathfrak{b}})_{\mathfrak{b}} \in M_k(\Gamma, \chi)$, and their components $f_{\mathfrak{b}} \in M_k(\Gamma_{\mathfrak{b}}, \chi)$, as elements of these spaces together with the data of a unit character ω , a precision bound $B > 0$, and finitely many Fourier coefficients in a computable subring $E \subset \mathbb{C}$.

Remark 3.1.1. We typically know what E is before we begin writing down the Fourier coefficients of any given Hilbert modular form. For example, by Proposition 1.7 of [Shimura], any space of modular forms with trivial nebentypus has an E' -basis where E' is the fixed field of the automorphisms fixing the weight k . In practice, we take E to be the compositum of this E' and the fields of definition of the nebentypus and unit character.

It remains to explain which Fourier coefficients of f are stored precisely. Recalling the indexing (2.4.5), let $\mathfrak{b}' \subset F$ be a fractional $\mathbb{Z}_F$ -ideal: in our application, $\mathfrak{b}' = \mathfrak{b}\mathbb{Z}_F^{\sharp}$ as in (2.4.2). We know that elements of $\mathbb{Z}_{F,>0}^{\times}$ act on the Fourier coefficients $\{a_{\nu}\}_{\nu}$ of f as specified by Eq. (2.4.9) or (??), depending on the type of standard congruence subgroup. Thus, in all cases, it will suffice to store the coefficients $\{a_{\nu}\}_{\nu}$ where ν ranges over (a finite subset of) representatives of $\mathfrak{b}'_{\geq 0}$ modulo the action of the full group $\mathbb{Z}_{F,>0}^{\times}$.

There is also a partial ordering on the monoid $\mathfrak{b}'_{\geq 0}$ that is natural for the purposes of multiplication, as we will see shortly: we say $\nu' \leq \nu$ if and only if $\nu'_i \leq \nu_i$ for all $i = 1, \dots, n$. For $\nu \in \mathfrak{b}'_{\geq 0}$, we write

$$\downarrow \nu := \{\nu' \in \mathfrak{b}'_{\geq 0} : \nu' \leq \nu\}.$$

Our design choice is to choose a finite precision B , and represent a form $f = (f_{\mathfrak{b}})_{\mathfrak{b}}$ in precision B by encoding each $f_{\mathfrak{b}}$ by its Fourier coefficients $a_{\nu}(f_{\mathfrak{b}}) \in E$, where ν runs through a finite set $\downarrow \mathfrak{b}'(B) \subset \mathfrak{b}'_{\geq 0}$ with the following two properties:

- (1) The image of the quotient map $\downarrow \mathfrak{b}'(B) \rightarrow \mathfrak{b}'_{\geq 0}/\mathbb{Z}_{F,>0}^{\times}$ is exactly the finite set

$$\{\nu \in \mathfrak{b}'_{\geq 0}/\mathbb{Z}_{F,>0}^{\times} : \text{Nm}(\nu \mathfrak{b}'^{-1}) \leq B\}.$$

- (2) The set $\downarrow \mathfrak{b}'(B)$ is initial with respect to the partial ordering $\leq$ on $\mathfrak{b}'_{\geq 0}$; this means that for any $\nu \in \downarrow \mathfrak{b}'(B)$, we have $\downarrow \nu' \subset \downarrow \mathfrak{b}'(B)$.

We will explicitly construct an index set $\downarrow \mathfrak{b}'(B)$ satisfying properties (??) and (??) in the next subsection ???. This is done by first constructing a fundamental set $\mathfrak{b}'(B)$ under the action of $\mathbb{Z}_{F,>0}^{\times}$, then taking its closure under the $\downarrow$ operation, hence the notation.

Property (??) guarantees that one can efficiently reconstruct $a_{\nu}(f)$ for any $\nu \in \mathfrak{b}'_{\geq 0}$ such that $\text{Nm}(\nu \mathfrak{b}^{-1})$ from our encoding, while property (??) guarantees that our encoding is compatible with multiplication, as the following lemma shows.

Lemma 3.1.2. *Let Γ and $\mathfrak{b}'$ be as above, and let $S \subset \mathfrak{b}'_{\geq 0}$ be initial for the partial ordering $\leq$. Then the additive group $\mathbb{C}^S$ carries a natural ring structure for which the map*

$$\begin{aligned} \mu: \quad M(\Gamma_{\mathfrak{b}}) &\longrightarrow \mathbb{C}^S \\ f &\longmapsto (a_{\nu}(f))_{\nu \in S} \end{aligned}$$

is a morphism of rings.

Proof. The ring structure on $\mathbb{C}^S$ is given by an identification with formal power series of the form $\sum_{\nu \in S} a_{\nu} q^{\nu}$; multiplication corresponds to taking the Fourier coefficients of a product of Hilbert modular forms, by the rule

$$a_{\nu}(fg) = \sum_{\substack{\nu' + \nu'' = \nu \\ \nu', \nu'' \in \mathfrak{b}'_{\geq 0}}} a_{\nu'}(f) a_{\nu''}(g).$$

For $\nu \in S$, any elements ν', ν'' appearing in the above sum also belong to $\downarrow \nu$, hence to S as S is initial, so this multiplication is indeed well-defined on $\mathbb{C}^S$. The map μ is then a morphism of rings by definition. $\square$

Remark 3.1.3. In contrast with the case of classical modular forms, there are several stratifications of the index set $\mathfrak{b}'_{\geq 0}$ that one could consider. Instead of ordering by norm, one could consider all elements ν such that $\text{Tr}(\nu) \leq B$ for some bound B . Despite some immediate advantages (such elements are straightforward to list for a given B , and there is only a finite number of them, so considering the action of units is not immediately necessary), using the norm ordering proved superior in the long run. Indeed, in the norm ordering, precision behaves nicely with respect to the action of Hecke and inclusion operators, which drastically reduces precision losses in some of our more advanced computations. Note that the trace and norm orderings on $\mathfrak{b}'_{\geq 0}$ are mutually cofinal up to units [RTV].

Remark 3.1.4. Van Hirtum [VanHirtum] considers adelic q -expansions (the paper was withdrawn, but the notion still makes sense) whose coefficients are indexed by ideals of $\mathbb{Z}_F$ instead of totally positive elements of F . Such an expansion accounts for all the components of a Hilbert modular form simultaneously, naturally takes the action of units into account, and does not involve choosing ideals $\mathfrak{b}$ to index the components of the Hilbert modular variety.

We prefer to use separate q -expansions on each component instead, as many elementary operations on HMFs are best performed componentwise. Indexing by ideals also has the disadvantage of not supporting nontrivial unit characters.

Going further, instead of storing the elements of $\downarrow \mathfrak{b}'(B)$ as elements of F , we encode those indices as tuples of nonnegative integers. To this end, we define a **nonnegative exponent map** on $\mathfrak{b}'$ to be a $\mathbb{Q}$ -vector space isomorphism $e: F \xrightarrow{\sim} \mathbb{Q}^n$ such that $e(\mathfrak{b}'_{\geq 0}) \subset \mathbb{Z}_{\geq 0}^n$. The reason for this change is that multiplication of Hilbert modular forms then corresponds to the usual multiplication of multivariate polynomials.

Proposition 3.1.5. *Keep notation from ??, and endow $\mathbb{C}^S$ with its ring structure. Let e be a nonnegative exponent map on $\mathfrak{b}'$, and let $x_1, \dots, x_n$ be formal variables. Then the following holds.*

- (1) *The subset $R \subset \mathbb{C}[x_1, \dots, x_n]$ consisting of polynomials supported on $e(\mathfrak{b}'_{\geq 0})$ is a subring of $\mathbb{C}[x_1, \dots, x_n]$.*
- (2) *Let I denote the ideal of R generated by the monomials $x_1^{e(\nu)_1} \dots x_n^{e(\nu)_n}$ for every $\nu \in \mathfrak{b}'_{\geq 0} \setminus S$. Then the map*

$$\begin{aligned} \mu' : \quad \mathbb{C}^S &\longrightarrow R/I \\ (a_\nu)_{\nu \in S} &\longmapsto \sum_{\nu \in S} a_\nu x_1^{e(\nu)_1} \dots x_n^{e(\nu)_n} \end{aligned}$$

is an isomorphism of rings. Its inverse is induced by the map $R \rightarrow \mathbb{C}^S$ that extracts the coefficients attached to exponent vectors in $e(S)$.

- (3) *Consequently, the encoding map $\mu' \circ \mu : M(\Gamma_b) \rightarrow R/I$ is a morphism of rings whose kernel exactly consists of those Hilbert modular forms f satisfying $a_\nu(f) = 0$ for each $\nu \in S$.*

Proof.

- (1) This comes from the fact that $e(\mathfrak{b}_{\geq 0})$ is a monoid in $\mathbb{Z}_{\geq 0}^n$ containing $(0, \dots, 0)$.
- (2) Since S is initial, I is in fact the ideal of polynomials that are supported on $\mathfrak{b}'_{\geq 0} \setminus S$. Next, we check that the coefficient extraction map $R \rightarrow \mathbb{C}^S$ is a ring morphism: since e is linear, the multiplication formula in $\mathbb{C}^S$ from ?? exactly corresponds to the usual multiplication formula of multivariate polynomials. It is evidently surjective with kernel I .
- (3) This is a consequence of the above two points and ??. □

Finally we explain how the nonnegative exponent map e is chosen. Denote by $\mathbb{Z}\mathfrak{b}'_{\geq 0} \subseteq \mathfrak{b}'$ the $\mathbb{Z}$ -module of finite index generated by $\mathfrak{b}'_{\geq 0}$. We have $(\mathbb{Z}\mathfrak{b}'_{\geq 0})^\# \supseteq (\mathfrak{b}')^\#$.

Lemma 3.1.6. *The nonnegative exponent maps are exactly those of the form*

$$(3.1.7) \quad \begin{aligned} e : F &\rightarrow \mathbb{Q}^n \\ \nu &\longmapsto (\mathrm{Tr}(\beta_i \nu))_i \end{aligned}$$

where $\beta_1, \dots, \beta_n$ are totally positive, $\mathbb{Q}$ -linearly independent elements in $(\mathbb{Z}\mathfrak{b}'_{\geq 0})^\#$.

Proof. Let e be a nonnegative exponent map. By nondegeneracy of the trace pairing, for each i , the projection $e_i: F \rightarrow \mathbb{Q}$ onto the i^{th} factor is of the form $e_i(\nu) = \mathrm{Tr}(\beta_i \nu)$ for some $\beta_i \in F$. Since $\mathbb{Q}_{>0}\mathfrak{b}'_{>0} = F_{>0}$, we have $e(F_{\geq 0}) = \mathbb{Q}_{\geq 0}^n$. Since $F_{>0} \hookrightarrow \mathbb{R}_{>0}^n$ is dense, we must

have $\beta_i \geq 0$ and since e is an isomorphism we conclude $\beta_i > 0$. Since $e_i(\nu) = \text{Tr}(\beta_i \nu) \in \mathbb{Z}$ for all $\nu \in \mathfrak{b}'_{\geq 0}$, we have that $\beta_i \in (\mathbb{Z}\mathfrak{b}'_{\geq 0})^\#$ for all i . Finally, since e is an isomorphism, $\{\beta_i\}_i$ is $\mathbb{Q}$ -linearly independent. Conversely, every such e defines a nonnegative exponent map. $\square$

The choice of nonnegative exponent function depends only on the component ideal $\mathfrak{b}$, so it can be considered a precomputation step. We can compute a nonnegative exponent map using Equation (3.1.12) as follows. Starting with any $\mathbb{Z}$ -basis $\beta_1, \dots, \beta_n$ for $(\mathfrak{b}')^\# = (\mathfrak{b}')^{-1}\mathfrak{d}_F^{-1}$, compute $b' \in (\mathfrak{b}')^\# \cap \mathbb{Z}_{>0}$ and add a $\mathbb{Z}$ -multiple of b' so that β_i is totally positive for all i .

For algorithmic purposes, it is helpful to choose a nonnegative exponent map which gives small exponent vectors in $??$. To this end, we can improve on the above by starting with a lattice-reduced basis (e.g. LLL-reduced) for $(\mathbb{Z}\mathfrak{b}'_{\geq 0})^\#$ under the positive definite quadratic form $\beta \mapsto \text{Tr}(\beta)^2$, enumerating short vectors in $(\mathbb{Z}\mathfrak{b}'_{\geq 0})^\#$ to find totally positive elements of small trace.

3.2. Computing the index set. Keep notation from subsection 3.1. We start by constructing a finite set $\mathfrak{b}'(B)$ of representatives, up to the multiplicative action of $\mathbb{Z}_{F,>0}^\times$, of elements $\nu \in \mathfrak{b}'_{\geq 0}$ such that $\text{Nm}(\nu\mathfrak{b}'^{-1}) \leq B$.

Consider the log-Minkowski embedding

$$(3.2.1) \quad \begin{aligned} \log: F_{>0}^\times &\rightarrow V := \mathbb{R}^n \\ \nu &\mapsto (\log \nu_1, \dots, \log \nu_n). \end{aligned}$$

Then $\log(\mathbb{Z}_{F,>0}^\times)$ is a lattice in $\mathbb{R}^n$ of rank $n-1$, contained in the trace zero hyperplane

$$V^0 := \{x \in \mathbb{R}^n : \sum_i x_i = 0\}.$$

A convenient fundamental domain for the (additive) action of $\log(\mathbb{Z}_{F,>0}^\times)$ on $\mathbb{R}^n$ is provided by the translates along $(1, \dots, 1)$ of a symmetric parallelepiped with respect to a choice of basis $\{\varepsilon_j\}_{j=1}^{n-1}$ for $\mathbb{Z}_{F,>0}^\times \simeq \mathbb{Z}^{n-1}$. More precisely, every $x \in V^0$ can be written uniquely as

$$(3.2.2) \quad x = \sum_{j=1}^{n-1} a_j \log(\varepsilon_j)$$

and we say $x \in V^0$ is **reduced** (with respect to the basis $\{\varepsilon_j\}_{j=1}^{n-1}$) if it belongs to the fundamental domain defined by $-\frac{1}{2} < a_j \leq \frac{1}{2}$ for all $j = 1, \dots, n-1$. We say $x \in V$ is **reduced** if its orthogonal projection x^0 to V^0 is reduced (note $x^0 = x - \bar{x}(1, \dots, 1)$ where $\bar{x} = (x_1 + \dots + x_n)/n$). In other words, an element $\nu \in F_{>0}^\times$ is reduced if and only if

$$(3.2.3) \quad \log(\nu) = \frac{\log \text{Nm}(\nu)}{n}(1, \dots, 1) + \sum_{j=1}^{n-1} a_j \log(\varepsilon_j)$$

with $-\frac{1}{2} < a_j \leq \frac{1}{2}$ for all j .

Given $\nu \in F_{>0}^\times$, it is straightforward to compute the unique $\eta \in \mathbb{Z}_{F,>0}^\times$ such that $\eta\nu$ is reduced: write ν^0 as in (3.1.5) and take $\eta = \prod_j \varepsilon_j^{-\lfloor a_j \rfloor}$ with exponents obtained by rounding coefficients towards zero ($\lfloor x \rfloor$ is defined such that $-\frac{1}{2} < x - \lfloor x \rfloor \leq \frac{1}{2}$).

With these definitions in hand, for $B > 0$ we define

$$(3.2.4) \quad \mathfrak{b}'(B) := \{\nu \in \mathfrak{b}'_{\geq 0} : \text{Nm}(\nu\mathfrak{b}'^{-1}) \leq B \text{ and } \nu \text{ is reduced}\}.$$

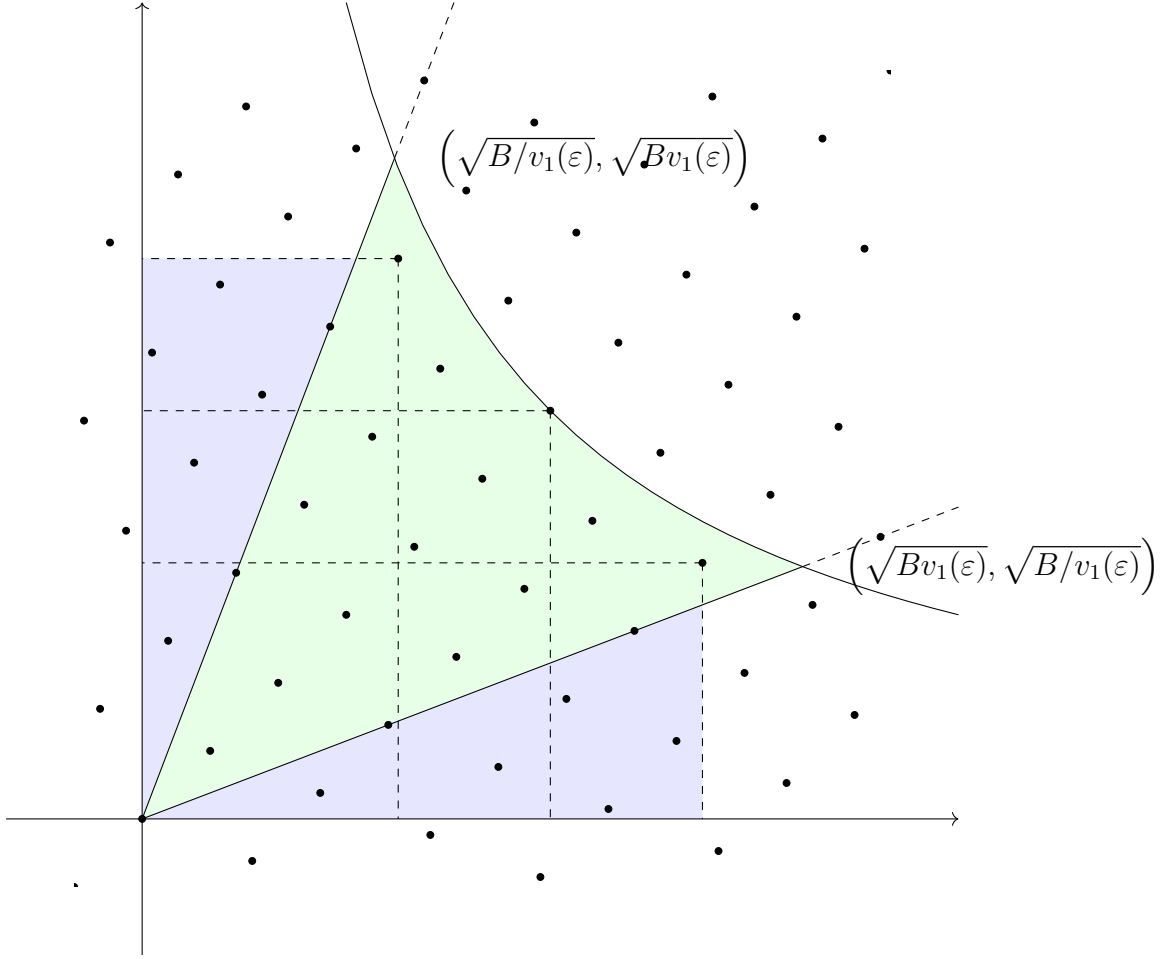


FIGURE 1. The sets $\mathfrak{b}'(B)$ and $\downarrow \mathfrak{b}'(B)$ for $F = \mathbb{Q}(\sqrt{5})$, $\mathfrak{b} = \frac{\sqrt{5}+5}{2}\mathbb{Z}_F$ (so that $\mathfrak{b}' = \mathbb{Z}_F$), and $B = 36$. The set $\mathfrak{b}'(B)$ consists of lattice points in the green region, and the set $\downarrow \mathfrak{b}'(B)$ consists of the points in $\mathfrak{b}'(B)$ as well as the points in the blue region.

Our index set will then be

$$(3.2.5) \quad \downarrow \mathfrak{b}'(B) = \{\nu \in \mathfrak{b}'_{\geq 0} : \nu \leq \nu' \text{ for some } \nu' \in \mathfrak{b}'(B)\} = \bigcup_{\nu \in \mathfrak{b}'(B)} \downarrow \nu.$$

A visual representation of these sets is provided on Fig. 1 in the case $F = \mathbb{Q}(\sqrt{5})$. In the rest of this section, we verify that these sets indeed satisfy the required properties, describe algorithms to compute them, and analyze their complexity.

Lemma 3.2.6. *Let $\varepsilon = \{\varepsilon_j\}_{j=1}^{n-1}$ be the basis of $\mathbb{Z}_{F,>0}^\times$ chosen above. For $1 \leq k \leq n$, define*

$$L_k(\varepsilon) = \sum_{j=1}^{n-1} |\log(\varepsilon_{j,k})|,$$

and let $L(\varepsilon) = \max_{1 \leq k \leq n} L_k(\varepsilon)$. Then for any $\nu \in \mathfrak{b}'(B)$, we have

$$\sum_{k=1}^n |\nu_k|^2 \leq n e^{L(\varepsilon)} (B \text{Nm}(\mathfrak{b}'))^{\frac{2}{n}}.$$

In particular $\mathfrak{b}'(B)$ is finite.

Proof. Let $\nu \in \mathfrak{b}'(B)$; we may assume $\nu \neq 0$. Then $\text{Nm}(\nu) \leq B \text{Nm}(\mathfrak{b}')$. By [KV10] with totally positive units assuming the role of the units, there exists a $\nu' \in (\nu)$ such that $(\nu') = (\nu)$, $\nu' \gg 0$, and

$$|\nu'_k| \leq e^{\frac{1}{2}L(\varepsilon)} \text{Nm}(\nu)^{\frac{1}{n}} \leq e^{\frac{1}{2}L(\varepsilon)} (B \text{Nm}(\mathfrak{b}'))^{\frac{1}{n}}$$

for all k . The result follows by squaring and summing over k . $\square$

As a consequence of ??, the set $\mathfrak{b}'(B)$ may be computed by listing all elements $\nu \in \mathfrak{b}'_{\geq 0}$ such that $\sum_{k=1}^n |\nu_k|^2$ is less than the provided explicit bound, and reducing them as explained above. Notice that this upper bound also holds for elements of $\downarrow \mathfrak{b}'(B)$, so we obtain a finite superset of $\downarrow \mathfrak{b}'(B)$ as a byproduct. We formalize this procedure in Algorithm 3.2.1.

Algorithm 3.2.7: ReducedRepresentatives($\varepsilon, \mathfrak{b}', B$)

Data: $\varepsilon = \{\varepsilon_j\}_j \in (\mathbb{Z}_{F, > 0}^\times)^{n-1}$, $\mathfrak{b}' \subset F$, $B \in \mathbb{Z}_{> 0}$

Result: $(\mathfrak{b}'(B), N)$ where N is a finite superset of $\downarrow \mathfrak{b}'(B)$.

for $k \in \{1, 2, \dots, n\}$ **do**

$L_k(\varepsilon) := \sum_{j=1}^{n-1} |\log(\varepsilon_{j,k})|$;

$L(\varepsilon) := \max_{1 \leq k \leq n} \{L_k(\varepsilon)\}$;

$N := \{\nu \in \mathfrak{b}'_{> 0} : \sum_{k=1}^n |\nu_k|^2 \leq n e^{L(\varepsilon)} (B \text{Nm}(\mathfrak{b}'))^{\frac{2}{n}}\}$;

$R(\varepsilon) := (\log(\varepsilon_{j,k}))_{j,k=1}^{n-1}$;

$S := \{0\}$;

for $\nu \in N$ **do**

if $\text{Nm}(\nu) > B \text{Nm}(\mathfrak{b}')$ **then**

 Remove ν from N ;

$a := R(\varepsilon)^{-1} \left(\log(\nu) - \frac{\log \text{Nm}(\nu)}{n} (1, \dots, 1) \right) \in \mathbb{R}^{n-1}$;

$S := S \cup \left\{ \nu \prod_j \varepsilon_j^{-\lfloor a_j \rfloor} \right\}$;

return (S, N) ;

To analyze this algorithm, we first consider the size of its output. We write $O_F(\cdot)$, $\Theta_F(\cdot)$ to indicate that the implied constants depend only on F .

Lemma 3.2.8. *If Reg_F denotes the regulator of $\mathbb{Z}_F$, then*

$$(3.2.9) \quad \#\mathfrak{b}'(B) = \Theta_F(B) = \Theta \left(\frac{2^{n-1} h(\mathbb{Z}_F) \text{Reg}_F}{h^+(\mathbb{Z}_F) \sqrt{d_F}} B \right) = \Theta \left(\frac{\text{Res}_{s=1}(\zeta_F(s))}{h^+(\mathbb{Z}_F)} B \right).$$

Proof. The elements of $\mathfrak{b}'(B)$ are in bijection with integral ideals of norm at most B lying in the narrow ideal class $[\mathfrak{b}']^{-1}$. By the Chebotarev density theorem (applied to the narrow

Hilbert class field) and [LowryDudaTaniguchiThorne],

$$\begin{aligned}\#\mathfrak{b}'(B) &= \Theta\left(\frac{1}{h^+(\mathbb{Z}_F)}\#\{\mathfrak{n} \in I_F : \text{Nm}(\mathfrak{n}) \leq B\}\right) \\ &= \Theta\left(\frac{2^{n-1}h(\mathbb{Z}_F)\text{Reg}_F}{h^+(\mathbb{Z}_F)\sqrt{d_F}}B\right).\end{aligned}$$

The last equality in Eq. (3.2.4) follows from the analytic class number formula. $\square$

complexity}

Proposition 3.2.10. *Algorithm 3.2.1 is correct, and its running time complexity is $O_F(B)$.*

Proof. By construction, the set S produced by **ReducedRepresentatives** has the following property: any $0 \neq \nu \in S$ is obtained as $\nu = \nu' \prod_j \varepsilon_j^{-\lfloor a_j \rfloor}$ for some $\nu' \in N \subseteq \mathfrak{b}'_{>0}$ with $\text{Nm}(\nu') \leq B \text{Nm}(\mathfrak{b}')$ and a_j is as in (3.1.6). In particular ν is reduced. Since $\text{Nm}(\nu) = \text{Nm}(\nu')$, it follows that $\text{Nm}(\nu \mathfrak{b}'^{-1}) \leq B$, and $\nu \in \mathfrak{b}'_{>0}$, hence $\nu \in \mathfrak{b}'(B)$. Conversely, if $0 \neq \nu \in \mathfrak{b}'(B)$, then $\nu \in N$ by ???. Since ν is already reduced, the algorithm makes no modification to this element and puts it in S . Thus $S = \mathfrak{b}'(B)$ at the end of the algorithm. Since N is also a superset of $\downarrow \mathfrak{b}'(B)$ by ???, we conclude that Algorithm 3.2.1 is correct.

We now analyze its running time. Let $d_1, \dots, d_n$ be a basis for $\mathbb{Z}_F^\sharp \hookrightarrow \mathbb{R}^n$ as a $\mathbb{Z}$ -module, and write $C(F) = \prod_{j=1}^n \|d_j\|_2$ for the product of their norms. Then, by [KV10] and [FP85], the set N can be computed in time $O(2^n n^{\frac{n}{2}+1} e^{\frac{1}{2}L(\varepsilon)} C(F) B)$. In addition, for each of the elements in $\mathfrak{b}'(B)$ we perform an $O(n^2)$ linear algebra operation, which by Eq. (3.2.3) has a running time of $O_F(B)$, hence the result. $\square$

To compute the final index set $\downarrow \mathfrak{b}'(B)$ from the output of Algorithm 3.2.1, it would be sufficient to test, for each $\nu \in N$, whether there exists $\nu' \in \mathfrak{b}'(B)$ satisfying $\nu \leq \nu'$. Naively, each such query would require $O(\#\mathfrak{b}'(B))$ time. This can be improved using range trees [CompGeo], leading to ???.

Algorithm 3.2.11: IndexSet($\varepsilon, \mathfrak{b}', B$)

Data: $\varepsilon = \{\varepsilon_j\}_j \in (\mathbb{Z}_{F,>0}^\times)^{n-1}$, $\mathfrak{b}' \subset F$, $B \in \mathbb{Z}_{>0}$

Result: $\downarrow \mathfrak{b}'(B)$.

$(S, N) := \text{ReducedRepresentatives}(\varepsilon, \mathfrak{b}', B)$;

$T := \text{RangeTree}(\log(\mathfrak{b}'(B)))$;

return $\{\mu \in N : \text{RangeQuery}(T, \mu)\}$;

imple-shadow}

In ???, $\text{RangeTree}(S)$ denotes a function which constructs a range tree from a set S , and $\text{RangeQuery}(T, \mu)$ denotes a function whose input is a range tree T constructed from a set S , and an element μ , returning true if and only if there exists $\nu' \in S$ with $\mu \leq \nu'$.

le-shadow}

Proposition 3.2.12. *??? is correct, and its running time is $O_F(B(\log B)^{n-1})$.*

Proof. Correctness follows directly from the above specifications of **RangeQuery**. By Equation (3.2.5), the first step of computing (S, N) has running time complexity $O_F(B)$. Constructing the range tree then takes time $O_F(B(\log B)^{n-1})$ [CompGeo]. Again by Equation (3.2.5), we have $\#N = O_F(B)$, and each range query costs $O(\log^{n-1}(\#\mathfrak{b}'(B)))$ [CompGeo]. Thus the last step costs $O_F(B(\log B)^{n-1})$ as well, hence the result. $\square$

-quadratic}

When F is quadratic, we can avoid range trees and save a factor of $\log(B)$. Indeed, taking a look at Fig. 1 suggests the following lemma.

Lemma 3.2.13. *Keeping notation as above, assume $n = 2$. For $i \in \{1, 2\}$, let $M_i = \max_{\nu \in \mathfrak{b}'(B)} v_i(\nu)$. Define N as in Algorithm 3.2.1. Then we have*

$$\downarrow \mathfrak{b}'(B) = \{\nu \in N : \text{Nm}(\nu \mathfrak{b}^{-1}) \leq B \text{ and } v_i(\nu) \leq M_i \text{ for each } i \in \{1, 2\}\}.$$

Proof. Let S be the set on the right hand side. The inclusion $\downarrow \mathfrak{b}'(B) \subset S$ is immediate from definitions. To show the converse inclusion, we will assume without loss of generality that the single element of the basis ε satisfies $v_1(\varepsilon) > 1$. Then we have by (3.1.6):

$$\mathfrak{b}'(B) \setminus \{0\} = \left\{ \nu \in \mathfrak{b}'_{>0} : \text{Nm}(\nu \mathfrak{b}'^{-1}) \leq B \text{ and } \frac{1}{v_1(\varepsilon)} < \frac{v_1(\nu)}{v_2(\nu)} \leq v_1(\varepsilon) \right\}.$$

Consider now a nonzero element $\nu \in S$. If $\nu \notin \mathfrak{b}'(B)$, then one of the two inequalities above fails. Assume for instance that $v_1(\nu)/v_2(\nu) > v_1(\varepsilon)$, and let $\nu' \in \mathfrak{b}'(B)$ be such that $v_1(\nu') = M_1$. Then $v_1(\nu) \leq v_1(\nu')$ by assumption, and

$$v_2(\nu) \leq v_1(\nu) \cdot \frac{1}{v_1(\varepsilon)} \leq v_1(\nu) \frac{v_2(\nu')}{v_1(\nu')} \leq v_2(\nu'),$$

so $\nu \leq \nu'$ and $\nu \in \downarrow \mathfrak{b}'(B)$. If $v_1(\nu)/v_2(\nu) \leq 1/v_1(\varepsilon)$, a similar argument goes through, taking $\nu' \in \mathfrak{b}'(B)$ such that $v_2(\nu') = M_2$. $\square$

Proposition 3.2.14. *If $n = 2$, then one can compute $\downarrow \mathfrak{b}'(B)$ in time $O_F(B)$.*

Proof. Starting from (S, N) as computed by Algorithm 3.2.1, one can compute M_1 and M_2 in time $O_F(B)$, and compute $\downarrow \mathfrak{b}'(B)$ in time $O_F(B)$ as well using the characterization from ?? $\square$

Remark 3.2.15. By Eq. (3.2.3) and Eq. (3.2.5), keeping track of Fourier coefficients indexed by $\downarrow \mathfrak{b}'(B)$ instead of $\mathfrak{b}'(B)$ only yields an increase in storage size by a constant factor. In the quadratic case for example, we can describe the asymptotic size of $\mathfrak{b}'(B)$ and $\downarrow \mathfrak{b}'(B)$ explicitly in terms of the precision B . Let $\text{Reg}_{>0}$ be the totally positive regulator of F . We quickly obtain as $B \rightarrow \infty$:

$$\#\mathfrak{b}'(B) = \text{Reg}_{>0} \cdot B + O(\sqrt{B}) \quad \text{and} \quad \#\downarrow \mathfrak{b}'(B) = \#\mathfrak{b}'(B) + B + O(\sqrt{B}).$$

As $\text{Reg}_{>0}$ grows, i.e. the difference between the sets $\mathfrak{b}'(B)$ and $\downarrow \mathfrak{b}'(B)$ thus becomes negligible, with $\#\mathfrak{b}'(B)/\#\downarrow \mathfrak{b}'(B) \rightarrow 1 - \frac{1}{1+\text{Reg}_{>0}}$. In the “worst” case $F = \mathbb{Q}(\sqrt{5})$, the totally positive regulator is $\log(\phi^2) = 0.96 \dots$, and the ratio tends to approximately 0.49 as $B \rightarrow \infty$.

arith-ops}

3.3. Arithmetic operations. With basic linear algebra operations needing no further explanation, in this section we discuss the remaining elementary operations on Hilbert modular forms, namely multiplication and (when possible) division.

We define these operations by component, working with forms represented in bits as in Section 3. We first discuss multiplication,

$$M_k(\Gamma_{\mathfrak{b}}, \chi) \times M_{k'}(\Gamma_{\mathfrak{b}}, \chi') \rightarrow M_{k+k'}(\Gamma_{\mathfrak{b}}, \chi\chi').$$

Let $f \in M_k(\Gamma_{\mathfrak{b}}, \chi)$ and $g \in M_{k'}(\Gamma_{\mathfrak{b}}, \chi')$ with Fourier coefficients $a_{\nu}(f)$ and $a_{\nu}(g)$ for $\nu \in \mathfrak{b}'_{\geq 0}$ where as before $\mathfrak{b}' = \mathfrak{b}\mathfrak{d}$. By ??, computing an encoding of the product fg amounts to computing the product of the encodings of f and g on each component as

multivariate polynomials, and extracting the monomials of the result whose exponent vectors lie in $e(\downarrow \mathfrak{b}'(B))$. For this, we use the built-in Magma methods for polynomial multiplication. If the Fourier coefficients of f and g lie in a fixed computable subring $E \subset \mathbb{C}$, then these computations take place in $E[x_1, \dots, x_n]$.

complexity}

Proposition 3.3.1. *Let B be a positive integer and let f and g be Hilbert modular forms given in bits as above to precision B . The product $fg = h$ can be computed to precision B in $O_F(B \log B)$ time.*

Proof. By Eq. (3.2.5), the set $\downarrow \mathfrak{b}'(B)$ has size $O_F(B)$. Fix elements $\beta_1, \dots, \beta_n$ defining a nonnegative exponent map $e : \nu \mapsto (\text{Tr}(\beta_i \nu))_i$ as in Eq. (3.1.12). We can then represent f and g as multivariate polynomials as in ???. Let $L(\varepsilon)$ be as in ??. For all $1 \leq i \leq n$ and $\nu \in \downarrow \mathfrak{b}'(B)$, we have

$$v_i(\nu) \leq B^{\frac{1}{n}} e^{\frac{1}{2}L(\varepsilon)} = O_F(B^{\frac{1}{n}})$$

Because the β_j are totally positive and depend only on F ,

$$\text{Tr}(\beta_j \nu) = \sum_i v_i(\beta_j) v_i(\nu) = O_F(B^{\frac{1}{n}}).$$

This is a bound on the partial degree (degree of any individual variable) of the encodings of f and g . Because the product of the partial degrees is $O_F(B^{1/n})$, it follows that f and g can be multiplied in $O_F(B \log B)$ time using methods for multiplying dense multivariate polynomials (e.g. Kronecker substitution [von·zur·Gathen·Gerhard]). $\square$

We finally turn to division of HMF components. Since division is simply an inversion followed by a multiplication, we focus on the former. Let f be an HMF component as above, and assume that a_0 is invertible. After multiplying f by a_0^{-1} , we may assume that $a_0 = 1$. Let R and I be as in ??? for the initial set $S = \downarrow \mathfrak{b}'(B)$. Let $P \in R$ be the multivariate polynomial encoding f . Then it is sufficient to compute the inverse of P in R/I . Write $P = 1 - u$ where the constant coefficient of u vanishes.

vanishes}

Lemma 3.3.2. *Let*

$$m := \left\lceil \frac{B}{\min_{\nu \in \mathfrak{b}'_{>0}} \text{Nm}(\nu)} \right\rceil + 1.$$

Then, $u^m \in I$.

Proof. Any exponent vector corresponding to a nonzero coefficient in u^m is of the form $e(\nu_1 + \dots + \nu_m)$, where $\nu_i \in \mathfrak{b}'_{>0}$ for each i . Since these elements are totally positive, we have

$$\text{Nm}(\nu_1 + \dots + \nu_m) \geq \sum_{i=1}^m \text{Nm}(\nu_i) \geq m \cdot \min_{\nu \in \mathfrak{b}'_{>0}} \text{Nm}(\nu) > B$$

by definition of m . Thus $\nu_1 + \dots + \nu_m \notin \downarrow \mathfrak{b}'(B)$, showing that $u^m \in I$. $\square$

Remark 3.3.3. We can even take $m = O_F(B^{\frac{1}{n}})$ in ??? using the bound on partial degrees from the proof of Eq. (3.3.3).

We then have $P^{-1} = 1 + u + \dots + u^{m-1}$ in R/I . We recall the standard fast algorithm to compute this sum in Algorithm 3.3.5; recall also that multiplications in R/I are performed

alg:division}

Algorithm 3.3.4: Inverse(f)

Data: $P \in R/I$, the encoding of f with constant coefficient 1

Result: $P^{-1} \in R/I$

$u := 1 - P$;

$C := 1 + u$;

$D := u^2$;

while $D \neq 0 \in R/I$ **do**

$C := C + DC$;

$D := D^2$;

by multiplying in R (i.e. in $E[x_1, \dots, x_n]$) then extracting the monomials whose exponent vectors lie in $e(\downarrow \mathfrak{b}'(B))$.

Note that the number of iterations of the loop is logarithmic in m , i.e. $O_F(\log B)$, so this is also the asymptotic number of multiplications involved in an application of **Inverse**.

{sec:hecke}

3.4. Hecke operators. We follow [Shimura]. Let $f = (f_{\mathfrak{b}})_{\mathfrak{b} \in \text{Cl}^+ R}$ be a Hilbert modular form with $f_{\mathfrak{b}} \in M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}}, \chi)$, where χ is a Hecke character with modulus $\mathfrak{N}$, and let $\mathfrak{n}$ be an integral ideal. Then the action of the Hecke operator $T_{\mathfrak{n}}$ on f is given by [Shimura].

$$(3.4.1) \quad a_{\mathfrak{n}}(T_{\mathfrak{m}}(f)) = \sum_{\mathfrak{n} + \mathfrak{m} \subset \mathfrak{d}} \chi(\mathfrak{d}) \text{Nm}(\mathfrak{d})^{k_0-1} a_{\mathfrak{n}\mathfrak{m}/\mathfrak{d}^2}(f),$$

where $k_0 = \max(k_i)$ is the maximum weight in k . In the case when $\mathfrak{m} = (0)$, we take the zero coefficient $a_{(0),\mathfrak{b}}$ of $f_{\mathfrak{b}}$ to be

$$(3.4.2) \quad a_{(0),\mathfrak{b}}(T_{\mathfrak{m}}(f)) = \sum_{\mathfrak{m} \subset \mathfrak{d}} \chi(\mathfrak{d}) \text{Nm}(\mathfrak{d})^{k_0-1} a_{(0),\mathfrak{b}\mathfrak{m}/\mathfrak{d}^2}(f)$$

If f is an HMF known to precision B , then we can compute $T_{\mathfrak{m}}(f)$ as an HMF to precision $\lfloor B/\text{Nm}(\mathfrak{m}) \rfloor$. In particular, we can compute the Hecke operator on the space by taking B large enough.

ec:incldeg}

3.5. Inclusion and degeneracy operators. There are also natural inclusion and degeneracy operators that further decompose spaces of Hilbert modular forms according to forms that arise from lower level. Let $k \in \mathbb{Z}_{\geq 0}^n$ be a weight.

{lem:incl}

Lemma 3.5.1. *Let $\mathfrak{N}, \mathfrak{M}$ be nonzero ideals of R such that $\mathfrak{M}$ is a multiple of $\mathfrak{N}$. Let χ be a Hecke character with modulus $\mathfrak{N}$. Then for each divisor $\mathfrak{D} \mid \frac{\mathfrak{M}}{\mathfrak{N}}$, there is an injective map*

$$(3.5.2) \quad \iota_{\mathfrak{D}} : M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}}, \chi) \rightarrow M_k(\Gamma_0(\mathfrak{M})_{\tilde{\mathfrak{b}}:=\mathfrak{D}\mathfrak{b}}, \tilde{\chi})$$

where $\tilde{\chi}$ is the extension of χ to modulus $\mathfrak{M}$. The image of a form $f \in M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}}, \chi)$ is $\iota_{\mathfrak{D}} f$ such that for all ideals $\mathfrak{n}$,

$$a_{\mathfrak{n}}(\iota_{\mathfrak{D}} f) := \begin{cases} 0, & \text{if } \mathfrak{D} \nmid \mathfrak{n}; \\ a_{\mathfrak{n}/\mathfrak{D}}(f), & \text{if } \mathfrak{D} \mid \mathfrak{n}. \end{cases}$$

Proof. See Shimura [Shimura] for an idelic proof. □

The oldspace $S_k^{\text{old}}(\Gamma_1(\mathfrak{N})) \subseteq S_k(\Gamma_1(\mathfrak{N}))$ is the subspace spanned by $\iota_{\mathfrak{m}}(S_k(\Gamma_1(\mathfrak{M})))$ for all $\mathfrak{M} \mid \mathfrak{N}$ and all $\mathfrak{m} \mid \mathfrak{N}\mathfrak{M}^{-1}$. The subspace orthogonal to the oldspace is the newspace $S_k^{\text{new}}(\Gamma_1(\mathfrak{N}))$, and there is an orthogonal decomposition

$$S_k(\Gamma_1(\mathfrak{N})) = S_k^{\text{old}}(\Gamma_1(\mathfrak{N})) \oplus S_k^{\text{new}}(\Gamma_1(\mathfrak{N})).$$

3.6. Actions by field automorphisms. Given a Hilbert modular form component $f = \sum_{\nu \in \mathfrak{b}'_{\geq 0}} a_{\nu} q^{\nu}$ with coefficients in a subring $E \subset \mathbb{C}$, there are two natural Galois actions on f . First, if $\sigma \in \text{Aut}(\mathbb{C})$, we can define

$$f^{\sigma} := \sum_{\nu \in \mathfrak{b}'_{\geq 0}} \sigma(a_{\nu}) q^{\nu}$$

Shimura [Shimura] proves that for any congruence subgroup Γ and any paritious weight k , $f^{\sigma} \in M_{\sigma(k)}(\mathfrak{N}, \sigma \circ \chi)$. We provide an implementation of f^{σ} when $\Gamma = \Gamma_0(\mathfrak{N})$.

Secondly, recall that we have fixed an ordering $(\nu_i)_i$ of the real places of F . Any $\sigma \in \text{Aut}(F|\mathbb{Q})$ induces a permutation on the real places and hence a permutation on the coordinates $(z_{\nu_i})_i \in \mathcal{H}$ as well as on tuples $k = (k_i)_i \in \mathbb{Z}_{\geq 0}^n$. Moreover, $\text{Aut}(F|\mathbb{Q})$ acts on the set of ideals and also induces maps on the group of Hecke characters of F . With these actions in hand, let

$$f_{\sigma} := \sum_{\nu \in \mathfrak{b}'_{\geq 0}} a_{\sigma(\nu)} q^{\nu},$$

and one can check that $f_{\sigma} \in M_{\sigma(k)}(\sigma(\mathfrak{N}), \chi \circ \sigma)$.

AM: [Given the number of conditions, I'm not sure if the following is worth including if we don't use it elsewhere in the paper or otherwise explain why it's useful – it sounds like we might use it in the models section to compute symmetric Hilbert modular varieties? But I think someone should confirm and explain why this is useful once we've written the application down.]

We provide an implementation of f_{σ} when k is parallel, $\chi = \chi \circ \sigma$, $\mathfrak{b}$ is Galois stable, and the unit character is trivial. Under these hypotheses, we can compute a basis for the subspace $M_{k, \text{sym}}(\Gamma_{\mathfrak{b}}, \chi) \subseteq M_k(\mathfrak{N}, \chi)$ that is invariant under $f \mapsto f_{\sigma}$.

Concretely, let $\{f_i\}_{i=1}^m$ be a basis for $M_k(\Gamma_{\mathfrak{b}}, \chi)$. For each $j \in \{1, \dots, m\}$, define

$$(3.6.1) \quad g_j := \frac{1}{r} \sum_{i=1}^r (f_j)_{\sigma_i}.$$

Then each g_j is invariant under $\text{Aut}(F|\mathbb{Q})$. Moreover, $\{g_1, \dots, g_m\}$ spans $M_{k, \text{sym}}(\Gamma_{\mathfrak{b}}, \chi)$: given any $g = \sum_{j=1}^m c_j f_j \in M_{k, \text{sym}}(\Gamma_{\mathfrak{b}}, \chi)$, it can be rewritten by (3.6.1) as

$$g = \frac{1}{r} \sum_{i=1}^r g_{\sigma_i} = \frac{1}{r} \sum_{i=1}^r \left(\sum_{j=1}^m (c_j f_j)_{\sigma_i} \right) = \sum_{j=1}^m c_j \left(\frac{1}{r} \sum_{i=1}^r (f_j)_{\sigma_i} \right) = \sum_{j=1}^m c_j g_j,$$

and our claim holds.

Edgar: [do we want to have a forward reference to an example?]

3.7. Restriction to modular curves. AM: [TODO there seems to be something wrong with the Fourier expansion of $g_{\lambda}(\tau)$ – I don't see why t_{0j} are integral yet.] Alex: [Isn't it explained right after its Fourier expansion is written down?] Given a Hilbert modular form, we can define different restrictions to classical modular forms. In this section, we mainly focus on restrictions of Hilbert modular forms to forms on a special class of Hirzebruch–Zagier

divisors for quadratic fields. Other restrictions can be defined, see for example [Azaiez]. We conclude this section by defining the restriction to the diagonal of forms of parallel weight in $M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}})$.

As before, let $f = (f_{\mathfrak{b}})_{\mathfrak{b} \in \text{Cl}^+ R}$ be a Hilbert modular form with $f_{\mathfrak{b}} \in M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}})$. We assume that F is a quadratic number field and fix a component $\mathfrak{b} \in \text{Cl}^+ R$. For any $\lambda \in \mathfrak{b}^{-1}$ define

$$\text{Res}_{\lambda}(f)(\tau) := f(\lambda_1 \tau, \lambda_2 \tau)$$

for $\tau \in \mathcal{H}^2$. Given $\alpha_{\lambda} := \begin{pmatrix} 0 & \lambda_1 \\ -\lambda_2 & 0 \end{pmatrix}$, let $\Gamma_{\lambda} := \{\gamma \in \Gamma_0(\mathfrak{N})_{\mathfrak{b}} : (\gamma_2)^T \alpha_{\lambda} \gamma_1 = \pm \alpha_{\lambda}\}$. One can check that $\text{Res}_{\lambda}(f)$ is a quaternionic modular form with respect to the subgroup Γ_{λ} . This subgroup may not be a subgroup of $M_2(\mathbb{Z})$, but it turns out that it is conjugate to a congruence subgroup. Let t_0 be a generator for the ideal $\sigma(\lambda)\mathfrak{b}^{-1} \cap \mathbb{Q}$, where σ is the nontrivial automorphism of F , and $\gamma_{\lambda} := \begin{pmatrix} t_0 & 0 \\ 0 & 1 \end{pmatrix}$. A direct calculation shows that $\Gamma_0(L) := \gamma_{\lambda}^{-1} \Gamma_{\lambda} \gamma_{\lambda}$ is a standard congruence subgroup and that

$$g_{\lambda}(\tau) := \text{Res}_{\lambda}(f)(\gamma_{\lambda} \tau) = \text{Res}_{\lambda}(f)(t_0 \tau)$$

is a classical modular form of weight $k_1 + k_2$ on $\Gamma_0(L)$.

We now compute the Fourier expansion of $\text{Res}_{\lambda}(f)$, by (2.4.5) we have

$$\text{Res}_{\lambda}(f)(\tau) = a_0 + \sum_{\nu \in (\mathfrak{b}\mathbb{Z}_F^{\sharp})_{>0}} a_{\nu} \exp(2\pi\sqrt{-1}(\text{Tr}(\lambda\nu)\tau)).$$

The fact that $\mathfrak{b} \subseteq \mathbb{Z}_F$ implies that $\text{Tr}(\nu) \in \mathbb{Z}_{>0}$. Thus, we can rewrite

$$\text{Res}_{\lambda}(f)(\tau) = a_0 + \sum_{j \geq 1} \left(\sum_{\substack{\nu \in (\mathfrak{b}\mathbb{Z}_F^{\sharp})_{>0} \\ \text{Tr}(\lambda\nu)=j}} a_{\nu} \right) \exp(2\pi\sqrt{-1}j\tau).$$

Setting $q := \exp(2\pi\sqrt{-1}\tau)$ and evaluating $g(\tau) = \text{Res}_{\lambda}(f)(t_0\tau)$, we get the fourier expansion

$$g_{\lambda}(\tau) = a_0 + \sum_{j \geq 1} \left(\sum_{\substack{\nu \in (\mathfrak{b}\mathbb{Z}_F^{\sharp})_{>0} \\ \text{Tr}(\lambda\nu)=j}} a_{\nu} \right) q^{t_0 j}.$$

AM: [I'm confused as to why this next sentence is necessary]. Alex: [I think it's just pointing out that the exponent of q is integral.] Juanita: [Indeed, tried to make it more clear.] The exponents of q are integers since the denominator of t_0 divides $\text{Tr}(\lambda\nu)$ for all $\nu \in \mathfrak{b}\mathbb{Z}_F^{\sharp}$. It is worth noting that the expansion depends on the initial choice of representative $\mathfrak{b}$ of the narrow class.

We can generalize the above procedure to give the restriction to the diagonal of any Hilbert modular form $f = (f_{\mathfrak{b}})_{\mathfrak{b} \in \text{Cl}^+ \mathbb{Z}_F}$ of parallel weight k over a number field F . We define the restriction $\text{Res}(\tau) := f(\tau, \dots, \tau)$ for $\tau \in \mathcal{H}^2$ (n times). By setting b as denominator of t_0 and evaluating at τ/b , we obtain the classical modular form $g(\tau) := \text{Res}\left(\frac{\tau}{b}\right)$ of weight nk on $\Gamma_0(N)$, where N is the (positive) generator of the ideal $\mathfrak{N} \cap \mathbb{Z}$. Alex: [The notation in this final paragraph looks a little scuffed.] Juanita: [Better?] Alex: [Also, why does the weight have

to be parallel?]

Juanita: [I think this should work in general, but we picked parallel weight to get something nice. I'll double check and make this explicit.]

Edgar: [do we want to have a forward reference to an example?]

4. SOURCES OF HILBERT MODULAR FORMS

We describe several algorithmic sources of Hilbert modular forms: Eisenstein series, forms obtained by quaternionic methods via Jacquet–Langlands transfer, and theta series associated to characters of quadratic extensions.

4.1. Eisenstein series. Recall that $n := [F : \mathbb{Q}]$. Let $\mathfrak{N} \subseteq \mathbb{Z}_F$ be a nonzero integral ideal, let $k \in \mathbb{Z}_{\geq 1}^n$ be a weight, and let χ be a finite-order Hecke character of F , as in Sections 2.5 and 2.6, whose conductor divides $\mathfrak{N}$. In this subsection, we give an algorithm for computing a basis for the space of Eisenstein series of weight k , level $\mathfrak{N}$, and nebentypus χ , which we denote by $\mathcal{E}_k(\mathfrak{N}, \chi)$.

The following theorem gives conditions on when Eisenstein series can exist.

Theorem 4.1.1 ([Goren] and [Dasgupta-Darmon-Pollack]). *Assume that the space $\mathcal{E}_k(\mathfrak{N}, \chi)$ is non-zero. Then the following two statements hold.*

- (1) *The weight k is parallel.*
- (2) *$\chi_v(-1) = (-1)^{k_v}$ for every infinite place v .*

Remark 4.1.2. The first condition of Eq. (4.1.1) implies that in the second condition, the k_v are all the same, and so $\chi_v(-1) = (-1)^k$ for all $v \mid \infty$.

When they exist, we turn to producing a basis for $\mathcal{E}_k(\mathfrak{N}, \chi)$. Let η and ψ be finite-order Hecke characters of modulus $\mathfrak{a}$ and $\mathfrak{b}$. For each weight k and each pair (η, ψ) , there is an Eisenstein series $E_k(\eta, \psi) \in \mathcal{E}_k(\mathfrak{a}\mathfrak{b}, \eta\psi)$ given by an explicit formula in terms of k , η , and ψ [Dasgupta-Darmon-Pollack].

Let $\text{char}(\mathfrak{N}, \chi)$ be the set of triples $(\eta, \psi, \mathfrak{t})$ such that η and ψ are primitive characters with $\eta\psi = \chi$ and $\mathfrak{t} \subseteq \mathbb{Z}_F$ is an ideal such that $\text{cond}(\eta)\text{cond}(\psi)\mathfrak{t} \mid \mathfrak{N}$, and let $\text{char}_1(\mathfrak{N}, \chi)$ be the same set modulo the relation $(\eta, \psi, \mathfrak{t}) = (\psi, \eta, \mathfrak{t})$. For $k \geq 2$, write also $\text{char}_k(\mathfrak{N}, \chi) := \text{char}(\mathfrak{N}, \chi)$. The following theorem, whose proof is given in Section A, describes a basis for the space.

Theorem 4.1.3 (Eq. (A.21)). *Assume $F \neq \mathbb{Q}$. Then the set*

$$\{E_k(\eta, \psi) \mid \mathfrak{t} : (\eta, \psi, \mathfrak{t}) \in \text{char}_k(\mathfrak{N}, \chi)\}$$

is a basis for $\mathcal{E}_k(\mathfrak{N}, \chi)$.

This means that we can produce a basis of $\mathcal{E}_k(\mathfrak{N}, \chi)$ by computing pairs (η, ψ) of modulus $\mathfrak{N}$ satisfying the conditions in Eq. (4.1.3).

Remark 4.1.4. Let $\mathbb{Q}(\zeta_n)$ be the minimal cyclotomic field of definition for (η, ψ) , i.e., the smallest n such that the images of η and ψ lie in $\mathbb{Q}(\zeta_n)$. The coefficient field of $E_k(\eta, \psi)$ is then contained in $\mathbb{Q}(\zeta_n)$. If (η, ψ) and (η', ψ') are Galois conjugate, i.e., if there is some $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ such that $\eta' = \sigma(\eta)$ and $\psi' = \sigma(\psi)$, $E_k(\eta, \psi)$ and $E_k(\eta', \psi')$ are also Galois conjugate, in particular, the $\mathbb{Q}$ -basis of the span of the Galois conjugates of $E_k(\eta, \psi)$ and of $E_k(\eta', \psi')$ yields the same vector space of forms over $\mathbb{Q}$. Since in the end we prefer to work with $\mathbb{Q}$ -bases for our spaces of modular forms, in practice we apply Galois descent to a single representative from each Galois conjugacy class of pairs to produce a basis for the Eisenstein series.

ubsec:quat}

4.2. Quaternionic methods. Given level $\mathfrak{N}$, a weight k , and a nebentypus χ , we primarily use quaternionic methods to produce q -expansions for a basis of $S_k(\mathfrak{N}, \chi)$ to precision B . These methods have two steps. First, we use the Jacquet–Langlands correspondence to compute matrices for the action of $\{T_p\}_{\text{Nm}(\mathfrak{p}) \leq B}$ on a space of quaternionic modular forms which is isomorphic as a Hecke module to $S_k(\mathfrak{N}, \chi)$. Second, we use these “Hecke matrices” to produce a basis of q -expansions spanning $S_k(\mathfrak{N}, \chi)$.

Algorithms for computing Hecke matrices on spaces with paritious weight and nebentypus character χ for which $\chi_0 = 1$ (trivial Dirichlet restriction) are presented in [DembeleVoight]. [DembeleDonnelly] give a procedure for producing a basis of q -expansions for $S_k(\mathfrak{N}, \chi)$ given these Hecke matrices (they only state their result for parallel weight 2 but it generalizes easily to general paritious weight and nebentypus). In [hmf-neb], the methods of [DembeleVoight] are extended to produce Hecke matrices on spaces with arbitrary nebentypus character. In [hmf-nonpar], all of the above methods are generalized to arbitrary (not necessarily paritious) weights when the narrow class number of F is 1.

cuspid-basis}

Theorem 4.2.1 ([DembeleVoight; DembeleDonnelly; hmf-neb]). *Let F be a totally real field. Given an integral ideal $\mathfrak{N}$ of F , a paritious weight $k \in \mathbb{Z}_{\geq 2}^{[F:\mathbb{Q}]}$, and a finite order Hecke character χ of F , there is an algorithm that computes a basis of q -expansions (to any given precision) spanning $S_k(\mathfrak{N}, \chi)$.*

Note that forms of (partial) weight one

theta-lift}

4.3. Theta series associated with characters of quadratic extensions. In this subsection, we give an alternative approach to constructing some cusp forms without going through the quaternionic methods of Section 4.2.

Let K/F be a quadratic extension with non-trivial automorphism τ and $\psi : K^\times \backslash \mathbb{A}_K^\times \rightarrow \mathbb{C}^\times$ be a Hecke character of K (as explicated in Section 2.5). Let v be a real place of F which splits into real places w, w' in K . We say that ψ is “good” if for every such place, $\varepsilon_w \neq \varepsilon_{w'}$ in the notation of Section 2.5.

l-aut-ind}

Theorem 4.3.1 ([JacquetLanglands]). *For any good Hecke character ψ of K , there is an associated Hilbert modular form θ_ψ on F .*

- (1) *The weight k of θ_ψ is determined by the infinity type ψ_∞ (using the notation of Section 2.5) as follows:*

$$k_v = \begin{cases} |a_w - b_w| + 1 & K_v \cong \mathbb{C} \text{ and } w|v \\ 1 & K_v \cong \mathbb{R} \times \mathbb{R}; \end{cases}$$

- (2) *The level of θ_ψ is $d_{K/F} \cdot \text{Nm}_{K/F}(\text{cond}(\psi))$, where $d_{K/F} \subseteq \mathcal{O}_F$ is the relative discriminant of K/F ;*
(3) *The nebentypus of θ_ψ is $\psi|_{\mathbb{A}_F^\times} \cdot \omega_{K/F}$, where $\omega_{K/F}$ is the quadratic character associated to K/F .*

Furthermore, θ_ψ is a cusp form if and only if ψ does not factor through $\text{Nm}_{K/F} : \mathbb{A}_K^\times \rightarrow \mathbb{A}_F^\times$.

Sketch. Jacquet and Langlands [JacquetLanglands] define an automorphic form θ_ψ of $\text{GL}_2(\mathbb{A}_F)$ called the automorphic induction of an automorphic form of $\text{GL}_1(\mathbb{A}_K)$ (a Hecke character of K). This automorphic induction has the property that $\Lambda(s, \theta_\psi) = \Lambda(s, \psi)$, where

Λ denotes the completed L -function (the product of the L -function and the gamma factors at the archimedean places). Formulas for $\Lambda(s, \psi)$ and $\Lambda(s, f)$ (f a Hilbert modular newform) can be found in [Miyake] and [Shimura] respectively. By comparing the gamma factors of $\Lambda(s, \psi)$ and $\Lambda(s, f)$, we deduce the condition on the weight of θ_ψ . For the remaining conditions, see [Harris]. $\square$

Remark 4.3.2. There is an associated automorphic form θ_ψ even when ψ fails the goodness condition above some infinite place v . However, the automorphic representation associated to θ_ψ will then be principal series at v .

The action of τ on F extends to an action of τ on $\mathbb{A}_F^\times$. Let ψ and ψ' be good Hecke characters. Then, $\theta_\psi = f_{\psi'}$ if and only if $\psi' \in \{\psi, \psi_\tau\}$, where $\psi_\tau(x) := \psi(\tau(x))$.

Remark 4.3.3. If K is itself totally real, then as noted above, any Hilbert modular forms induced from K are parallel weight 1. If K is mixed-signature, then any Hilbert modular form induced from K must be weight 1 at the places below the real places of K . However, by [patrikis], the infinity-type of any Hecke character of K descends to the maximal subfield of K which is either CM or totally real. Because K is mixed-signature, this means the infinity-type must descend the totally real field F . As noted in the discussion after Eq. (2.5.4), ψ must then be finite order up to twisting by the norm character, meaning that the induced form must be parallel weight 1.

Definition 4.3.4. A Hilbert modular cuspidal eigenform f of weight k is CM if there exists a CM extension K/F and a Hecke character ψ of K such that $f = \theta_\psi$. If $k = 1$ (parallel weight 1), then f is **dihedral** if there exists a quadratic extension K/F (not necessarily CM) and a finite order Hecke character ψ of K such that $f = \theta_\psi$.

The name dihedral refers to the projective image of the associated Artin representation in the parallel weight one case.

Theorem 4.3.5. Let ψ be a good Hecke character. The ideal coefficient (Eq. (2.4.10)) of θ_ψ at a prime $\mathfrak{p} \subset \mathbb{Z}_F$ is

$$a_{\mathfrak{p}} = \begin{cases} 0 & \mathfrak{p}\mathbb{Z}_K \text{ prime} \\ \psi(\mathfrak{q}) + \psi(\mathfrak{q}') & \mathfrak{p}\mathbb{Z}_K = \mathfrak{q}\mathfrak{q}', \mathfrak{q} \neq \mathfrak{q}' \\ \psi(\mathfrak{q}) & \mathfrak{p}\mathbb{Z}_K = \mathfrak{q}^2. \end{cases}$$

Proof. By the equality $L(s, \psi) = L(s, \theta_\psi)$, we have

$$\begin{aligned} \prod_{\mathfrak{q}|\mathfrak{p}} (1 - \psi(\mathfrak{q}) \text{Nm}(\mathfrak{q})^{-s})^{-1} &= \prod_{\mathfrak{q}|\mathfrak{p}} L_{\mathfrak{q}}(s, \psi) \\ &= L_{\mathfrak{p}}(s, \theta_\psi) \\ &= (1 - a_{\mathfrak{p}} \text{Nm}(\mathfrak{p})^{-s} + \chi(\mathfrak{p}) \text{Nm}(\mathfrak{p})^{k_0-1-2s})^{-1}. \end{aligned}$$

The result follows. $\square$

Corollary 4.3.6. There is an algorithm which, given a totally real field F , a level $\mathfrak{N} \subset \mathbb{Z}_F$, a weight k , and a Hecke character χ on F , produces a basis for the subspace of $S_k(\mathfrak{N}, \chi)$ spanned by automorphically induced forms.

Proof. By Eq. (4.3.1), the relative discriminant of K/F must divide $\mathfrak{N}$. Let $\mathbb{Z}_{F,\mathfrak{N}}^\times$ denote the S -unit group associated to (the primes dividing) $\mathfrak{N}$. By Kummer theory, any valid extension K has the form $F(\sqrt[\varepsilon]{\varepsilon})$, where ε is a representative of the quotient $\mathbb{Z}_{F,\mathfrak{N}}^\times/\mathbb{Z}_{F,\mathfrak{N}}^{\times 2}$. For any valid K , we can then loop through the Hecke characters ψ with infinity type corresponding to k and for which $\chi = \psi|_{\mathbb{A}_F^\times} \cdot \omega_{K/F}$, and use Eq. (4.3.5) to produce a q -expansion for θ_ψ for every possible ψ . The span of these θ_ψ , by definition, is a basis for the subspace of $S_k(\mathfrak{N}, \chi)$ we are interested in. $\square$

sec:moduli}

4.4. Pullbacks of Siegel modular forms. Angie: [In the following, we seem to use interchangeably $Y_0^1(1)_\mathfrak{b}$, $Y(\Gamma_0^1(1))_\mathfrak{b}$, and $Y_0^1(1)(\mathbb{C})_\mathfrak{b}$. Does it make sense to keep all three, or should we stick with one?] In this section, we first describe a map ϕ from $Y_0^1(1)_\mathfrak{b} := \Gamma_0^1(1)_\mathfrak{b} \backslash \mathcal{H}$ (where $\Gamma_0^1(1)_\mathfrak{b}$ is as defined in (??)) to a moduli space $\mathcal{A}$ of n -dimensional abelian varieties with polarization compatible with $\mathfrak{b}$. Intuitively, ϕ is the forgetful map which sends pairs $(A, \iota: \mathbb{Z}_F \hookrightarrow \text{End}(A))$ – where A is a polarized abelian variety – to A . Given a Siegel modular form on $\mathcal{A}$, it will turn out that its pullback to $Y_0^1(1)_\mathfrak{b}$ is a Hilbert modular form for the group $\Gamma_0^1(1)_\mathfrak{b}$. The Proj of the graded ring generated by the Hilbert modular forms in the image of the pullback will be isomorphic to the image of $Y_0^1(1)_\mathfrak{b}$ under ϕ .

In the case of abelian surfaces, this image is called a Humbert surface. By computing the aforementioned pullback map explicitly, we can compute equations for Humbert surfaces. Additionally, by computing the Igusa–Clebsch invariants [Igusa], (invariants associated to an abelian surface analogous to the j -invariant for elliptic curves) in terms of Siegel modular forms on $\mathcal{A}$, we can compute models for universal genus 2 curves with real multiplication [Mestre; LauterYang]. We give examples of the graded ring computation in Section 5.3.4, and in this subsection describe the construction and computation of the pullback map.

Each point of $Y_0^1(1)_\mathfrak{b}(\mathbb{C})$ corresponds to a complex abelian variety A with an embedding $\iota: \mathbb{Z}_F \subseteq \text{End}(A)$ and a chosen polarization. Because A is a point on $Y_0^1(1)_\mathfrak{b}(\mathbb{C})$, $H^1(A, \mathbb{Z}) \cong \mathbb{Z}_F \oplus \mathfrak{b}$ as a $\mathbb{Z}_F$ -module. Therefore, the module of polarizations on A is $(\wedge^2(\mathbb{Z}_F \oplus \mathfrak{b}))^* \cong (\mathfrak{d}\mathfrak{b})^{-1}$, and the totally positive elements $((\mathfrak{d}\mathfrak{b})^{-1})_{>0}$ form the polarization cone.

For a sequence of positive integers $m = (m_1, \dots, m_n) \in \mathbb{Z}_{>0}^n$ satisfying $m_1 \mid m_2 \mid \dots \mid m_n$, we denote

$$\text{Sp}(m_1, \dots, m_n) := \text{Sp}_{2n}(\mathbb{Q}) \cap \begin{pmatrix} I & 0 \\ 0 & m^* \end{pmatrix} \text{GL}_{2n}(\mathbb{Z}) \begin{pmatrix} I & 0 \\ 0 & m^* \end{pmatrix}^{-1}.$$

When $m = 1 = (1, \dots, 1)$, we write $\text{Sp}_{2n}(\mathbb{Z}) := \text{Sp}(1, \dots, 1)$. Let $\mathfrak{H}_n$ be the Siegel upper half-space. The symplectic group $\text{Sp}_{2n}(\mathbb{Q})$ acts on $\mathfrak{H}_n$ by fractional linear transformations. There is a bijection from the Siegel modular variety $\text{Sp}(m_1, \dots, m_n) \backslash \mathfrak{H}_n$ to the moduli space $\mathcal{A}(m_1, \dots, m_n)$ of abelian varieties with an $(m_1, \dots, m_n)$ -polarization.

Continue to let A be an abelian variety corresponding to a point in $Y_0^1(1)_\mathfrak{b}(\mathbb{C})$. If we forget about the $\mathbb{Z}_F$ -module structure on $H^1(A, \mathbb{Z})$, then $H^1(A, \mathbb{Z})$ is a symplectic $\mathbb{Z}$ -lattice preserved by $\text{Sp}(m_1, \dots, m_n)$, where $m_1 \mid \dots \mid m_n$ are the elementary divisors of $(\mathfrak{d}\mathfrak{b})^{-1}$. Then, there is a natural forgetful map $\phi: Y_0^1(1)_\mathfrak{b} \rightarrow \mathcal{A}(m_1, \dots, m_n)$. Let $k \in \mathbb{Z}$ denote a (parallel) weight. We now describe the pullback map.

A Siegel modular form $h(\tau) = h(\{\tau_{ij}\}_{i,j=1}^n) \in M_k(\mathrm{Sp}(m_1, \dots, m_n))$ has a Fourier expansion
 Angie: [What does the sum below $\sum_{i=1,j}^n T_{ij}\tau_{ji}$ mean?]

(4.4.1)

$$h(\tau) = \sum_{T \geq 0} a_h(T) q^{\mathrm{tr}(T\tau)} = \sum_{T \geq 0} a_h(T) q^{\sum_{i=1,j}^n T_{ij}\tau_{ji}} = \sum_{T \in \mathcal{T}} a_h(T) q^{\sum_{i=1}^n T_{ii}\tau_{ii} + \sum_{i < j} (1 + \frac{m_j}{m_i}) T_{ij}\tau_{ij}},$$

where $\mathcal{T}$ is the set of positive semidefinite symmetric $n \times n$ matrices such that for $i < j$, $T_{ij} + \frac{m_j}{m_i} T_{ji} \in \mathbb{Z}$.

-pullback}

Proposition 4.4.2 ([Geer88]). *For h as in (4.4.1), the pullback $\phi^*(h)$ is a Hilbert modular form for $\Gamma_0^1(1)_{\mathfrak{b}} = \mathrm{SL}_2(\mathbb{Z}_F \oplus \mathfrak{b})$, with Fourier expansion given by*

$$(\phi^*h)(z) := h(\phi(z)) = \sum_{T \geq 0} a_h(T) q^{\mathrm{tr}(TQz^*Q^t)} = a_h(0) + \sum_{\nu \in \mathfrak{d}^{-1}\mathfrak{b}} \left(\sum_{\substack{T \in \mathcal{T} \\ \mathrm{tr}(TQz^*Q^t) = \nu}} a_h(T) \right) q^{\mathrm{Tr}(\nu z)},$$

where $Q \in \mathrm{GL}_n(\mathbb{R})$ is such that

$$\mathbb{Z}^n \oplus \bigoplus_{i=1}^n m_i \mathbb{Z} = \begin{pmatrix} Q & 0 \\ 0 & Q^{-t} \end{pmatrix} (\mathbb{Z}_F \oplus \mathfrak{b}).$$

Remark 4.4.3. The pullbacks given by Eq. (4.4.2) have full level and have nebentypen with trivial conductor but possibly nontrivial ideal class characters. In the language of Section 2.5, $\chi_D = 1$ but χ_I may be nontrivial. This first occurs for the quadratic field of discriminant $D = 40$, which has $\# \mathrm{Cl} F = \# \mathrm{Cl}^+ F = 2$.

4.4.1. *Humbert surfaces.* For the remainder of Section 4.4, we assume that F is a totally real quadratic field, so $n = 2$. Let $M_k(\mathrm{Sp}_4(\mathbb{Z}))$ be the $\mathbb{C}$ -vector space of Siegel modular forms of weight k . Igusa proved [Igusa62] that the graded $\mathbb{C}$ -algebra of scalar-valued even-weight Siegel modular forms in genus 2 is given by

$$M^{\mathrm{even}}(\mathrm{Sp}_4(\mathbb{Z})) := \bigoplus_{k \in 2\mathbb{Z}_{\geq 0}} M_k(\mathrm{Sp}_4(\mathbb{Z})) = \mathbb{C}[\psi_4, \psi_6, \chi_{10}, \chi_{12}],$$

where $\psi_4, \psi_6, \chi_{10}$, and χ_{12} are Siegel modular forms whose Fourier expansions are given by explicit formulas – the ψ_k are Siegel Eisenstein series and the χ_k are given by linear combinations of weight- k products of Siegel Eisenstein series.

If we choose $\mathfrak{b} = \mathfrak{d}^{-1}$, then $m_1 = \dots = m_n = 1$. Then, the graded ring generated by the pullbacks of $\psi_4, \psi_6, \chi_{10}$, and χ_{12} under the forgetful map $\phi: Y_0^1(1)_{\mathfrak{d}^{-1}} \rightarrow \mathcal{A}_2$ can be used to compute a model for the Humbert surface $\mathrm{im} \phi$. We remind the reader that the pullbacks are forms for $\mathrm{SL}(\mathbb{Z}_F \oplus \mathfrak{d}^{-1})$. As discussed in Eq. (2.7.7), there is a decomposition

$$(4.4.4) \quad M_k(\Gamma_0^1(1)_{\mathfrak{d}^{-1}}) \cong M_k(\Gamma_0^1(1)_{\mathfrak{d}^{-1}}, \omega_0) \oplus M_k(\Gamma_0^1(1)_{\mathfrak{d}^{-1}}, \omega_0 \eta),$$

η is the nontrivial quadratic character of $\mathbb{Z}_{F, > 0}^\times / \mathbb{Z}_F^{\times 2}$. Note that $M_k(\Gamma_0^1(1)_{\mathfrak{d}^{-1}}, \omega_0) \cong M_k(\Gamma_0(1)_{\mathfrak{d}^{-1}})$ is the subspace of $\mathrm{GL}^+(\mathbb{Z}_F \oplus \mathfrak{d}^{-1})$ forms within the space of $\mathrm{SL}(\mathbb{Z}_F \oplus \mathfrak{d}^{-1})$ forms. The details of our implementation follow [KiefferPageRobert; LauterYang].

In Section 5.3.4, we use these methods in conjunction with the algorithm for computing graded rings presented in Section 5 to compute models for Humbert surfaces.

5. COMPUTING GRADED RINGS

Edgar: [I will try to fix this section]

Fix a totally real field F of degree n , a level $\mathfrak{N} \subset \mathbb{Z}_F$, and a narrow class representative $\mathfrak{b}$. Throughout this section, we abuse notation use k to denote a nonnegative integer or the n -tuple $(k, \dots, k)$. When the weight is parallel, $M_k(\Gamma_0(\mathfrak{N}))_{\mathfrak{b}}$ has a basis of forms with coefficients in $\mathbb{Q}$ (see e.g. [hmf-nonpar]). Therefore, we can define

$$S := S(\mathfrak{N}, F)_{\mathfrak{b}} := \bigoplus_{k \in 2\mathbb{Z}_{\geq 0}} M_k(\Gamma_0(\mathfrak{N}))_{\mathfrak{b}}$$

to be the graded $\mathbb{Q}$ -algebra of Hilbert modular forms of level $\mathfrak{N}$. By [Geer88], S is a finitely generated $\mathbb{Q}$ -algebra. In this section, we give an algorithm for computing a presentation of S as a $\mathbb{Q}$ -algebra.

Using the algorithms of Eq. (4.1.3) and ??, we can compute the graded pieces $M_k(\Gamma_0(\mathfrak{N}))_{\mathfrak{b}}$ and also multiply elements in S . Let S_D denote the finitely presented ring whose generators are the generators of S of degree at most D and whose relations are the relations of S of degree at most D . We can think of this as an “approximation of S to degree D ”. Because we can compute arbitrary graded pieces of S , we can compute any S_D by iteratively computing each graded piece and using linear algebra to detect any new relations and determine if we need to include any new generators. We denote by `ApproximateGradedRing`($\mathfrak{N}, \mathfrak{b}, D, B$) the algorithm which, given a level $\mathfrak{N}$, a narrow class representative $\mathfrak{b}$, a degree D , and a precision B , returns S_D . There is a natural map $\phi_D: S_D \rightarrow S$. Because S_D may be missing higher-degree relations between the existing generators as well as higher-degree generators, S_D is not expected to be a subring of S and we cannot say too much about ϕ_D in general. However, because S is finitely presented, ϕ_D is an isomorphism (i.e. $S_D = S$) for D large enough. The crux of our algorithm will be certifying that an S_D we have computed is indeed equal to S .

Remark 5.0.1. In what follows, we frequently need to certify linear relations between forms $f_1, \dots, f_m \in M_k(\Gamma_0(\mathfrak{N}))_{\mathfrak{b}}$ computed to finite precision. The Sturm bound provides a bound on the precision needed to certify linear relations between classical modular forms, but the known equivalents for Hilbert modular forms are too large to be practically useful [BurgosGilPacetti]. Instead, we can certify linear relations by first computing a basis of $M_k(\Gamma_{\mathfrak{b}})$. All of our methods for computing bases are “certified” in the sense that they come with a guarantee that the basis forms should be linearly independent and that there are no additional basis vectors. If we can compute a certified basis for $M_k(\Gamma_{\mathfrak{b}})$ at some precision – that is, if the forms we know form a basis for $M_k(\Gamma_{\mathfrak{b}})$ are linearly independent when truncated to this precision – then we know that any linear relation between the $\{f_i\}_i$ observed at this precision must be genuine.

5.1. Preparatory lemmas and functions. Consider the Hilbert series of S ,

$$\text{Hilb}_S(T) := \sum_{k \geq 0} \dim M_k(\Gamma_0(\mathfrak{N}))_{\mathfrak{b}} T^k \in \mathbb{Z}[[T]]$$

The Hilbert series of any finitely generated graded ring is known to be a rational function. However, even though we can compute arbitrary graded pieces of S – and hence compute arbitrary coefficients of $\text{Hilb}_S(T)$ – it is not immediately clear how to turn this into an

algorithm for computing $\text{Hilb}_S(T)$ as a rational function. In particular, it is difficult to know when we have computed enough graded pieces to see all the generators and relations. Nonetheless, $\text{Hilb}_S(T)$ can be computed as a rational function using the trace formula.

Theorem 5.1.1 ([HMSDB]). *There exists an algorithm which, given a level $\mathfrak{N}$ and a narrow class representative $\mathfrak{b}$, computes polynomials $P(T), Q(T) \in \mathbb{Z}[T]$ such that $\text{Hilb}_S(T) = \frac{P(T)}{Q(T)}$.*

We denote such an algorithm by `HilbertSeriesFromTraceFormula`($\mathfrak{N}, \mathfrak{b}$). The following lemma shows that we can compute the Hilbert series of a finitely presented ring.

Lemma 5.1.2. *Let $S' := \mathbb{Q}[x_1, \dots, x_m]/I$ for some ideal $I \subset R$ given in terms of finitely many polynomial relations $r_1, \dots, r_n$. There is an algorithm `HilbertSeries` which computes polynomials $P(T)$ and $Q(T)$ such that $\text{Hilb}_{S'}(T) = \frac{P(T)}{Q(T)}$.*

Proof. Let $R' := \mathbb{Q}[x_1, \dots, x_n]$, and let $I_{>}$ be the monomial ideal defined by the leading monomials (with respect to, say, the lexicographic ordering) of polynomials in I . By the theory of Gröbner degenerations ([Eisenbud]), there is a scheme $\text{Spec } R'[t]/\mathcal{I}$, flat over $\text{Spec } \mathbb{Q}[t]$, such that $\text{Spec } R'/I$ and $\text{Spec } R'/I_{>}$ are the fibres of this scheme at $t = 0$ and $t = 1$, respectively. The Hilbert series is constant in flat families, so it's enough to compute the Hilbert series of $R'/I_{>}$. Because $I_{>}$ is a monomial ideal, the graded piece $(R'/I_{>})_k$ is spanned by the monomials of degree k not contained in $I_{>}$, and these can be easily enumerated. Furthermore, because we know the degrees of the largest generators and relations of R' , we have a bound on how many graded pieces we need to compute before we can realize $\text{Hilb}_{S'}(T)$ as a rational function. $\square$

Next, we give a procedure to check if a given set of Hilbert modular forms contains an algebraically independent set that is as large as possible.

Lemma 5.1.3. *Let $\{f_i\}_{0 \leq i \leq m}$ be elements in S_D of the same degree – i.e., Hilbert modular forms of the same weight – with coefficients in $\mathbb{Q}$. Let $r := \min(n, m)$. There is an algorithm `MaximallyAlgIndependent` which, given the q -expansions of the $\{f_i\}_i$ to some finite precision (as defined in Section 3), satisfies the following properties:*

- (1) `MaximallyAlgIndependent` returns true only if $\{f_i\}_i$ contains an algebraically independent (over $\mathbb{Q}$) subset of cardinality $r + 1$.
- (2) If $\{f_i\}$ contains an algebraically independent (over $\mathbb{Q}$) subset of cardinality $r + 1$ and the precision D is large enough, `MaximallyAlgIndependent` return true.

Furthermore, if `MaximallyAlgIndependent` returns true, then $\dim \phi_D(S_D) \geq r$.

Proof. First, we claim that it is enough to prove Eq. (5.1.3) with algebraic independence over $\mathbb{C}$ instead of over $\mathbb{Q}$. This is because – by the q -expansion principle and because the f_i have Fourier coefficients in $\mathbb{Q}$ – some subset of the $\{f_i\}_i$ satisfy an algebraic equation over $\mathbb{C}$ if and only if they satisfy an algebraic equation over $\mathbb{Q}$.

If X is a projective variety and f is a homogeneous element of its coordinate ring, write X_f for the open affine subset of X corresponding to inverting f . Similarly, if R is a graded ring and f a homogeneous element of R , we write $R_f^{(0)}$ for the degree zero elements of R_f .

Let $U \subset \text{Proj } S = X_0(\mathfrak{N})$ and $V \subset \text{Proj } \phi_D(S_D)$ denote the respective open affine subschemes corresponding to inverting the homogeneous element $f_0 \in \phi_D(S_D) \hookrightarrow S$. Equivalently,

$U := X_0(\mathfrak{N})_{f_0} = \text{Spec } S_{f_0}^{(0)}$ and $V := (\text{Proj } \phi_D(S_D))_{f_0} = \text{Spec } \phi_B(S_B)_{f_0}^{(0)}$. Consider the map

$$(5.1.4) \quad \begin{aligned} \Phi: U &\longrightarrow \mathbb{A}^m \\ z &\longmapsto \left(\frac{f_i}{f_0}(z) \right)_{1 \leq i \leq m}. \end{aligned}$$

This is a map of affine algebraic varieties because, by the definition of U , each function $\frac{f_i}{f_0}$ is a regular function on U .

Then, Φ and its pullback $\Phi^\sharp$ fit into the following commutative diagrams.

$$\begin{array}{ccc} U := X_0(\mathfrak{N})_{f_0} & \longrightarrow & V := (\text{Proj } \phi_D(S_D))_{f_0} \\ & \searrow & \downarrow \\ \Phi: z \longmapsto \left(\frac{f_i}{f_0}(z) \right)_{0 \leq i \leq m} & & \mathbb{A}^m \end{array}$$

$$\begin{array}{ccc} S_{f_0}^{(0)} & \xleftarrow{\quad} & \phi_D(S_D)_{f_0}^{(0)} \\ & \nwarrow \Phi^\sharp & \uparrow x_i \mapsto \frac{f_i}{f_0} \\ & & \mathbb{C}[x_1, \dots, x_n] \end{array}$$

We see that Φ must factor through V because $\Phi^\sharp$ factors through $\mathbb{C}[V]$.

The image $\text{im } \Phi$ is not generally an algebraic variety. However, we can consider its Zariski closure $\overline{\text{im } \Phi}$. By definition, it is $\text{Spec } \mathbb{C}[x_1, \dots, x_n]/I$, where I is the ideal of algebraic relations satisfied by the components $\{\frac{f_i}{f_0}\}_i$. This is the same as $\text{Spec } \mathbb{C}[\{\frac{f_i}{f_0}\}_i]$. In particular, the dimension of $\overline{\text{im } \Phi}$ is the same as the transcendence degree of the function field of $\overline{\text{im } \Phi}$, which in turn is exactly the size of the largest algebraically independent subset of $\{\frac{f_i}{f_0}\}_{1 \leq i \leq m}$. This also explains why there cannot be more than $n + 1$ algebraically independent elements in $\{f_i\}_i$, as the dimension of U is n .

The point of all this is that we can study $\overline{\text{im } \Phi}$ (which captures algebraic independence) by studying $\text{im } \Phi$ (which we can access analytically by computing q -expansions). If the dimension of $\text{im } \Phi$ (as a complex analytic manifold) is t , then the analytic dimension (and hence algebraic dimension, by GAGA) of $\overline{\text{im } \Phi}$ is at least t . Therefore, (perhaps after reindexing the $\{f_i\}$) there exists a subset $T \subset \{1 \dots, m\}$ of cardinality $t + 1$ such that there are no algebraic relations among $\{\frac{f_i}{f_0}\}_{i \in T}$, and hence (because the $\{f_i\}_{0 \leq i \leq m}$ are all of the same weight) no algebraic relations among $\{f_0\} \cup \{f_i\}_{i \in T}$.

To finish the proof of the first part of the lemma, it remains to show that we can compute the analytic dimension of $\text{im } \Phi$. The analytic dimension of a smooth map from U to $\mathbb{A}^n$ is the same as the rank of its Jacobian at a generic point. Because we are given the f_i explicitly as q -expansions to some precision, we can compute an approximation to the Jacobian of Φ as a function of q . The quality of this approximation depends on the precision – if the precision is too low (e.g. if the precision is 0), then the matrix we compute may have rank smaller than $\dim(\text{im } \Phi)$, but for sufficiently high precision this is not an issue.

If the Jacobian is rank r at some point, then analytically there is a local submersion from $U(\mathbb{C})$ to $\mathbb{C}^r$. Therefore, there is a local submersion from $V(\mathbb{C})$ to $\mathbb{C}^r$, and the analytic dimension of $V(\mathbb{C})$ is at least r . By GAGA, the analytic dimension and algebraic dimension agree. Therefore, $\dim V \geq r$. $\square$

Remark 5.1.5. The condition that the f_i have the same degree is a mild one. For example, as long as degrees of all the f_i are divisible by the degree of f_0 , we could have taken i^{th} coordinate to instead be the meromorphic form $f_i f_0^{-\frac{\deg f_i}{\deg f_0}}$, and an analogous argument goes through. We can take this even further, but we do not need to for our purposes.

If we take $m = n (= \dim S - 1)$ in Eq. (5.1.3), then **MaximallyAlgIndependent** checks that the given q -expansions actually give (at least locally) a modular parameterization of the Hilbert modular variety.

5.2. An algorithm for computing graded rings. We will now introduce an algorithm (Algorithm 5.2.2) which computes S_D for larger and larger D , until $\text{Hilb}_{S_D} = \text{Hilb}_S$ (computed as in Eq. (5.1.2)) and $\dim \phi(S_D) = \dim S = n + 1$ (checked using Eq. (5.1.3)). Let **Basis** be a function which, given a level $\mathfrak{N}$, a weight k , a narrow class representative $\mathfrak{b}$, and a precision B , returns a basis for $M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}})$ using the methods of Section 4. As the degree D increases, we need to increase the precision B we use so that we have enough coefficients to disambiguate the entire space of forms of degree D . We will take $B := g(D)$ for some function $g: \mathbb{N} \rightarrow \mathbb{N}$ which we choose as appropriate. Heuristically, because $\dim M_k(\Gamma_0(\mathfrak{N})_{\mathfrak{b}})$ grows linearly in nk , we can take g to be some linear function.

Remark 5.2.1. In practice, rather than changing the precision throughout the execution of the algorithm, we chose a sufficiently large initial precision B_0 and do not update it.

}graded-ring}

Algorithm 5.2.2: GradedRing($\mathfrak{N}$)

Data: $\mathfrak{N} \subset \mathbb{Z}_F$

/ Find a D_0 such that S_{D_0} has the same Hilbert series as S */*

$n := [F : \mathbb{Q}];$

$H_S := \text{HilbertSeriesFromTraceFormula}(\mathfrak{N}, \mathfrak{b});$

for $D \in \mathbb{Z}_{\geq 0}$ **do**

$S_D := \text{ApproximateGradedRing}(\mathfrak{N}, \mathfrak{b}, D, g(D));$

if $\text{HilbertSeries}(S_D) = H_S$ **then**

$D_0 = D;$

/ Find a $D \geq D_0$ such that S_D has $n + 1$ algebraically independent elements which are still algebraically independent in $\phi(S_D)$ */*

for $D \geq D_0$ **do**

if $\text{MaximallyAlgIndependent}(\text{Basis}(k, \mathfrak{N}, \mathfrak{b}, g(D)))$ **then**

return $S_D;$

We remind the reader that $(S_D)_D$ denotes the D^{th} graded piece of the ring S_D , or in this case the space of Hilbert modular forms of level $\mathfrak{N}$ and weight D .

o-comm-alg}

Lemma 5.2.3. *Let R and S be graded rings such that*

- *$\text{Proj } R$ is an integral scheme;*
- *$\text{Hilb}_R = \text{Hilb}_S$;*
- *There is a morphism $\phi: R \rightarrow S$ such that $\dim \phi(R) = \dim R$.*

Then, $R = S$.

Proof. Because $\text{Hilb}_R = \text{Hilb}_S$, we certainly have $\dim R = \dim S$ [Stanley78]. Let I be an ideal of R such that $\phi(R) \cong R/I$. Since $\text{Proj } R$ is integral and $\text{Proj } R/I \rightarrow \text{Proj } R$ is a closed immersion, $\dim R = \dim R/I$ implies that $R = R/I$ and hence $\phi: R \hookrightarrow S$ is an inclusion of graded rings. But now, since $\text{Hilb}_R = \text{Hilb}_S$, we must have $R = S$. $\square$

Remark 5.2.4. Eq. (5.2.3) is why we are working with the components of the Hilbert modular variety separately, as otherwise the scheme would not be integral.

graded-ring}

Theorem 5.2.5. *Algorithm 5.2.2 returns a ring S_D , such that $S_D = S$.*

Proof. The first loop in Algorithm 5.2.2 will always succeed in producing a D_0 such that $\text{Hilb}_{S_{D_0}} = \text{Hilb}_S$. If there is some $D \geq D_0$ such that **MaximallyAlgIndependent** succeeds, then this means that $\dim \phi_D(S_D) = n + 1 = \dim S$. By Eq. (5.2.3), this means that $S_D = S$.

In Algorithm 5.2.2, we apply **MaximallyAlgIndependent** to elements $\{f_0, \dots, f_n\} \subset S_D$ which do not satisfy any relations in degree at most D (this is the output of **CandidateAlgIndependentElements**) in order to ensure that they do not satisfy any relations in higher degree. Even though **MaximallyAlgIndependent** is not guaranteed to accept even when the $\{f_i\}_i$ are algebraically independent, Algorithm 5.2.2 is still guaranteed to terminate. This is because as D increases, the precision used in **ApproximateGradedRing** to compute S_D also increases, so we will inevitably reach whatever precision is necessary to certify that the set of $\{f_i\}_i$ – which does not change even as D increases if it is indeed algebraically independent – are algebraically independent. $\square$

ementation}

5.3. Examples. JK: [In general it would be great if we could point out, for each example, a Magma file that contains the relevant computations!] Edgar: [We should just make clear here, that every example is reproducible in the repository] JK: [+1]

We illustrate Algorithm 5.2.2 and its verification in dimensions one and two: the modular curve $X_0(11)$ and the first Hilbert modular surfaces. We also include two proof-of-concept computations over a totally real cubic field, showing how product spaces and Hecke translates can produce full graded pieces in higher dimension, and how coefficient conditions at the cusps witness a Kodaira-dimension bound for a Hilbert modular threefold.

5.3.1. Modular curve $X_0(11)$. We start with a classical example where $F = \mathbb{Q}$. Let $\Gamma = \Gamma_0(11 \cdot \mathbb{Z})$. The Hilbert series can be computed directly from dimension formulas, and one obtains

$$\text{Hilb}_{M(\Gamma)}(T) = 1 + \frac{2T^2}{(1 - T^2)^2} = 1 + 2T^2 + 4T^4 + 6T^6 + 8T^8 + 10T^{10} + \dots$$

Choose fixed generators

$$E_2 \in E_2(\Gamma), \quad f_2 \in S_2(\Gamma), \quad E_4 \in E_4(\Gamma)$$

with the normalizations used in the displayed q -expansions below; E_2 and E_4 are Eisenstein series of level 11, and f_2 is a cusp form. We have $M_2(\Gamma) = \langle E_2, f_2 \rangle$, $M_4(\Gamma) = \langle E_4 \rangle + M_2(\Gamma)^2$. There are nine weighted monomials of weight 8 in E_2, f_2, E_4 , while $\dim M_8(\Gamma) = 8$; hence there is at least one relation among them in weight 8, explicitly $R_8 = 0$, where

$$R_8 = E_2^4 + 240E_2^3f_2 + 2592E_2^2f_2^2 + 8640E_2f_2^3 + 9216f_2^4 \\ - 122E_2^2E_4 - 240E_2f_2E_4 + 288f_2^2E_4 + 121E_4^2.$$

With $S := \mathbb{Q}[E_2, f_2, E_4]/(R_8)$, direct computation verifies the equality $\text{Hilb}_{M(\Gamma)} = \text{Hilb}_S$.

To verify that $S = M(\Gamma)$, we apply the strategy of Eq. (5.1.3) to the elements $\{f_2, E_2\} \in S$. Let U be the open subset of $X_0(11)$ where E_2 is nonzero. The map $\Phi : U \rightarrow \mathbb{C}$ of Eq. (5.1.4) is given by $z \mapsto f_2(z)/E_2(z)$.

As power series in q , we find that

$$\frac{f_2}{E_2} = q - 2q^2 - 13q^3 + 14q^4 + 169q^5 + 2q^6 - 2042q^7 - 2040q^8 + \dots$$

and its derivative with respect to the local parameter q is

$$\frac{d}{dq} \left(\frac{f_2}{E_2} \right) = 1 - 4q - 39q^2 + 56q^3 + 845q^4 + 12q^5 - 14294q^6 - 16320q^7 + 203454q^8 + \dots$$

It follows that the Jacobian of Φ is nonvanishing at the cusp, so the algebraic-independence condition in Eq. (5.1.3) holds. A direct factorization check shows that R_8 is irreducible over $\mathbb{Q}$, hence S is an integral domain. Together with the Hilbert-series equality above, Eq. (5.2.3) gives $S = M(\Gamma)$.

As an independent check, the degree-4 Veronese embedding of this weighted plane curve gives a genus-one curve over $\mathbb{Q}$ with a rational cusp; a minimal Weierstrass model computation identifies its Jacobian with the conductor-11 elliptic curve.

5.3.2. First cases of Hilbert modular surfaces. We apply Algorithm 5.2.2 to the real quadratic fields of smallest discriminant with trivial narrow class group. For $D = 5, 8, 13$, the algorithm applied to $\Gamma_0(1)$ reproduces the classical presentations of Gundlach and Hirzebruch [Gundlach; Hirzebruch77] ($D = 5$), Hirzebruch [Hirzebruch77] ($D = 8$), and van der Geer–Zagier [GeerZagier] ($D = 13$). For $D = 5$ ($\mathbb{Q}(\sqrt{5})$), the presentation has generators in weights 2, 6, 10, 20 and a single relation. For $D = 8$ ($\mathbb{Q}(\sqrt{2})$), Hirzebruch realizes the corresponding even-weight graded ring as a double cover of a rational surface, with generators in weights 2, 4, 6, 14 and a single relation of degree 28. For $D = 13$ ($\mathbb{Q}(\sqrt{13})$), van der Geer–Zagier gives generators in weights 2, 4, 6, 6, 8 and two relations of degrees 12 and 16.

To compare two weighted-graded presentations $A = \mathbb{Q}[x_i]/I$ and $B = \mathbb{Q}[y_j]/J$, we first find candidate weighted-homogeneous maps in both directions by writing the image of each generator as a homogeneous element of the same weight. The certificate is exact and algebraic: one checks that the two maps carry I into J and J into I , and that both composites are the identity on generators modulo the respective ideals. Since the maps are weighted-homogeneous, these are finite ideal-membership checks over $\mathbb{Q}$. Hence the displayed formulas below certify a graded $\mathbb{Q}$ -isomorphism of the quotient rings.

We spell out the isomorphism for $D = 13$ as the representative worked example. The $D = 5$ and $D = 8$ identifications are analogous; we omit them here because they add no new method, and they are available in [ourcode].

Edgar: [better name for the subsection?] JK: [Perhaps this subsection could just be named "Hilbert modular surface for $\mathbb{Q}(\sqrt{13})$ ". We could conclude it with a small remark to say that $D=5,8$ are similar.]

Let $\Gamma = \Gamma_0(1)$ be the Hilbert modular group for $\mathbb{Q}(\sqrt{13})$, and let $P = \mathbb{Q}[X_2, X_4, X_6, Y_6, X_8]$, with degrees indicated by the subscripts. In the next two presentations the same symbols denote abstract weighted polynomial generators; the automorphism ϕ below identifies the two copies of P .

Van der Geer–Zagier [GeerZagier] determine $M(\Gamma)$ as $P/(R_{12}, R_{16})$, where

$$R_{12} = X_4^3 - X_6 Y_6$$

$$R_{16} = X_8^2 - (256X_2^5 X_6 - 128X_2^4 X_4^2 + 16X_2^3 X_4 Y_6 - 656X_2^3 X_4 X_6 + 776X_2^2 X_4^3 \\ - 261X_2 X_4^2 Y_6 + 27X_4 Y_6^2 - 27X_2^2 X_6^2 + 495/2 X_2 X_4^2 X_6 - 947/16 X_4^4 + 54X_4 X_6^2).$$

Applying Algorithm 5.2.2 to Γ yields instead the presentation $M(\Gamma) = P/(R'_{12}, R'_{16})$ where

$$R'_{12} = X_4^3 - X_2 X_4 Y_6 - 4X_6 Y_6 + 4Y_6^2$$

$$R'_{16} = X_2^4 X_4^2 - 4X_2^3 X_4 X_6 - 816X_2 X_4^2 X_6 + 16X_2^2 X_6^2 + 3456X_4 X_6^2 - X_2^5 Y_6 + 4X_2^3 X_4 Y_6 \\ - 2932X_2 X_4^2 Y_6 + 784X_2^2 X_6 Y_6 + 9872X_4 X_6 Y_6 + 2948X_2^2 Y_6^2 - 11600X_4 Y_6^2 \\ + 912X_4^2 X_8 - 64X_2 X_6 X_8 - 912X_2 Y_6 X_8 + 64X_8^2.$$

The two presentations agree over $\mathbb{Q}$: the graded ring automorphism ϕ of P given by

$$\begin{aligned} \phi(X_2) &= \frac{1}{4}X_2, & \phi(X_4) &= -2X_4, & \phi(X_6) &= 4Y_6, \\ \phi(Y_6) &= -2(X_2 X_4 + 4X_6 - 4Y_6), & \phi(X_8) &= 8X_8 - 4X_2 X_6 + 57X_4^2 - 57X_2 Y_6. \end{aligned}$$

For example, a direct computation gives $\phi(R_{12}) = -8R'_{12}$ and $\phi(R_{16}) \in (R'_{12}, R'_{16})$; applying the inverse automorphism gives the reverse inclusion. This ϕ is not the only such isomorphism: since R_{16} is quadratic in the weight-8 generator, precomposing with the involution $X_8 \mapsto -X_8$ (a graded automorphism of $P/(R_{12}, R_{16})$) gives another, as do the weighted rescalings $X_j, Y_j \mapsto \lambda^j X_j, \lambda^j Y_j$ for $\lambda \in \mathbb{Q}^\times$.

5.3.3. *Hilbert modular threefolds.* Edgar: [the examples below need to be carefully reread, and I am not fully convinced of their correctness.]

We finish with two proof-of-concept computations over a totally real cubic field. Unlike the surface examples above, neither reproduces a known graded-ring presentation: the first demonstrates that the pipeline can produce and test full graded pieces in higher dimension by enlarging product spans with Hecke translates, and the second witnesses a Kodaira-dimension bound of Logan directly on q -expansions.

JK: [Maybe avoid using "example" environments if we only have one example per subsection?]

Example 5.3.1. Let $F = \mathbb{Q}(\zeta_7 + \zeta_7^{-1}) = \mathbb{Q}(\alpha)$, where α is a root of $x^3 - x^2 - 2x + 1$, and let $\Gamma = \mathrm{SL}_2(\mathbb{Z}_F)$. In each of the parallel weights 2, 4, 6, 8, and 10 the Eisenstein subspace is one-dimensional; write E_k for a generator in weight k . The trace-formula dimensions used below are complemented by direct linear-algebra checks on q -expansions.

In weight 8, the five products

$$E_2^4, \quad E_2^2 E_4, \quad E_2 E_6, \quad E_4^2, \quad E_8$$

are linearly independent. Since $\dim_{\mathbb{C}} M_8(\Gamma) = 5$, they form a basis for $M_8(\Gamma)$.

In weight 10, the product span generated by

$$E_2^5, \quad E_2^3 E_4, \quad E_2^2 E_6, \quad E_2 E_4^2, \quad E_2 E_8, \quad E_4 E_6, \quad E_{10}$$

has dimension 7, while $\dim_{\mathbb{C}} M_{10}(\Gamma) = 9$. The rational prime 13 splits as $13\mathbb{Z}_F = \mathfrak{p}_1 \mathfrak{p}_2 \mathfrak{p}_3$. Applying Hecke operators at primes above 13 produces the two missing directions: for two of these primes, say $\mathfrak{p}_1$ and $\mathfrak{p}_2$, the nine forms

$$E_2^5, E_2^3 E_4, E_2^2 E_6, E_2 E_4^2, E_2 E_8, E_4 E_6, E_{10}, T_{\mathfrak{p}_1}(E_2^5), T_{\mathfrak{p}_2}(E_2^5)$$

are linearly independent. Adding $T_{\mathfrak{p}_3}(E_2^5)$ yields one further linear relation, consistent with $\dim_{\mathbb{C}} M_{10}(\Gamma) = 9$.

The same field also lets us make contact, at the level of q -expansions, with the study of Kodaira dimensions of Hilbert modular threefolds by Logan [**LoganThreefolds**]. **JK:** [\[This example has been moved to Section 7\]](#)

t-examples}

5.3.4. *Models for universal genus 2 curves with RM.* We now show some examples of pulling back Siegel modular forms to Hilbert modular forms, giving rise to universal genus 2 curves with real multiplication (RM).

Example 5.3.2. Consider the case $F = \mathbb{Q}(\sqrt{12})$. Applying our algorithm, the graded ring for $\Gamma = \mathrm{GL}^+(\mathbb{Z}_F \oplus \mathfrak{d}^{-1})$ has the presentation

$$M(\Gamma) = \mathbb{C}[X_2, X_4, X_6].$$

Following (4.4.4), given $g \in M_k(\Gamma_0^1(1)_{\mathfrak{d}^{-1}})$, we write $g = g_+ + g_-$, where $g_+ \in M_k(\Gamma_0^1(1)_{\mathfrak{d}^{-1}}, \omega_0) = M_k(\Gamma_0(1)_{\mathfrak{d}^{-1}})$ and $g_- \in M_k(\Gamma_0^1(1)_{\mathfrak{d}^{-1}}, \omega_0 \eta)$.

We abuse notation and define $\psi_4, \psi_6, \chi_{10}$, and χ_{12} to be the pullbacks (to $M_k(\Gamma_0^1(1)_{\mathfrak{d}^{-1}})$ of the corresponding Siegel modular forms. Each $+$ -part is a Hilbert modular form for Γ , and can be written in terms of X_2, X_4 , and X_6 :

$$\begin{aligned} \psi_{4,+} &= \frac{1}{4}X_2^2 + 72X_4, & \psi_{6,+} &= \frac{1}{8}X_2^3 - 162X_2X_4 - 1728X_6 \\ \chi_{10,+} &= -\frac{1}{8}X_4X_6, & \chi_{12,+} &= \frac{1}{8}X_4^3 - \frac{1}{24}X_2X_4X_6 + X_6^2. \end{aligned}$$

On the other hand, because the product of two forms has unit character equal to the product of their unit characters,

$\psi_{4,-}^2, \psi_{4,-}\psi_{6,-}, \psi_{4,-}\chi_{10,-}$ and $\psi_{4,-}\chi_{12,-}$ are also Hilbert modular forms over Γ , and we have

$$\begin{aligned} \psi_{4,-}^2 &= 7200X_2X_6, & \psi_{4,-}\psi_{6,-} &= -7560X_2^2X_6 \\ \psi_{4,-}\chi_{10,-} &= \frac{15}{2}X_2X_6^2, & \psi_{4,-}\chi_{12,-} &= -\frac{15}{2}X_2X_4^2X_6 + \frac{5}{2}X_2^2X_6^2. \end{aligned}$$

In particular, for $\Gamma^1 = \mathrm{SL}_2(\mathbb{Z}_F \oplus \mathfrak{d}^{-1})$, we get

$$M(\Gamma^1) = \frac{\mathbb{C}[X_2, X_4, Y_4, X_6]}{(Y_4^2 - 2X_2X_6)}.$$

with

$$\begin{aligned}\psi_{4,-} &= 60Y_4, & \psi_{6,-} &= -63X_2Y_4 \\ \chi_{10,-} &= \frac{1}{16}X_6Y_4, & \chi_{12,-} &= -\frac{1}{16}X_4^2Y_4 + \frac{1}{48}X_2X_6Y_4.\end{aligned}$$

Using formulas for the Igusa–Clebsch invariants in terms of ψ_4 , ψ_6 , χ_{10} , and χ_{12} , we can express the Igusa–Clebsch invariants of a generic point on the Humbert surface. Using the techniques of [CowanFrengleyMartin], one may write the corresponding Mestre conic and simplify it, leading to the universal genus 2 curve with RM by $\mathbb{Q}(\sqrt{12})$, as in [CowanFrengleyMartin].

5.3.5. *A symmetric Hilbert modular variety, TODO.* AM: [A symmetric Hilbert modular variety should be added here.]

Edgar: [We decided that we would like to find the first example with trivial level that is not rational. JV says discriminant 101 might be the first example. See <https://www.lmfdb.org/ModularForm/GL2/TotallyReal/2.2.24.1/holomorphic/2.2.24.1-1.1-a> and <https://arxiv.org/pdf/1209.3527>. Did anyone else do this? Even if it is rational, we it would be a good example to do.]

-stability}

6. HECKE STABILITY

AM: [add a theorem saying algorithm blah]

In this section, we discuss the Hecke stability method, which gives another method to compute spaces of Hilbert modular forms. As our main application, we use it to compute spaces of parallel and partial weight one Hilbert modular forms, since these cannot be obtained through Jacquet–Langlands transfer (as they do not arise in cohomology).

6.1. **The method.** The basic idea behind Hecke stability is that subspaces of meromorphic Hilbert modular forms which are stable under the Hecke action should consist of holomorphic forms. For example, the order of a pole of a meromorphic classical modular form at a cusp will increase when we apply Hecke operators [Schaeffer]. This idea was first formalized by Schaeffer [Schaeffer] in the case of classical modular forms. In the Hilbert case, it was used as a heuristic by Van Hirtum [VanHirtum] (though the paper was withdrawn, with the example not claimed to be proven) to compute parallel weight one Hilbert modular forms and Moy–Specter [MoySpecter] to compute non-CM partial weight one Hilbert modular forms.

Suppose we want to compute the space $S_k(\Gamma_0(\mathfrak{N}), \chi)$ of holomorphic cusp forms of weight k , level $\mathfrak{N}$, and character χ . For $l \in \mathbb{Z}_{\geq 1}$, let $\underline{l} := (l, \dots, l) \in \mathbb{Z}_{\geq 1}^n$. Let ψ be a finite-order Hecke character with finite conductor dividing $\mathfrak{N}$ and compatible (in the sense of Eq. (4.1.1)) with the weight $\underline{l}$. Define the space

$$(6.1.1) \quad M_k^{\underline{l}, \psi}(\Gamma_0(\mathfrak{N}), \chi) := \{E_{\underline{l}, \psi}^{-1} f : f \in S_{k+\underline{l}}(\Gamma_0(\mathfrak{N}), \chi\psi)\}$$

Assuming that the constant term of $E_{\underline{l}, \psi}$ is nonzero at each component (and in practice we can always choose an $\underline{l}$ such that there exists an E for which this is true), we can define the action of the Hecke operator $T_{\mathfrak{p}}$ on $M_k^{\underline{l}, \psi}(\Gamma_0(\mathfrak{N}), \chi)$ by the usual formula on Fourier expansions (Eq. (3.4.1), Eq. (3.4.2)) for weight k and nebentypus χ . This arises from the considering the action on meromorphic differentials given by the usual geometric correspondences.

Observe then that the intersection

$$S_k(\Gamma_0(\mathfrak{N}), \chi) \cap M_k^{l,\psi}(\Gamma_0(\mathfrak{N}), \chi) \subseteq M_k^{l,\psi}(\Gamma_0(\mathfrak{N}), \chi)$$

contains $S_k(\Gamma_0(\mathfrak{N}), \chi)$. It is not necessarily the same as $S_k(\Gamma_0(\mathfrak{N}), \chi)$ because there may be an Eisenstein series $E' \in M_k(\Gamma_0(\mathfrak{N}), \chi)$ such that E and E' are nonzero on disjoint sets of cusps.

For a prime $\mathfrak{p}$, consider the following $T_{\mathfrak{p}}$ -stable subspace of $M_k^{l,\psi}(\Gamma_0(\mathfrak{N}), \chi)$:

$$(6.1.2) \quad M_k^{l,\psi}(\Gamma_0(\mathfrak{N}), \chi)^{\langle T_{\mathfrak{p}} \rangle} := \ker(T_{\mathfrak{p}}) \oplus \bigcap_{a \geq 0} T_{\mathfrak{p}}^a(M_k^{l,\psi}(\Gamma_0(\mathfrak{N}), \chi)).$$

The intersection in Eq. (6.1.2) stabilizes for $a \geq \dim S_{k+l}(\Gamma_0(\mathfrak{N}), \chi\psi)$, we can compute $M_k^{l,\psi}(\Gamma_0(\mathfrak{N}), \chi)^{\langle T_{\mathfrak{p}} \rangle}$ by repeatedly applying Hecke operators to the space of quotients.

In the case of modular forms, Schaeffer [Schaeffer] proves that

$$S_k(\Gamma_0(N), \chi) = M_k^{l,\psi}(\Gamma_0(N), \chi)^{\langle T_p \rangle}.$$

Instead of generalizing this result to Hilbert modular forms, we use the $T_{\mathfrak{p}}$ -stable space $M_k^{l,\psi}(\Gamma_0(\mathfrak{N}), \chi)^{\langle T_{\mathfrak{p}} \rangle}$ as a *candidate* for a subspace of $M_k(\Gamma_0(\mathfrak{N}), \chi)$, and verify that it is correct using the following lemma.

Lemma 6.1.3. *Let $f \in M_k^{l,\psi}(\Gamma_0(\mathfrak{N}), \chi)$ and recall that $f \cdot E_{l,\psi} \in S_{k+l}(\Gamma_0(\mathfrak{N}), \chi\psi)$. If there exists an $h \in S_{2k}(\Gamma_0(\mathfrak{N}), \chi^2)$ such that $f^2 \cdot E_{l,\psi}^2 \in S_{2(k+l)}(\Gamma_0(\mathfrak{N}), \chi^2\psi^2)$ is equal to $h \cdot E_{l,\psi}^2$, then $f \in S_k(\Gamma_0(\mathfrak{N}), \chi)$.*

Proof. We need to check that f is holomorphic and that f is a cusp form. It suffices to check that f^2 is a holomorphic cusp form. This is not as simple as seeing if $f^2 \in S_{2k}(\Gamma_0(\mathfrak{N}), \chi^2)$, since we cannot rule out the q -expansion f^2 agreeing with some element $h \in S_{2k}(\Gamma_0(\mathfrak{N}), \chi^2)$ up to the precision we are working with and disagreeing at some higher-order term. We also cannot rely on some version of the Sturm bound to conclude that f^2 is holomorphic; as remarked in [VanHurtum], the Sturm bound is too crude to be useful in practice. Instead, we ask if there exists an $h \in S_{2k}(\Gamma_0(\mathfrak{N}), \chi^2)$ such that

$$(6.1.4) \quad f^2 \cdot E_{l,\psi}^2 = h \cdot E_{l,\psi}^2.$$

If so, then $f^2 = h$ is a holomorphic cusp form. The point here is that $f^2 \cdot E_{l,\psi}^2 = (fE_{l,\psi})^2$ and $E_{l,\psi}^2$ are known to be Hilbert modular forms. As such, we can use a basis $\{h_i\}$ for $S_{2k}(\Gamma_0(\mathfrak{N}), \chi^2)$ given by Eq. (4.2.1) to provably check if there is an h satisfying Eq. (6.1.4) by seeing if there is a linear relation $\sum_i c_i (h_i E_{l,\psi}^2 - f^2 E_{l,\psi}^2) = 0$ in $S_{2(k+l)}(\Gamma_0(\mathfrak{N}), \chi^2\psi^2)$ (see the remark in Section 5). $\square$

Remark 6.1.5. In practice, it is enough to compute that $f^2 \in S_{2k}(\Gamma_0(\mathfrak{N}), \chi^2)$. However, we still need to ensure that the chosen precision is enough to compute a basis of $S_{2(k+l)}(\Gamma_0(\mathfrak{N}), \chi^2\psi^2)$.

In the current implementation, we always take $\psi = \chi^{-1}$. We take $l = 1$, unless $E_{1,\chi^{-1}}$ is cuspidal on one of the components, in which case we use Eisenstein series of higher weights.

6.2. A non-dihedral form of weight $(1, 1)$. Parallel weight one newforms have Galois representations which can be realized in $\mathrm{GL}_2(\mathbb{C})$, i.e. as Artin representations [carayol]. In particular, their Galois representations have finite image. The corresponding projective representations are classified by the following theorem.

pgl2-subst}

Theorem 6.2.1 (Klein). *The only finite subgroups of $\mathrm{PGL}_2(\mathbb{C})$ are C_n , D_{2n} (the dihedral group with $2n$ elements), A_4 , S_4 , and A_5 .*

Given a parallel weight one newform f , let $\rho_f: G_F \rightarrow \mathrm{GL}_2(\mathbb{C})$ be its Galois representation and $\mathbb{P}(\rho_f): G_F \rightarrow \mathrm{PGL}_2(\mathbb{C})$ the projectivization of ρ_f . We call f a G form if $\mathbb{P}(\rho_f)$ has image isomorphic to G . Because f is a cusp form, ρ_f is irreducible, so G will never be C_n . The two-dimensional irreducible representations of D_{2n} are all induced from one-dimensional characters on an index two subgroup. This implies that if f is a dihedral form, then its Galois representation is induced from a finite-order character of G_K for some quadratic extension K/F .

Empirically, the "majority" of parallel weight one forms are dihedral. Constructing non-dihedral forms in parallel weight one is both of intrinsic interest and also has applications in various works related to Stark's conjecture [darmon-pozzi-vonk]. Throughout this section, we follow techniques from [buzzard-lauder].

The following lemma makes it easy to guess what the projective image of a modular form is based on its coefficients.

Lemma 6.2.2 ([buzzard-lauder]). *For any matrix $M \in \mathrm{GL}_2(\mathbb{C})$, the order of the image $M' \in \mathrm{PGL}_2(\mathbb{C})$ is determined by the value of $\frac{(\mathrm{tr} M)^2}{\det M}$.*

order	1	2	3	4	5
$\frac{(\mathrm{tr} M)^2}{\det M}$	4	0	1	2	$\frac{3 \pm \sqrt{5}}{2}$

For a parallel weight 1 newform f and $\mathfrak{p}$ not dividing the level of f , we have $\det(1 - \rho_f(\mathrm{Frob}_{\mathfrak{p}})T) = 1 - a_{\mathfrak{p}}T + \chi(\mathfrak{p})T^2$, so Eq. (6.2.2) lets us compute the distribution of orders of projective images of Frobenii for any given newform. The Chebotarev density theorem tells us that the projective images of Frobenius elements in a G form should be equidistributed amongst the elements of G .

As we noted earlier, dihedral forms over F have Galois representations which are induced from one-dimensional characters of quadratic extensions K/F . As such, we can construct the space of dihedral forms of a given level and nebentypus as in Section 4.3, by computing the spaces of Hecke characters for K/F which could give rise to an induced representation with the correct determinant character and conductor.

Among the exotic (non-dihedral) forms, A_5 and S_4 are easy to identify. Because A_4 and S_4 have no order 5 elements, any form which is not dihedral and contains an order 5 element must be A_5 . Similarly, a form is S_4 if it is not dihedral and contains an order 4 element.

Proving that a form is A_4 is somewhat more annoying – even if we do not see elements of order 4 or order 5, it could simply be because convergence to the limiting distribution is very slow. We can rule out A_5 forms in practice if the coefficient field of f does not contain $\sqrt{5}$. To rule out S_4 , suppose that f is actually an S_4 form, and that $\mathbb{P}(\rho_f)$ factors through the finite extension L/F with $\mathrm{Gal}(L/F) \cong S_4$. There exists a quadratic extension K/F such that $\mathrm{Gal}(L/K) \cong A_4$. This means that for every $\mathfrak{p} \subset \mathbb{Z}_F$ inert in K/F (i.e. such that $\mathrm{Frob}_{\mathfrak{p}} \notin \mathrm{Gal}(L/K)$) the order of $\mathbb{P}(\rho_f)$ must be either 2 or 4. If we think that f is an A_4 form, we can rule the possibility of S_4 out by looking through the quadratic extensions of F with conductor dividing $\mathfrak{N}$ and checking that none of them have the aforementioned property.

Theorem 6.2.3. *There exists a Hilbert modular newform f over $F = \mathbb{Q}(\sqrt{5})$ of weight $k = (1, 1)$, level $\mathfrak{N} = (18\sqrt{5})$, and order 6 nebentypus of conductor $\mathfrak{N}$, such that the projective*

hm:hmf-a4}

Galois representation associated to f has image isomorphic to A_4 . Moreover, f is not the base change to F of a classical modular form.

Proof. The form f is produced using the Hecke stability method and verified via Eq. (6.1.3). The coefficients $a_p(f)$ and the orders of projective Frobenius elements $\mathbb{P}(\rho_f(\text{Frob}_p))$ are in Table 1. The coefficient field $E = \mathbb{Q}(\zeta_{12})$ of f does not contain a square root of 5, so this form cannot be an A_5 form by Eq. (6.2.2). For every quadratic extension K/F with conductor dividing the level of f , there is some prime of F with projective Frobenius order 3 that is inert in K . Therefore, this form is neither dihedral nor S_4 , meaning it can only be A_4 . Because the level of f is not of form $N\mathbb{Z}_F$ for $N \in \mathbb{Z}$, it cannot be obtained via base change. $\square$

Given an Artin representation, we can ask what number field it corresponds to. The projective representation factors through an extension L/F such that $\text{Gal}(L/F) \cong A_4$. Let $K'/\mathbb{Q}$ be the degree 8 extension $\mathbb{Q}[x]/(x^8 - 3x^7 + 4x^6 - 7x^5 + 6x^4 + 5x^3 + 5x^2 - 30x + 20)$ (8.0.1640250000.1). It contains $F/\mathbb{Q}$, and its splitting field $L/\mathbb{Q}$ has Galois group S_4 . The extension L/F appears to be the A_4 extension through which $\mathbb{P}(\rho_f)$ factors. In Eq. (6.2.4), we describe a criterion that gives evidence for this.

In what follows, for any extension of fields F_2/F_1 , if $\mathfrak{p}$ is a prime ideal of F_1 unramified in F_2/F_1 , we say that the residue field tuple associated to $\mathfrak{p}\mathbb{Z}_{F_2}$ is $(f_1, \dots, f_g)$, where $f_1 \geq f_2 \geq \dots \geq f_g$ are the residue field degrees of primes of F_2 above $\mathfrak{p}$. We write $(f^{(g)})$ for the tuple consisting of g entries all equal to f .

Lemma 6.2.4. *Let $\mathfrak{p} \subset \mathbb{Z}_F$ be a prime not dividing the discriminant of L/F . The residue field tuple associated to $\mathfrak{p}\mathbb{Z}_{K'}$ is one of $(1, 1, 1, 1)$, $(2, 2)$, or $(3, 1)$. These cases correspond to $\text{Frob}_{\mathfrak{p}} \in \text{Gal}(L/F)$ having order 1, 2, or 3.*

Proof. Because L/F is an A_4 -extension and L/K' is a $\mathbb{Z}/3\mathbb{Z}$ -extension, if $\mathfrak{q}$ is a prime of $\mathbb{Z}_{K'}$ unramified in L/K' with residue field degree $f_{\mathfrak{q}}$, then the residue field tuple of $\mathfrak{q}\mathbb{Z}_L$ is either $(f_{\mathfrak{q}}, f_{\mathfrak{q}}, f_{\mathfrak{q}})$ or $(3f_{\mathfrak{q}})$. In particular, if the residue field tuple of $\mathfrak{p}\mathbb{Z}_K$ were $(2, 1, 1)$, then the residue field tuple of $\mathfrak{p}\mathbb{Z}_L$ could not have all its entries be the same. This is a contradiction since L/F is Galois.

Suppose that the residue field tuple of $\mathfrak{p}\mathbb{Z}_L$ is $(3, 3, 3, 3)$. Then, writing $\{\mathfrak{P}_i\}_{i \in \{1, \dots, 4\}}$ for the primes above $\mathfrak{p}$, the decomposition groups of $\mathfrak{P}_i$ in $\text{Gal}(L/F) \cong A_4$ are (up to reordering) $\langle (123) \rangle$, $\langle (134) \rangle$, $\langle (124) \rangle$, and $\langle (234) \rangle$. In particular, if we restrict to the subgroup $\text{Gal}(L/K')$, which without loss of generality is $\langle (123) \rangle$, we see that the residue field degrees of the $\mathfrak{P}_i$ in the extension L/K' are 1, 1, and 1, and 3. Therefore, the residue field degree of $\mathfrak{p}\mathbb{Z}_{K'}$ must be $(3, 1)$ – the first of these two primes in $\mathbb{Z}_{K'}$ splits completely in L/K' and the latter is inert.

Therefore, if the residue field tuple of $\mathfrak{p}\mathbb{Z}_{K'}$ is $(1, 1, 1, 1)$ then the possible residue field tuples of $\mathfrak{p}\mathbb{Z}_L$ are $(1^{(12)})$ and $(3, 3, 3, 3)$, but as we have just ruled out the latter case it must be $(1^{(12)})$. If the residue field tuple of $\mathfrak{p}\mathbb{Z}_{K'}$ is $(2, 2)$ then the only possible residue field tuple of $\mathfrak{p}\mathbb{Z}_L$ is $(2^{(6)})$ (since A_4 does not have elements of order 6), so in this case $\text{Frob}_{\mathfrak{p}}$ has order 2 in $\text{Gal}(L/F)$. If the residue field tuple of $\mathfrak{p}\mathbb{Z}_{K'}$ is $(3, 1)$, then the only possible residue field tuple of $\mathfrak{p}\mathbb{Z}_L$ is $(3, 3, 3, 3)$, so in this case $\text{Frob}_{\mathfrak{p}}$ has order 3 in $\text{Gal}(L/F)$. $\square$

We can use Eq. (6.2.4) and Eq. (6.2.2) to check that the order of $\text{Frob}_{\mathfrak{p}}$ in A_4 agrees with the order of $\mathbb{P}(\rho_f(\text{Frob}_{\mathfrak{p}}))$ for as many primes $\mathfrak{p} \subset \mathbb{Z}_F$ as we like (we checked this for primes of norm up to 1000).

We expect that similar methods would allow us to produce S_4 and A_5 extensions as well.

$\text{Nm}(\mathfrak{p})$	Generator of $\mathfrak{p}$	$a_{\mathfrak{p}} = \text{tr } \rho_f(\text{Frob}_{\mathfrak{p}})$	Order of $\mathbb{P}(\rho_f)(\text{Frob}_{\mathfrak{p}})$
11	$(-3\sqrt{5} + 1)/2$	$-\zeta^2$	3
11	$(-3\sqrt{5} - 1)/2$	$\zeta^2 - 1$	3
19	$-2\sqrt{5} - 1$	$-\zeta^3 + \zeta$	3
19	$-2\sqrt{5} + 1$	$-\zeta$	3
29	$(\sqrt{5} + 11)/2$	$-\zeta^3 + \zeta$	3
29	$(-\sqrt{5} + 11)/2$	$-\zeta$	3
31	$(-5\sqrt{5} - 1)/2$	$\zeta^2 - 1$	3
31	$(-5\sqrt{5} + 1)/2$	$-\zeta^2$	3
41	$(\sqrt{5} + 13)/2$	$-\zeta^2$	3
41	$(-\sqrt{5} + 13)/2$	$\zeta^2 - 1$	3
49	7	0	2
59	$(7\sqrt{5} - 3)/2$	$-\zeta$	3
59	$(7\sqrt{5} + 3)/2$	$-\zeta^3 + \zeta$	3
61	$(7\sqrt{5} - 1)/2$	0	2
61	$(7\sqrt{5} + 1)/2$	0	2
71	$(\sqrt{5} - 17)/2$	$-\zeta^2 + 1$	3
71	$(\sqrt{5} + 17)/2$	ζ^2	3
79	$4\sqrt{5} + 1$	0	2
79	$4\sqrt{5} - 1$	0	2
89	$(\sqrt{5} - 19)/2$	$\zeta^3 - \zeta$	3
89	$(\sqrt{5} + 19)/2$	ζ	3
101	$(9\sqrt{5} + 1)/2$	$-\zeta^2 + 1$	3
101	$(9\sqrt{5} - 1)/2$	ζ^2	3
109	$(\sqrt{5} - 21)/2$	ζ	3
109	$(\sqrt{5} + 21)/2$	$\zeta^3 - \zeta$	3
131	$(\sqrt{5} - 23)/2$	$-\zeta^2$	3
131	$(\sqrt{5} + 23)/2$	$\zeta^2 - 1$	3
139	$\sqrt{5} - 12$	$\zeta^3 - \zeta$	3
139	$\sqrt{5} + 12$	ζ	3
149	$(11\sqrt{5} + 3)/2$	0	2
149	$(11\sqrt{5} - 3)/2$	0	2
151	$(11\sqrt{5} + 1)/2$	$\zeta^2 - 1$	3
151	$(11\sqrt{5} - 1)/2$	$-\zeta^2$	3
169	13	$2\zeta^3$	1
179	$6\sqrt{5} - 1$	$\zeta^3 - \zeta$	3
179	$6\sqrt{5} + 1$	ζ	3
181	$(\sqrt{5} - 27)/2$	$-\zeta^2 + 1$	3
181	$(\sqrt{5} + 27)/2$	ζ^2	3

TABLE 1. Traces and orders of $\mathbb{P}(\rho_f)(\text{Frob}_{\mathfrak{p}})$ for the candidate A_4 form f . Here, ζ denotes a twelfth root of unity.

{tab:1}

6.3. Non-CM partial weight one forms. A major step in extending the methods of the previous subsection to weight $(1, 3)$ forms is that computing dihedral forms is significantly more challenging. In parallel weight 1, the Hecke characters of interest are of finite order and are not too far from characters of $(\mathbb{Z}_F/\mathfrak{N})^\times$ for various $\mathfrak{N}$. Once the weight is not parallel, we need to treat Hecke characters with nontrivial infinity type.

The analogues of dihedral forms in this setting are CM forms. In [MoySpecter], Moy and Specter gave the first example of a non-CM partial weight 1 form – theirs was a weight $(1, 5)$ form with coefficients in a degree 4 extension of $F = \mathbb{Q}(\sqrt{5})$.

Theorem 6.3.1. *There exists a non-CM Hilbert modular newform f over $F = \mathbb{Q}(\sqrt{5})$ of weight $(1, 3)$, level $\mathfrak{N} = 18\mathbb{Z}_F$, quadratic Hecke nebentypus χ with ray conductor $3\mathbb{Z}_F \cdot \infty_1 \infty_2$, and with coefficients in $E := \mathbb{Q}[\alpha]/(\alpha^4 + 3\alpha^2 + 1) \simeq F[\alpha]/(\alpha^2 - (\sqrt{5} - 3)/2)$.*

This is the first known example of a non-CM weight $(1, 3)$ form. It has coefficients in a CM extension of F .

Lemma 6.3.2. *Any non-CM paritious weight Hilbert modular normalized newform over a totally real field F with nontrivial quadratic nebentypus has coefficients in a totally imaginary field containing F .*

Proof. Let E be the field of coefficients of F . By an extension of Theorem 5.5.3 in [diamond-shurman] to Hilbert modular forms (everything goes through the same way), under any embedding of the Hecke matrices into the complex numbers, $T_{\mathfrak{p}}^* = \chi(\mathfrak{p})T_{\mathfrak{p}}$. If f is a normalized newform, then this implies that under any embedding of E , $\overline{a_{\mathfrak{p}}} = \chi(\mathfrak{p})a_{\mathfrak{p}}$. Therefore, whenever $\chi(\mathfrak{p}) = -1$, $\overline{a_{\mathfrak{p}}} = -a_{\mathfrak{p}}$. Because f is non-CM, the coefficient $a_{\mathfrak{p}}$ will be nonzero for a density 1 subset of primes. For any prime $\mathfrak{p}$ with $\chi(\mathfrak{p}) = -1$ and $a_{\mathfrak{p}} \neq 0$, $a_{\mathfrak{p}}$ must be purely imaginary, and the result follows. $\square$

Therefore, Eq. (6.3.1) gives in some sense a “simplest possible” partial weight 1 form.

7. COMPUTING PLURIGENERA

Edgar: [A proposal: What about we leave this for a later paper? At the moment, this section is quite narrow with the goal of making some examples practical, but we perhaps should write it generically in a new paper, and then specialize for the example.]JK: [OK. To make this section more general, the wishlist could be:

- (1) Review the construction of the toroidal compactification for general F and review the explicit conditions to extend pluricanonical forms. Is this done explicitly somewhere or do we need to specialize Mumford’s general construction?
- (2) Have some tools to compute the “jet functionals” at elliptic points. Some exact arithmetic to show that they’re zero and/or interval arithmetic to show that they’re nonzero, using explicit bounds on Fourier coefficients?
- (3) Have some tools to compute Fourier expansions at other cusps.
- (4) Review “well-known” algebraic geometry stuff about the classification of surfaces and threefolds.

There’s room for a new paper for sure. (Some stuff about jet functionals is in comments.)]

We conclude with an application of the preceding algorithms to the computation of plurigenera of connected Hilbert modular surfaces of type $\overline{Y}_0(\mathfrak{N})$. This section treats the case

in which sections of the pluricanonical bundle can be detected from Hilbert modular forms using a finite collection of exact linear conditions on their Fourier expansions. Throughout this section we impose the following hypotheses.

- {quadratic} (H1) F is real quadratic.
- {number one} (H2) $h^+(F) = 1$.
- {free level} (H3) $\mathfrak{N} \subseteq \mathbb{Z}_F$ is squarefree.

Since $h^+(F) = 1$, the Hilbert modular surface $\overline{Y}_0(\mathfrak{N})$ has only one component in the sense of (2.2.5); thus the notation $\Gamma_0(\mathfrak{N})$ below means $\Gamma_0(\mathfrak{N})_{\mathbb{Z}_F}$. We let $X_0(\mathfrak{N}) \rightarrow \overline{Y}_0(\mathfrak{N})$ be a toroidal compactification, as constructed in [Geer88]. In particular $X_0(\mathfrak{N})$ is a smooth and proper algebraic surface that is birational to $Y_0(\mathfrak{N})$. For $m \geq 1$, we denote by

$$(7.0.1) \quad P_m(X_0(\mathfrak{N})) := \dim_{\mathbb{C}} H^0(X_0(\mathfrak{N}), K_{X_0(\mathfrak{N})}^{\otimes m})$$

its m -th plurigenus. Since plurigenera are birational invariants (see e.g. [hartshorneAlgebraicGeometry1]), this number is independent of our particular choice of smooth compactification of $\overline{Y}_0(\mathfrak{N})$. We would like to compute $P_m(X_0(\mathfrak{N}))$ exactly under (H1)–(H3).

To this end, we explicitly relate sections of pluricanonical bundles to Hilbert modular forms, following [Geer88]. Let $m \geq 1$, and let $f \in M_{2m}(\Gamma_0(\mathfrak{N}))$ be a modular form of parallel weight $2m$. Then

$$\omega_f := f(dz_1 \wedge dz_2)^{\otimes m}$$

is a section of the m -th power of the canonical bundle on the smooth part of $\overline{Y}_0(\mathfrak{N})$ [Geer88]. To decide whether ω_f extends to a section of $K_{X_0(\mathfrak{N})}^{\otimes m}$, one then needs to study whether ω_f extends holomorphically to neighborhoods of the resolutions of cusps and elliptic points in $X_0(\mathfrak{N})$. In addition to (H1)–(H3), we will add the assumption that no obstruction to extend these pluricanonical forms occurs at elliptic points. This additional assumption is in fact of a combinatorial nature: see ?? below.

{elliptic-pts}

7.1. The assumption on elliptic points. Let Γ denote the image of $\Gamma_0(\mathfrak{N})$ in $\mathrm{PGL}_2^+(F)$. A point $z \in \mathcal{H}$ is called an *elliptic point* for $\Gamma_0(\mathfrak{N})$ if the stabilizer of z in Γ has order at least 2. This notion only depends on the image of z in $Y_0(\mathfrak{N})$, and there are only finitely many elliptic points in $Y_0(\mathfrak{N})$ [Geer88].

Following the convention of [Geer88], we say that an elliptic point is of type $(n; 1, q)$ if, in suitable local coordinates (w_1, w_2) , its faithful stabilizer is isomorphic to μ_n acting by

$$(w_1, w_2) \mapsto (\zeta w_1, \zeta^q w_2), \quad \zeta^n = 1.$$

The number of elliptic points of each type can be computed explicitly given F and $\mathfrak{N}$ as explained in [HMSDB].

Given an elliptic point type $(n; 1, q)$, we introduce the following combinatorial data which appears in [Geer88]. Let

$$N_{q,n} = \mathbb{Z}^2 + \mathbb{Z} \cdot \frac{1}{n}(1, q), \quad M_{q,n} = \{(u_1, u_2) \in \mathbb{Z}^2 : u_1 + qu_2 \equiv 0 \pmod{n}\}.$$

The boundary of the convex hull of $N_{q,n} \cap \mathbb{R}_{\geq 0}^2$ facing the origin is finite polygonal line starting in $(1, 0)$ and ending in $(0, 1)$; let $\Sigma_{q,n}$ be the finite set of points of $N_{q,n}$ lying on this boundary, $(0, 1)$ excluded. Finally, let

$$\mathcal{B}_m(q, n) := \{\lambda \in \mathbb{Z}_{\geq 0}^2 : \lambda + m(1, 1) \in M_{q,n} \text{ and } \langle \lambda + m(1, 1), v \rangle < m \text{ for some } v \in \Sigma_{q,n}\}.$$

In complement of our above hypotheses (H1)–(H3), we add the following assumption, which depends on the order $m \geq 1$ of the plurigenus we wish to compute:

(H4) For any elliptic point type $(n; 1, q)$ occurring on $Y_0(\mathfrak{N})$, we have $B_m(q, n) = \emptyset$.

In particular, ?? holds if all elliptic points on $Y_0(\mathfrak{N})$ have $n = 2$, by [Geer88]. Depending on the value of m , some other types of elliptic points can be allowed. For instance, the assumption always holds for $m = 1$ [Geer88]

Under (H1), (H2) and ??, the relation between modular forms and sections of pluricanonical bundles on $X_0(\mathfrak{N})$ only involves the resolution of cusps in $X_0(\mathfrak{N})$. We now make this statement precise. By [Geer88], the preimage of the cusp ∞ of $\bar{Y}_0(\mathfrak{N})$ in $X_0(\mathfrak{N})$ is a divisor, denoted by Δ_∞ , which is a finite union of lines arranged in a circle. The same holds for any cusp c of $\bar{Y}_0(\mathfrak{N})$, as one can see after conjugating by a suitable element in $\mathrm{GL}_2^+(F)$; let Δ_c be the corresponding divisor. The cuspidal divisor in $X_0(\mathfrak{N})$ is

$$\Delta = \sum_{c \in \Gamma_0(\mathfrak{N}) \backslash \mathbb{P}^1(F)} \Delta_c.$$

As above, let $K_{X_0(\mathfrak{N})}$ denote the canonical sheaf of $X_0(\mathfrak{N})$. Write $K_{X_0(\mathfrak{N})}(\Delta) = K_{X_0(\mathfrak{N})} \otimes \mathcal{L}(\Delta)$ for the sheaf of differential forms having a pole of order at most 1 along Δ , using notation from [hartshorneAlgebraicGeometry1977].

Proposition 7.1.1. *Assume (H1), (H2), ?. Then we have an isomorphism*

$$\begin{array}{ccc} M_{2m}(\Gamma_0(\mathfrak{N})) & \longrightarrow & H^0(X_0(\mathfrak{N}), K_{X_0(\mathfrak{N})}(\Delta)^{\otimes m}) \\ f & \longmapsto & \omega_f. \end{array}$$

Proof. Given ??, the computation on [Geer88] shows that for any $f \in M_{2m}(\Gamma_0(\mathfrak{N}))$, the form ω_f extends holomorphically to a neighborhood of the resolution of any elliptic point. The result then follows from [Geer88] applied at each cusp. $\square$

7.2. Linear conditions at infinity. When ?? applies, we can compute $P_m(X_0(\mathfrak{N}))$ by finding out which elements $f \in M_{2m}(\Gamma_0(\mathfrak{N}))$ induce sections of $K_{X_0(\mathfrak{N})}^{\otimes m}$, and not only of $K_{X_0(\mathfrak{N})}(\Delta)^{\otimes m}$. Write $\Delta' = \Delta - \Delta_\infty$. Again following [Geer88], we can express whether f induces a section of $K_{X_0(\mathfrak{N})}(\Delta')^{\otimes m}$, in other words whether ω_f extends holomorphically to a neighborhood of Δ_∞ in $X_0(\mathfrak{N})$, using exact linear conditions on the Fourier expansion of f . We will next treat the other cusps with the help of Atkin–Lehner involutions and (H3). We first introduce notation related to the resolution of the cusp at infinity in $X_0(\mathfrak{N})$, following [Geer88].

Embed $\mathbb{Z}_F$ as a lattice in $\mathbb{R}^2$ through the two real embeddings of F . Let ε be a generator of $\mathbb{Z}_{F, >0}^\times$ chosen with $\varepsilon_1 > 1$. The boundary of the convex hull of $\mathbb{Z}_F \cap \mathbb{R}_{>0}^2$ facing the origin is an infinite polygonal line. We number the points of $\mathbb{Z}_F \cap \mathbb{R}_{>0}^2$ lying on this boundary as

$$\dots, A_{-1}, A_0, A_1, \dots \in \mathbb{Z}_F \cap \mathbb{R}_{>0}^2.$$

so that the abscissas of A_j for $j \in \mathbb{Z}$ form an increasing sequence. There is a positive integer r such that

$$(7.2.1) \quad \varepsilon A_j = A_{j+r} \quad \text{for all } j \in \mathbb{Z}.$$

The finite list $A_0, \dots, A_{r-1}$ is computed from the continued fraction attached to the real quadratic lattice $\mathbb{Z}_F$. We can compute it as presented in [HMSDB]. From this data, [Geer88] provides an explicit description of the divisor Δ_∞ .

For $m \geq 1$, define the finite obstruction set

$$(7.2.2) \quad \mathcal{L}_m := \left\{ \nu \in (\mathbb{Z}_F^\#)_{>0} : \text{Tr}(\nu A_j) < m \text{ for some } 0 \leq j \leq r-1 \right\}.$$

This set is finite: for fixed A_j , both real embeddings of A_j are positive, so the inequality $\text{Tr}(\nu A_j) < m$ puts ν in a bounded subset of $\mathbb{R}^2$, which contains only finitely many elements of the lattice $\mathbb{Z}_F^\#$. Let

$$(7.2.3) \quad U_m(\infty) := \{f \in M_{2m}(\Gamma_0(\mathfrak{N})) : a_\nu(f) = 0 \text{ for all } \nu \in \mathcal{L}_m\}.$$

The criterion we are looking for is then the following [Geer88].

Proposition 7.2.4. *Assume (H1), (H2), ??, and let $f \in M_{2m}(\Gamma_0(\mathfrak{N}))$. Then ω_f induces a section of $K_{X_0(\mathfrak{N})}(\Delta')^{\otimes m}$ if and only if $f \in U_m(\infty)$.*

7.3. Linear conditions at other cusps via Atkin–Lehner involutions. We will now use hypothesis (H3) to relate any cusp of $\bar{Y}_0(\mathfrak{N})$ to the cusp at infinity via Atkin–Lehner involutions. Because the effect of such involutions on Fourier expansions can be described effectively, this will lead us to a computational criterion for a Hilbert modular form to induce a section of $K_{X_0(\mathfrak{N})}^{\otimes m}$.

Since $h^+(F) = 1$, every divisor Ω of $\mathfrak{N}$ has a totally positive generator. As in [GreenbergVoight], choose a standard Atkin–Lehner matrix $W_\Omega = W_\Omega^{(\mathfrak{N})} \in \text{GL}_2^+(F)$ normalizing $\Gamma_0(\mathfrak{N})$, whose determinant generates Ω , and such that for every $f \in M_{2m}(\Gamma_0(\mathfrak{N}))$, we have $f|_{2m}W_\Omega^2 = f$. The matrix W_Ω is unique up to right multiplication by $\Gamma_0(\mathfrak{N})$ and scalars, so the involution of $Y_0(\mathfrak{N})$ induced by W_Ω and the action $f \mapsto f|_{2m}W_\Omega$ on modular forms are well-defined. We have the relation

$$(7.3.1) \quad W_\Omega^*(\omega_f) = \omega_{f|_{2m}W_\Omega}.$$

Proposition 7.3.2. *Assume (H1), (H2), ??. For $f \in M_{2m}(\Gamma_0(\mathfrak{N}))$, the form ω_f extends holomorphically over the resolution of $W_\Omega\infty$ in $X_0(\mathfrak{N})$ if and only if $f|_{2m}W_\Omega \in U_m(\infty)$.*

Proof. Indeed, W_Ω identifies a neighborhood of ∞ with a neighborhood of $W_\Omega\infty$, and we can choose the cusp resolution at $W_\Omega\infty$ to be the translate by W_Ω of the resolution at ∞ . Thus ω_f extends holomorphically to the resolution of $W_\Omega\infty$ if and only if $W_\Omega^*(\omega_f)$ does so at ∞ . We conclude by Eq. (7.2.3) and (?). $\square$

Crucially, our hypotheses ensure that any cusp is of the form $W_\Omega\infty$.

Proposition 7.3.3. *Suppose $\mathfrak{N}$ is squarefree and $h^+(F) = 1$. Then the Atkin–Lehner group $\mathcal{W}_\mathfrak{N} = \{W_\Omega\}_{\Omega|\mathfrak{N}}$ acts transitively on the cusps of $X_0(\mathfrak{N})$.*

Proof. Let $(a : c) \in \mathbb{P}^1(F)$ represent a cusp of $X_0(\mathfrak{N})$. Since $h(F) = 1$, we may assume $a, c \in \mathbb{Z}_F$ and $a\mathbb{Z}_F + c\mathbb{Z}_F = \mathbb{Z}_F$, hence there exist $\lambda, \mu \in \mathbb{Z}_F$ with $\lambda a + \mu c = 1$. Set $\mathfrak{M} = \mathfrak{N} + c\mathbb{Z}_F$ and $\Omega = \mathfrak{N}\mathfrak{M}^{-1}$; note that $c \in \mathfrak{M}$.

We first adjust (λ, μ) so that $\lambda \in \Omega$. For each prime $\mathfrak{p} \mid \Omega$, as $\mathfrak{N}$ is squarefree, we have $\mathfrak{p} \nmid \mathfrak{M}$, hence $c \notin \mathfrak{p}$, and c is invertible in $\mathbb{Z}_F/\mathfrak{p}$. By CRT we may choose $s \in \mathbb{Z}_F$ with $\lambda + sc \equiv 0 \pmod{\mathfrak{p}}$ for every $\mathfrak{p} \mid \Omega$. Replacing

$$(\lambda, \mu) \mapsto (\lambda', \mu') := (\lambda + sc, \mu - sa)$$

preserves the relation $\lambda'a + \mu'c = 1$, keeps $\mu' \in \mathbb{Z}_F$, and (as $\mathfrak{N}$ is squarefree) achieves $\lambda' \in \Omega$.

Since $h^+(F) = 1$, there exists $q \in F_{>0}^\times$ with $q\mathbb{Z}_F = \mathfrak{Q}$. Set

$$W_{\mathfrak{Q}} := \begin{pmatrix} aq & -\mu' \\ cq & \lambda' \end{pmatrix}.$$

Then $aq \in \mathfrak{Q}$ and $\lambda' \in \mathfrak{Q}$; $-\mu' \in \mathbb{Z}_F$; and $cq \in \mathfrak{M} \cdot \mathfrak{N}\mathfrak{M}^{-1} = \mathfrak{N}$. Moreover

$$\det W_{\mathfrak{Q}} = q(a\lambda' + \mu'c) = q \in F_{>0}^\times, \quad (\det W_{\mathfrak{Q}}) = \mathfrak{Q}.$$

So $W_{\mathfrak{Q}}$ is an Atkin–Lehner matrix of type $\mathfrak{Q}$, and

$$W_{\mathfrak{Q}} \cdot (1 : 0) = (aq : cq) = (a : c).$$

Hence $W_{\mathfrak{Q}}\infty = (a : c)$, proving the lemma. $\square$

Remark 7.3.4. Above, we only used that $h(F) = 1$ and that $\mathfrak{Q}$ is narrowly principal for all $\mathfrak{Q} \mid \mathfrak{N}$, a weaker condition than $h^+(F) = 1$. As a consequence, we could in fact compute plurigenera of connected components of $X_0(\mathfrak{N})$ under this weaker version of (H2).

Corollary 7.3.5. *Assume (H1)–??. For $f \in M_{2m}(\Gamma_0(\mathfrak{N}))$, the form ω_f induces a section of $K_{X_0(\mathfrak{N})}^{\otimes m}$ if and only if $f|_{2m}W_{\mathfrak{Q}} \in U_m(\infty)$ for all $\mathfrak{Q} \mid \mathfrak{N}$.*

Proof. By Eq. (7.3.2), under (H2) and (H3), the cusps of $\Gamma_0(\mathfrak{N})$ are represented by $W_{\mathfrak{Q}}\infty$ for $\mathfrak{Q} \mid \mathfrak{N}$. Apply Eq. (7.3.1) to conclude. $\square$

In particular, any Hilbert modular form f satisfying the conditions of Eq. (7.3.4) is a cusp form. Let

$$V_m := \{f \in S_{2m}(\Gamma_0(\mathfrak{N})) : f|_{2m}W_{\mathfrak{Q}} \in U_m(\infty) \text{ for all } \mathfrak{Q} \mid \mathfrak{N}\}.$$

Under our hypotheses, Eq. (7.3.4) states that $V_m \simeq H^0(X_0(\mathfrak{N}), K_{X_0(\mathfrak{N})}^{\otimes m})$ under the map $f \mapsto \omega_f$. In particular $P_m(X_0(\mathfrak{N})) = \dim_{\mathbb{C}} V_m$ for any $m \geq 1$.

In order to explicitly use Equation (7.3.4) in computations, it is useful to diagonalize the Atkin–Lehner action. For a character $\vartheta : \mathcal{W}_{\mathfrak{N}} \rightarrow \{\pm 1\}$, let

$$(7.3.6) \quad S_{2m}^{\vartheta}(\Gamma_0(\mathfrak{N})) := \{f \in S_{2m}(\Gamma_0(\mathfrak{N})) : f|_{2m}W = \vartheta(W)f \text{ for all } W \in \mathcal{W}_{\mathfrak{N}}\}.$$

Since the Atkin–Lehner involutions commute and have eigenvalues ± 1 , we have a direct sum decomposition

$$(7.3.7) \quad S_{2m}(\Gamma_0(\mathfrak{N})) = \bigoplus_{\vartheta : \mathcal{W}_{\mathfrak{N}} \rightarrow \{\pm 1\}} S_{2m}^{\vartheta}(\Gamma_0(\mathfrak{N})).$$

Proposition 7.3.8. *With notation as above,*

$$V_m = \bigoplus_{\vartheta : \mathcal{W}_{\mathfrak{N}} \rightarrow \{\pm 1\}} (S_{2m}^{\vartheta}(\Gamma_0(\mathfrak{N})) \cap U_m(\infty)).$$

Proof. Let $f = \sum_{\vartheta} f_{\vartheta}$ be the decomposition of f according to (7.4.2). If each f_{ϑ} belongs to $U_m(\infty)$, then for every $W \in \mathcal{W}_{\mathfrak{N}}$ we have $f|_{2m}W = \sum_{\vartheta} \vartheta(W)f_{\vartheta} \in U_m(\infty)$. Thus $f \in V_m$.

Conversely, suppose $f \in V_m$. Then $f|_{2m}W \in U_m(\infty)$ for every $W \in \mathcal{W}_{\mathfrak{N}}$. The projection of f onto the ϑ -eigenspace is

$$e_{\vartheta}(f) = \frac{1}{\#\mathcal{W}_{\mathfrak{N}}} \sum_{W \in \mathcal{W}_{\mathfrak{N}}} \vartheta(W) f|_{2m}W.$$

Hence $f_{\vartheta} = e_{\vartheta}(f) \in U_m(\infty)$ for every ϑ . $\square$

g:plurigenus}

Algorithm 7.3.9: $\text{Pl}(m, \mathfrak{N})$: the m -th plurigenus of $X_0(\mathfrak{N})$

Data: A real quadratic field F satisfying $h^+(F) = 1$, a squarefree ideal $\mathfrak{N} \subseteq \mathbb{Z}_F$, and an integer $m \geq 1$ such that ?? holds.

Result: $P_m(X_0(\mathfrak{N}))$.

Compute the cusp fan period $A_0, \dots, A_{r-1}$ for the cusp ∞ using [HMSDB];

Compute the finite set $\mathcal{L}_m$ of (7.2.2);

Compute a q -expansion basis $\mathcal{F} = \{f_1, \dots, f_d\}$ of $S_{2m}(\Gamma_0(\mathfrak{N}))$ to precision $\max_{\nu \in \mathcal{L}_m} \text{Nm}(\nu)$ using Equation (4.2.1).

For every $\mathfrak{Q} \mid \mathfrak{N}$, compute the action of $W_{\mathfrak{Q}}$ on $\mathcal{F}$.

$P_m := 0$;

for each character $\vartheta : \mathcal{W}_{\mathfrak{N}} \rightarrow \{\pm 1\}$ **do**

 Compute a basis $g_1, \dots, g_t$ of $S_{2m}^{\vartheta}(\Gamma_0(\mathfrak{N}))$;

$C_{\vartheta} := (a_{\nu}(g_j))_{\nu \in \mathcal{L}_m, 1 \leq j \leq t}$;

$P_m := P_m + \dim \ker C_{\vartheta}$;

return P_m ;

algorithm}

Theorem 7.3.10. *Given F , an ideal $\mathfrak{N} \subseteq \mathbb{Z}_F$, and an integer $m \geq 1$ such that (H1)–?? hold, Algorithm 7.4.6 returns $P_m(X_0(\mathfrak{N}))$.*

JK: [We may want to state this theorem differently. What we really do is compute a basis of the space of pluricanonical forms rather than just its dimension, and we use this more refined version in the examples to follow. So maybe the focus of this section should be on computing pluricanonical forms rather than computing plurigenera?]

Proof. First we must explain how the action of $W_{\mathfrak{Q}}$ on $S_{2m}(\Gamma_0(\mathfrak{N}))$ is computed in Algorithm 7.4.6. In practice, the action of Atkin–Lehner involutions is computed at the same time as the basis of Hilbert modular forms. One can write down the Atkin–Lehner action on the same exact automorphic or cohomological model used to compute the cusp space, by adding the corresponding normalizer double-coset operators, and then transport it to the chosen basis of $S_{2m}(\Gamma_0(\mathfrak{N}))$. In particular, no q -expansions at non-infinity cusps are used in the algorithm. *Edgar:* [The old draft proposed using Horawa, *Motivic action on coherent cohomology of Hilbert modular varieties*, Theorems 6.25 and 6.26, on the new subspace and an Asai-style formula for oldforms. Decide whether to state this implementation route here.]

The matrix C_{ϑ} records the conditions for membership in $U_m(\infty)$ on the Atkin–Lehner eigenspace S_{2m}^{ϑ} . Therefore, by Eq. (7.4.3) the algorithm returns $\dim V_m$, which is exactly $P_m(X_0(\mathfrak{N}))$ by Eq. (7.3.4). $\square$

Remark 7.3.11. The hypotheses made at the start of the section isolate the case where pluricanonical forms can be detected from the q -expansion at ∞ and its Atkin–Lehner translates. Several extensions are natural, but each requires additional work beyond the proof above.

- (H1) In higher-dimensional Hilbert modular varieties, the same philosophy should apply to forms of parallel weight $2m$ and the tensor power of the top differential form. The cusp condition, however, is no longer governed by a continued fraction polygon: one

must choose and compute regular admissible toroidal fans in higher-dimensional cones, and quotient singularities at elliptic points become correspondingly more complicated.

- (H2) If $h^+(F) > 1$, then $\overline{Y}_0(\mathfrak{N})$ has several components, and Hilbert modular forms are tuples indexed by narrow ideal classes as in (2.2.5). One must compute cusp representatives, polyhedral decompositions, Fourier expansions, and Atkin–Lehner operators component by component, including the possibility that an Atkin–Lehner operator moves between components. See also Eq. (7.3.3).
- (H3) For nonsquarefree levels, Atkin–Lehner involutions no longer act transitively on the cusps in general. The algorithm would need to compute q -expansion directly at different cusps. This requires an exact algorithm for the slash transform by a scaling matrix for the cusp, together with the translation lattice, unit action, coefficient field, and precision bookkeeping for the conjugate congruence group.

7.4. Examples. We conclude with two examples in which computing pluricanonical forms brings a finer understanding of the geometry of Hilbert modular varieties.

JK: [I still don't understand everything about the examples (but maybe that's just me).

- I don't understand the proof of Eq. (7.5.1). What does this determinant mean? The code turns Q into a 3×3 symmetric matrix. So we want to compute the points in $\mathbb{P}^1$ where the intersection of the two quadrics Q and f_2^2 becomes singular or something? What are we applying adjunction to?
- We are missing references on elliptic surfaces without sections to define e, χ , and the Kodaira types. The reference I know is the book by Schütt and Shioda but they assume a section is present.
- Eq. (7.5.3): the code only computes the pole order of j , and that's not sufficient. We need to reconstruct Weierstrass equations locally and run Tate's algorithm.
- §5.3.2: is the Knöller type relevant? If yes, we need to define it. Where does the Atkin–Lehner action on Fourier coefficients come from? Is it related to the article by Horawa above? All the stuff we explained above was only for surfaces, so it's hard to make sense of this example.

]

7.4.1. *The Hilbert surface $X_0(\mathfrak{p}_{31})$ over $\mathbb{Q}(\sqrt{5})$.* Let $F = \mathbb{Q}(\sqrt{5})$, and let $\mathfrak{p} = \mathfrak{p}_{31}$ be a prime of norm 31. The surface $X_0(\mathfrak{p})$ has geometric genus $p_g = \dim S_2(\mathfrak{p}) = 1$, raising the question of whether this is a K3-surface. Moreover, applying the methods of [HMSDB], one can compute its Hodge diamond, obtaining the same Hodge numbers as a K3-surface.

Then assumptions (H1)–(H3) are satisfied. The elliptic points are of types $(2; 1, 1)$, $(3; 1, 1)$, $(3; 1, 2)$, $(5; 1, 2)$, $(5; 1, 3)$. For these types, ?? holds for $m = 2$. We apply Algorithm 7.4.6 to compute that $P_2(X_0(\mathfrak{p})) = 2$. In particular, the Kodaira dimension of $X_0(\mathfrak{p})$ is $\kappa(X_0(\mathfrak{p})) = 1$, resolving its classification question that remained open in [HMSDB].

In fact, we can find out that $X_0(\mathfrak{p})$ is an elliptic surface: JK: [need a reference for various properties of elliptic surfaces without sections. Kodaira?]

Proposition 7.4.1. *The line bundle $K_{X_0(\mathfrak{p})}^{\otimes 2}$ induces an elliptic fibration $\varphi_2 : X_0(\mathfrak{p}) \rightarrow \mathbb{P}^1$ having a unique multiple fiber of multiplicity 2; the reduced fiber has arithmetic genus 1.*

Proof. We compute that $\dim M_2(\mathfrak{p}) = 3$ and $\dim M_4(\mathfrak{p}) = 6$. We check that there exist no quadratic relations between weight 2 forms; since $6 = \binom{3+1}{2}$, we have $\text{Sym}^2 M_2(\mathfrak{p}) = M_4(\mathfrak{p})$.

Let $e_{2,1}, e_{2,2}, f_2$ be a basis for $M_2(\mathfrak{p})$, with f_2 a cusp form. Computing the global sections of the bicanonical $H^0(X_0(\mathfrak{p}), K_{X_0(\mathfrak{p})}^{\otimes 2}) \subseteq M_4(\mathfrak{p})$, we find that it is spanned by f_2^2 and a cusp form $f_4 = Q(e_{2,1}, e_{2,2}, f_2)$, where Q is a quadric in $\mathbb{P}^3$. JK: [Shouldn't it be $\mathbb{P}^2$?]

JK: [I don't understand this paragraph (or at least not yet).] One computes that $\det(sQ + tf_2^2) = \ell \cdot s^2$, with ℓ vanishing along a conic of rank 2. It follows that the bicanonical map $\varphi_2 : X_0(\mathfrak{p}) \rightarrow \mathbb{P}^1$ given by $(f_2^2 : f_4)$ has a unique multiple fiber of multiplicity 2 given by the vanishing locus of f_2 . By adjunction, one computes that it has arithmetic genus 1. $\square$

Corollary 7.4.2. *The canonical ring $\bigoplus_{m \geq 0} H^0(X_0(\mathfrak{p}), K_{X_0(\mathfrak{p})}^{\otimes m})$ of $X_0(\mathfrak{p})$ is $\mathbb{C}[f_2, f_4]$.*

Proof. Since $X_0(\mathfrak{p})$ is an elliptic surface over $\mathbb{P}^1$ with Euler characteristic $\chi(X_0(\mathfrak{p})) = 2$ and a single multiple fiber with multiplicity 2, we have $P_n = 1 + \lfloor \frac{n}{2} \rfloor$, JK: [need a reference on Hilbert series of elliptic surfaces without sections] same as the Hilbert function of $\mathbb{C}[f_2, f_4]$. Since $\kappa(X_0(\mathfrak{p})) = 1$, its canonical ring $R(K_{X_0(\mathfrak{p})})$ is 2-dimensional. Therefore, the two homogeneous generators $f_2 \in H^0(K)$ and $f_4 \in H^0(K^{\otimes 2})$ are algebraically independent. $\square$

We further find the precise reduction type of the singular fibers. JK: [Need to check this. The code doesn't apply Tate's algorithm and only computes the polder order of j , that's not sufficient to tell the type of singular fiber in general. I think we need to reconstruct a local Weierstrass equation of the Jacobian of the genus 1 curve around each singular fiber to find out what the type is. Knowing j is enough for this so that should be possible.]

op:fibers}

Proposition 7.4.3. *The singular fibers of $\varphi_2 : X_0(\mathfrak{p}) \rightarrow \mathbb{P}^1$ are*

$$I_6 + 2I_4 + I_2 + 8I_1 \quad (\text{twelve fibers}),$$

together with the multiple fiber $2I_0$ of Proposition 7.5.1. In particular $\sum_v e(F_v) = 6 + 2 \cdot 4 + 2 + 8 \cdot 1 = 24 = e(X_0(\mathfrak{p}))$ and $\deg \Delta_{\text{ell}} = 24 = 12\chi$, and there are no additive fibers.

Proof. Fix the base coordinate $t = f_4/f_2^2$, so that the fiber X_t is the preimage under the degree-4 map $X_0(\mathfrak{p}) \rightarrow \mathbb{P}^2 = \mathbb{P}(M_2(\mathfrak{p}))$ of the conic $\{f_4 - tf_2^2 = 0\}$. We compute the map $j : \mathbb{P}^1 \rightarrow \mathbb{P}^1$, $t \mapsto j(X_t)$ as follows.

Using Algorithm 5.2.2, we compute a model for $\bar{Y} = \bar{Y}_0(\mathfrak{p})$ having 13 generators - 3 generators in weight 2, 6 in weight 6 and 4 in weight 10. For every t , adjoining the equation $f_4 - tf_2^2$ cuts the one-dimensional fiber $\bar{Y}_t$, which agrees with X_t away from finitely many points over the singularities of $X = X_0(\mathfrak{p})$.

We compute that $\bar{Y}_t$ is birational to a double cover of the conic $Q(e_{2,1}, e_{2,2}, f_2) - tf_2^2 = 0$. The conic has a rational point, and parameterizing, we obtain a hyperelliptic model $y^2 = h(s)$ with $\deg h = 4$. Its Jacobian is an elliptic curve with j -invariant $j(X_t) \in \mathbb{Q}$, which we obtain without having to compute a rational point on X_t . Sampling and interpolating allows us to reconstruct the degree 24 rational function $j(t) \in \mathbb{Q}(t)$ from 49 linear relations.

Having obtained $j(X_t)$ as a rational fraction, the singular fibers of φ_2 can be obtained by running Tate's algorithm. The poles of $j = j_N/j_D$ are the multiplicative fibers, of type I_n with $n = \text{ord}_t j_D(t)$ of the pole; j_D factors with pole multiplicities $e = \{6^1, 4^2, 2^1, 1^8\}$, giving the stated list. As $\sum e_v = 24 = e(X_0(\mathfrak{p}))$ is already exhausted by these fibers, no additive fibers can occur (each would contribute $e \geq 2$, and an I_n^* at a simple pole would contribute $e \geq 7$). $\square$

7.4.2. *The Kodaira dimension of a Hilbert threefold.* As in Eq. (5.3.1), take $F = \mathbb{Q}(\zeta_7 + \zeta_7^{-1})$, which has narrow class number 1. Consider the level $\mathfrak{N} = (2)$, inert with $\text{Nm}(\mathfrak{N}) = 8$, and the group $\Gamma_0(\mathfrak{N})$ of GL_2^+ -type in parallel weight 4. Then $\dim_{\mathbb{C}} M_4(\Gamma_0(\mathfrak{N})) = 6$, with a 2-dimensional Eisenstein subspace and a 4-dimensional space of cusp forms. Let X be a smooth resolution of the Hilbert modular threefold for $\Gamma_0(\mathfrak{N})$. For this field and level, Logan [LoganThreefolds] records a total cusp defect $\sum_P \delta_P(2) = 4$, giving the lower bound

$$h^0(X, K_X^{\otimes 2}) \geq \dim_{\mathbb{C}} M_4(\Gamma_0(\mathfrak{N})) - 4 = 2$$

for the bicanonical bundle, and hence Kodaira dimension at least 1. We realize the four defect conditions as explicit coefficient functionals on q -expansions and compute their rank, so that the bound is witnessed directly on Fourier expansions rather than through the defect count alone.

The group $\Gamma_0(\mathfrak{N})$ has two cusps. At the cusp ∞ the two conditions are the constant term $L_0(f) = a_0(f)$ and the coefficient $L_1(f) = a_{\nu_1}(f)$ at $\nu_1 = \frac{\alpha^2 - \alpha + 1}{7}$, a totally positive generator of the codifferent $\mathbb{Z}_F^\sharp$. The second cusp has Knöller type $((2), \mathbb{Z}_{F, > 0}^\times)$. JK: [Need to define this.

Is the following the same as (??)?] Writing $W_{(2)} = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$ for the Fricke involution and using the unitary slash $(f|\gamma)(z) = \text{Nm}(\det \gamma)^{k/2} \text{Nm}(cz + d)^{-k} f(\gamma z)$ in weight $k = 4$ (so that $W_{(2)}^2 = 1$), the coefficient identity JK: [Where does this come from?]

$$a_\nu(f|W_{(2)}) = \text{Nm}((2))^{k/2} b_{\nu/2} = 64 b_{\nu/2},$$

in which $b_\bullet$ are the coefficients of the second-cusp expansion $f|\sigma$ with $\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, identifies the two conditions at the second cusp with the same functionals L_0, L_1 evaluated on $f|W_{(2)}$.

Take the basis $(E_1, E_2, A, B, H_1, H_2)$ of $M_4(\Gamma_0(\mathfrak{N}))$, where E_1, E_2 are the two degeneracy images of the level-1 Eisenstein series, H_1, H_2 the two degeneracy images of the unique normalized level-1 weight-4 cusp eigenform, and A, B the two rational newforms, whose Atkin–Lehner signs $\varepsilon \in \{-1, +1\}$ are read off their $U_{(2)}$ -eigenvalues (via $a_{(2)} = -8\varepsilon$). On this basis $W_{(2)}$ exchanges each degeneracy pair with scalars 64 and $\frac{1}{64}$ and acts on A, B through their signs, so the four functionals assemble into the 4×6 matrix

$$D = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ \frac{1680}{79} & 0 & 1 & 1 & 1 & 0 \\ 64 & \frac{1}{64} & 0 & 0 & 0 & 0 \\ 0 & \frac{105}{316} & -1 & 1 & 0 & \frac{1}{64} \end{pmatrix},$$

whose top two rows are the conditions at ∞ and whose bottom two are the conditions at the second cusp. The matrix D has rank 4, so the four defect conditions are linearly independent and the space of forms satisfying all of them has dimension exactly $6 - 4 = 2$. Logan’s lower bound is therefore attained, and $h^0(X, K_X^{\otimes 2}) \geq 2$ JK: [=?] is witnessed on q -expansions; in particular the Kodaira dimension of this threefold is at least 1, matching the Kodaira dimension 1 recorded by Logan [LoganThreefolds]. The accompanying code assembles D from the computed q -expansions and verifies the displayed entries and that D has rank 4; the check is exact rational linear algebra on q -expansions computed to precision 100.

APPENDIX A. EISENSTEIN SERIES

Let $\mathfrak{N} \subseteq \mathbb{Z}_F$ be a nonzero integral ideal, let $k \in \mathbb{Z}_{\geq 1}$ be a parallel weight, and let χ be a finite-order Hecke character of F , as in Sections 2.5 and 2.6, whose conductor divides $\mathfrak{N}$. In this appendix, we prove Eq. (4.1.3), providing a basis for $\mathcal{E}_k(\mathfrak{N}, \chi)$.

We begin by establishing linear independence.

Theorem A.1. *Assume $F \neq \mathbb{Q}$. Then the set $\{E_k(\eta, \psi) | \mathfrak{t} : (\eta, \psi, \mathfrak{t}) \in \text{char}_k(\mathfrak{N}, \chi)\}$ is linearly independent.*

Proof. For $k \geq 2$, this is [AtwillLinowitz]. For $k = 1$, one follows the same argument, but instead of [AtwillLinowitz], use linear independence of characters to show that $\chi_1 + \chi_2 = \chi_3 + \chi_4$ would imply either $\chi_1 = \chi_3$ and $\chi_2 = \chi_4$, or $\chi_1 = \chi_4$ and $\chi_2 = \chi_3$. $\square$

We claim that the set $\{E_k(\eta, \psi) | \mathfrak{t} : (\eta, \psi, \mathfrak{t}) \in \text{char}_k(\mathfrak{N}, \chi)\}$ actually forms a basis for $\mathcal{E}_k(\mathfrak{N}, \chi)$. For this, we relate the dimension of $\mathcal{E}_k(\mathfrak{N}, \chi)$ to the number of orbits of cusps of $\overline{Y}_1(\mathfrak{N})$ satisfying some properties. In [DasguptaKakde], the authors prove a similar statement about the relevant $\Gamma_1(\mathfrak{N})$ spaces for $k \geq 2$. We need the refinement by nebentypus, hence show their bijection is equivariant with respect to certain group actions. We refer the reader to [DasguptaKakde] for the notion of admissible cusps and for the definition of the canonical action of the narrow ray class group $\text{Cl}_F^{\mathfrak{N}}$ on the cusps.

In this subsection we will follow [DasguptaKakde] and will recall below the notation we will be using. Note the two main differences from [DasguptaKakde] - our $\mathfrak{b}$ is their $\mathfrak{t}_\lambda \mathfrak{d}$, and our Γ_1 has $a \equiv 1 \pmod{\mathfrak{N}}$ while theirs has $d \equiv 1 \pmod{\mathfrak{N}}$.

Fix a choice of a set of representatives B of the narrow class group $\text{Cl}^+ \mathbb{Z}_F$. Eran: [Rethink the choice of B here.] Let $\text{Cl}_F^{\mathfrak{M}}$ be the narrow ray class group of modulus $\mathfrak{M}$. Explicitly, if $F_{\mathfrak{M},1}^\times := \{x \in F_{>0}^\times : x \equiv 1 \pmod{\mathfrak{M}}\}$, and $I_F^{\mathfrak{M}}$ is the group of fractional ideals coprime to $\mathfrak{M}$, then $\text{Cl}_F^{\mathfrak{M}} = I_F^{\mathfrak{M}} / F_{\mathfrak{M},1}^\times$.

By definition, the cusps of the Hilbert modular variety $\overline{Y}_1(\mathfrak{N})$ are the set of orbits

$$(A.2) \quad \text{cusp}_1(\mathfrak{N}) := \bigsqcup_{\mathfrak{b} \in B} \text{cusp}(\Gamma_1(\mathfrak{N})_{\mathfrak{b}}), \quad \text{cusp}(\Gamma) := \Gamma \backslash \mathbb{P}^1(F).$$

Lemma A.3. *There is a natural surjection $[\cdot] : \text{GL}_2^+(F) \times \text{Cl}^+ \mathbb{Z}_F \rightarrow \text{cusp}_1(\mathfrak{N})$.*

Proof. The group $\text{GL}_2^+(F)$ acts transitively on $\mathbb{P}^1(F)$, hence if we write P for the stabilizer of $\infty \in \mathbb{P}^1(F)$, we have a bijection $\Gamma \backslash \text{GL}_2^+(F) / P \leftrightarrow \text{cusp}(\Gamma)$, mapping $A \mapsto A\infty$. The class group element indicates which component to take. Explicitly, if $\mathfrak{b} \in B$, and

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2^+(F),$$

then $\mathcal{A} = (A, \mathfrak{b})$ is mapped to $(a : c)_{\mathfrak{b}}$. $\square$

Let $\mathcal{A} \in \text{GL}_2^+(F) \times \text{Cl}^+ \mathbb{Z}_F$, so that $[\mathcal{A}] = (a : c)_{\mathfrak{b}}$ is a cusp. Define

$$\mathfrak{s}_{\mathcal{A}} = \mathfrak{s}_{\mathfrak{b}}(a, c) := a\mathbb{Z}_F + c\mathfrak{b}^{-1}, \quad \mathfrak{M}_{\mathcal{A}} = \mathfrak{M}_{\mathfrak{b}}(a, c) := \mathfrak{N} + c(\mathfrak{s}_{\mathcal{A}}\mathfrak{b})^{-1}.$$

Note that by construction $c \in \mathfrak{s}_{\mathcal{A}}\mathfrak{b}$, hence $c(\mathfrak{s}_{\mathcal{A}}\mathfrak{b})^{-1} \subseteq \mathbb{Z}_F$ is integral, so that $\mathfrak{M}_{\mathcal{A}}$ is an integral ideal dividing $\mathfrak{N}$. For a fractional ideal $\mathfrak{s}$ and an integral ideal $\mathfrak{M}$, define

$$\mathcal{R}_{\mathfrak{M}}^{\mathfrak{s}} := (\mathfrak{s} / \mathfrak{M}\mathfrak{s})^\times / \mathbb{Z}_{F,>0}^\times,$$

where $(\mathfrak{s}/\mathfrak{M}\mathfrak{s})^\times$ denotes the subset of elements that generate $\mathfrak{s}/\mathfrak{M}\mathfrak{s}$ as a $\mathbb{Z}_F/\mathfrak{M}$ -module.

In [DasguptaKakde], the authors exhibit a decomposition of the set of cusps of $\bar{Y}_1(\mathfrak{N})$ as a disjoint union of sets $\mathfrak{Q}_{\mathfrak{M},\mathfrak{N}}$, indexed by ideals $\mathfrak{M}$ dividing $\mathfrak{N}$. These sets are in bijection with equivalence classes of certain quadruples, which we proceed to define. Let

$$\mathcal{P}_b(\mathfrak{N}) := \{(\mathfrak{s}, \mathfrak{M}, a, c) : \mathfrak{s} \in I_F, \quad \mathfrak{M} \mid \mathfrak{N}, \quad a \in \mathcal{R}_{\mathfrak{M}}^{\mathfrak{s}}, \quad c \in \mathcal{R}_{\mathfrak{N}/\mathfrak{M}}^{\mathfrak{M}\mathfrak{b}\mathfrak{s}}\},$$

and

$$\mathcal{P}_b(\mathfrak{M}, \mathfrak{N}) = \{(\mathfrak{s}, \mathfrak{M}', \bar{a}, \bar{c}) \in \mathcal{P}_b(\mathfrak{N}) : \mathfrak{M}' = \mathfrak{M}\}.$$

Let

$$(A.4) \quad \mathfrak{Q}_{\mathfrak{M},\mathfrak{N}} = \{\Gamma_1(\mathfrak{N})_b(a : c)_b \in \text{cusp}_1(\mathfrak{N}) : \mathfrak{M}_b(a, c) = \mathfrak{M}\}.$$

Lemma A.5 ([DasguptaKakde]). *The map $\varphi_b : \Gamma_1(\mathfrak{N})_b \backslash (F^2 \setminus \{(0, 0)\}) \rightarrow \mathcal{P}_b(\mathfrak{N})$ given by*

$$(a, c) \longmapsto (\mathfrak{s}_b(a, c), \mathfrak{M}_b(a, c), \bar{a}, \bar{c})$$

is a well-defined bijection.

There is a natural action of $F^\times$ on $\mathcal{P}_b(\mathfrak{N})$ given by

$$(A.6) \quad x \cdot (\mathfrak{s}, \mathfrak{M}, a, c) = (\mathfrak{s}(x), \mathfrak{M}, ax, cx).$$

An immediate corollary is

Corollary A.7 ([DasguptaKakde]). *For every $\mathfrak{b} \in B$, there is a canonical bijection*

$$\Gamma_1(\mathfrak{N})_b \backslash \mathbb{P}^1(F) \rightarrow \mathcal{P}_b(\mathfrak{N})/F^\times.$$

It also induces canonical bijections

$$(A.8) \quad \varphi : \text{cusp}_1(\mathfrak{N}) \rightarrow \bigsqcup_b \mathcal{P}_b(\mathfrak{N})/F^\times,$$

as well as

$$\mathfrak{Q}_{\mathfrak{M},\mathfrak{N}} \rightarrow \bigsqcup_b \mathcal{P}_b(\mathfrak{M}, \mathfrak{N})/F^\times.$$

We begin by describing a natural action of $I_F^{\mathfrak{M}} \times I_F^{\mathfrak{N}/\mathfrak{M}}$ on the set $\bigsqcup_b \mathcal{P}_b(\mathfrak{M}, \mathfrak{N})/F^\times$.

Let $I_1 \in I_F^{\mathfrak{M}}$ and $I_2 \in I_F^{\mathfrak{N}/\mathfrak{M}}$. Let $(\mathfrak{s}, \mathfrak{M}, \bar{a}, \bar{c})_b \in \mathcal{P}_b(\mathfrak{M}, \mathfrak{N})$. Let $\mathfrak{b}_{I_1, I_2} \in B$, $x \in F_{>0}^\times$ be such that $\mathfrak{b}_{I_1, I_2} = x\mathfrak{b}I_2I_1^{-1}$. Let $a_{I_1} \in I_1/\mathfrak{M}I_1$, $d_{I_2} \in xI_2/x(\mathfrak{N}/\mathfrak{M})I_2$ be such that $a_{I_1} \equiv 1 \pmod{\mathfrak{M}}$ and $d_{I_2} \equiv x \pmod{x\mathfrak{N}\mathfrak{M}^{-1}}$. Then

$$(A.9) \quad \langle I_1, I_2 \rangle (\mathfrak{s}, \mathfrak{M}, \bar{a}, \bar{c})_b = (\mathfrak{s}I_1, \mathfrak{M}, a_{I_1} \cdot \bar{a}, d_{I_2} \cdot \bar{c})_{\mathfrak{b}_{I_1, I_2}}.$$

Then, we define the action $\mathbb{Z}_{F, \mathfrak{N}, 1} = \{(x) : x \in \mathbb{Z}_F, x - 1 \in \mathfrak{N}\}$ embeds into both $\text{Cl}_F^{\tilde{\mathfrak{M}}}$ and $\text{Cl}_F^{\mathfrak{N}/\mathfrak{M}}$. Let $Q_{\mathfrak{M}, \mathfrak{N}} = (\text{Cl}_F^{\tilde{\mathfrak{M}}} \times \text{Cl}_F^{\mathfrak{N}/\mathfrak{M}})/\mathbb{Z}_{F, \mathfrak{N}, 1}$, with the diagonal embedding.

Lemma A.10. *The action of $I_F^{\mathfrak{M}} \times I_F^{\mathfrak{N}/\mathfrak{M}}$ on $\bigsqcup_b \mathcal{P}_b(\mathfrak{M}, \mathfrak{N})/F^\times$ via (A.9) is well-defined and factors through an action of $Q_{\mathfrak{M}, \mathfrak{N}}$.*

Proof. First note that by choice of a_{I_1}, d_{I_2} , we have

$$a_{I_1} \cdot \bar{a} \in (\mathfrak{s}I_1/\mathfrak{M}\mathfrak{s}I_1)^\times, \quad d_{I_2} \cdot \bar{c} \in (x\mathfrak{M}\mathfrak{b}\mathfrak{s}I_2/x\mathfrak{N}\mathfrak{b}\mathfrak{s}I_2)^\times = (\mathfrak{M}\mathfrak{b}_{I_1, I_2}\mathfrak{s}I_1/\mathfrak{N}\mathfrak{b}_{I_1, I_2}\mathfrak{s}I_1)^\times,$$

and if a'_{I_1}, d'_{I_2} is any other such choice, then $a_{I_1} - a'_{I_1} \in I_1\mathfrak{M}$ and $d_{I_2} - d'_{I_2} \in xI_2(\mathfrak{N}/\mathfrak{M})$, showing that $a_{I_1} \cdot \bar{a} = a'_{I_1} \cdot \bar{a}$ and $d_{I_2} \cdot \bar{c} = d'_{I_2} \cdot \bar{c}$.

Moreover, if $u \in \mathbb{Z}_{F,>0}^\times$, then $a_{I_1} \cdot (u\bar{a}) = u \cdot (a_{I_1}\bar{a})$ and $d_{I_2} \cdot (u\bar{c}) = u \cdot (d_{I_2}\bar{c})$, showing that the result is independent of representatives under the action of $\mathbb{Z}_{F,>0}^\times$. Furthermore,

$$\langle I_1, I_2 \rangle(x\mathfrak{s}, \mathfrak{M}, x\bar{a}, x\bar{c})_{\mathfrak{b}} = (x\mathfrak{s}I_1, \mathfrak{M}, a_{I_1} \cdot x\bar{a}, d_{I_2} \cdot x\bar{c})_{\mathfrak{b}_{I_1, I_2}} = x \langle I_1, I_2 \rangle(\mathfrak{s}, \mathfrak{M}, \bar{a}, \bar{c})_{\mathfrak{b}},$$

for any $x \in F^\times$, showing that this action commutes with the action of $F^\times$, hence is well defined on the quotient $\bigsqcup_{\mathfrak{b}} \mathcal{P}_{\mathfrak{b}}(\mathfrak{M}, \mathfrak{N})/F^\times$.

If $I_1 = I_2 = \mathbb{Z}_F$, then $\mathfrak{b}_{\mathbb{Z}_F, \mathbb{Z}_F} = \mathfrak{b}$, and we can choose $x = 1$, $a_{I_1} = 1$ and $d_{I_2} = 1$, so that the action is trivial. If $I'_1 \in I_F^{\mathfrak{M}}$ and $I'_2 \in I_F^{\mathfrak{N}/\mathfrak{M}}$, then there exist $x' \in F_{>0}^\times$ and $(\mathfrak{b}_{I_1, I_2})_{I'_1, I'_2} \in B$ such that

$$(\mathfrak{b}_{I_1, I_2})_{I'_1, I'_2} = x' \mathfrak{b}_{I_1, I_2} I'_2 (I'_1)^{-1} = x x' \mathfrak{b} (I_2 I'_2) (I_1 I'_1)^{-1}.$$

Therefore $\mathfrak{b}_{I_1 I'_1, I_2 I'_2} = (\mathfrak{b}_{I_1, I_2})_{I'_1, I'_2}$, and we can set $a_{I_1 I'_1} = a_{I_1} a_{I'_1}$ and $d_{I_2 I'_2} = d_{I_2} d_{I'_2}$, to show that $\langle I_1 I'_1, I_2 I'_2 \rangle = \langle I_1, I_2 \rangle \langle I'_1, I'_2 \rangle$, hence this is a group action.

If $y_1 \in F_{\mathfrak{M}, 1}^\times$ and $y_2 \in \widetilde{F_{\mathfrak{N}/\mathfrak{M}, 1}^\times}$, and $I_1 = y_1 \mathbb{Z}_F, I_2 = y_2 \mathbb{Z}_F$, then we have $\mathfrak{b}_{I_1, I_2} = \mathfrak{b}$ and $x = y_1 y_2^{-1}$. Since $y_1 \equiv 1 \pmod{\mathfrak{M}}$ and $y_2 \equiv 1 \pmod{\mathfrak{N}/\mathfrak{M}}$, we can choose $a_{I_1} = y_1$, and $d_{I_2} = y_1 = x y_2$. It follows that

$$\langle I_1, I_2 \rangle(\mathfrak{s}, \mathfrak{M}, \bar{a}, \bar{c})_{\mathfrak{b}} = (y_1 \mathfrak{s}, \mathfrak{M}, y_1 \bar{a}, y_1 \bar{c})_{\mathfrak{b}} = y_1 (\mathfrak{s}, \mathfrak{M}, \bar{a}, \bar{c})_{\mathfrak{b}},$$

showing that (I_1, I_2) acts trivially on the quotient, hence the action factors through an action of $\mathrm{Cl}_F^{\mathfrak{M}} \times \mathrm{Cl}_F^{\mathfrak{N}/\mathfrak{M}}$. Applying the same argument with $y_1 = y_2 = y \equiv 1 \pmod{\mathfrak{N}}$, we see that the action further factors through $Q_{\mathfrak{M}, \mathfrak{N}}$. $\square$

transitive}

Lemma A.11. *The action of $Q_{\mathfrak{M}, \mathfrak{N}}$ on $\bigsqcup_{\mathfrak{b}} \mathcal{P}_{\mathfrak{b}}(\mathfrak{M}, \mathfrak{N})/F^\times$ is free and transitive.*

Proof. Let $(\mathfrak{s}, \mathfrak{M}, \bar{a}, \bar{c})_{\mathfrak{b}} \in \bigsqcup_{\mathfrak{b}} \mathcal{P}_{\mathfrak{b}}(\mathfrak{M}, \mathfrak{N})$, and assume $(I_1, I_2) \in Q_{\mathfrak{M}, \mathfrak{N}}$ is such that $(I_1, I_2) = (y\mathfrak{s}, \mathfrak{M}, y\bar{a}, y\bar{c})_{\mathfrak{b}}$ for some $y \in F^\times$. Then $I_1 = y\mathbb{Z}_F$, and $\mathfrak{b}_{I_1, I_2} = \mathfrak{b}$, showing that there is $x \in F_{>0}^\times$ such that $I_2 = x^{-1}y\mathbb{Z}_F$. Moreover, we must have $y \equiv 1 \pmod{\mathfrak{M}}$. In addition, $x^{-1}y \equiv 1 \pmod{\mathfrak{N}/\mathfrak{M}}$ implies that $y \equiv x \pmod{x\mathfrak{N}/\mathfrak{M}}$.

By weak approximation, there exists $z \in F$ such that $z \equiv 1 \pmod{\mathfrak{N}}$ and $zy > 0$. Then $zI_1 = zy\mathbb{Z}_F$ with $zy \equiv 1 \pmod{\mathfrak{M}}$, and $zI_2 = zx^{-1}y\mathbb{Z}_F$ with $zx^{-1}y > 0$ and $zx^{-1}y \equiv 1 \pmod{\mathfrak{N}/\mathfrak{M}}$, showing that (zI_1, zI_2) is trivial in $\mathrm{Cl}_F^{\mathfrak{M}} \times \mathrm{Cl}_F^{\mathfrak{N}/\mathfrak{M}}$, hence (I_1, I_2) is trivial in $Q_{\mathfrak{M}, \mathfrak{N}}$. This shows that the action is free, hence by [DasguptaKakde] it is also transitive. $\square$

The homomorphism $I_F^{\mathfrak{N}} \rightarrow I_F^{\mathfrak{M}} \times I_F^{\mathfrak{N}/\mathfrak{M}}$ given by $J \mapsto (J, J^{-1})$ induces a homomorphism $\mathrm{Cl}_F^{\mathfrak{N}} \rightarrow \mathrm{Cl}_F^{\mathfrak{M}} \times \mathrm{Cl}_F^{\mathfrak{N}/\mathfrak{M}}$, hence a homomorphism $\mathrm{Cl}_F^{\mathfrak{N}} \rightarrow Q_{\mathfrak{M}, \mathfrak{N}}$, through which we also obtain an induced action of $\mathrm{Cl}_F^{\mathfrak{N}}$ on $\bigsqcup_{\mathfrak{b}} \mathcal{P}_{\mathfrak{b}}(\mathfrak{M}, \mathfrak{N})/F^\times$. We denote

$$\langle J \rangle(\mathfrak{s}, \mathfrak{M}, \bar{a}, \bar{c}) = \langle J, J^{-1} \rangle(\mathfrak{s}, \mathfrak{M}, \bar{a}, \bar{c}).$$

equivariance}

Lemma A.12. *The bijection $\varphi : \mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}} \leftrightarrow \bigsqcup_{\mathfrak{b}} \mathcal{P}_{\mathfrak{b}}(\mathfrak{M}, \mathfrak{N})/F^\times$ given in Eq. (A.8) is $\mathrm{Cl}_F^{\mathfrak{N}}$ -equivariant.*

Proof. Let $z_{\mathfrak{b}} \in \mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}$ be a cusp on the $\mathfrak{b}$ -th component, and let $J \in I_F^{\mathfrak{N}}$ be an integral ideal coprime to $\mathfrak{N}$. Let $\mathfrak{b}_J \in B$ and $x \in F_{>0}^\times$ be such that $\mathfrak{b}_J = x\mathfrak{b}J^{-2}$. We write $\langle J \rangle$ for the diamond operator, denoted by $S(J)$ in [DasguptaKakde]. Let α be as in [DasguptaKakde], so that $\langle J \rangle z_{\mathfrak{b}} = (\alpha z)_{\mathfrak{b}_J}$. Let $(z_1, z_2) \in F^2$ be a representative for

$z \in \mathbb{P}^1(F)$, and $\varphi_{\mathfrak{b}}(z) = (\mathfrak{s}, \mathfrak{M}, \bar{z}_1, \bar{z}_2)$. Then $(az_1 + bz_2, cz_1 + dz_2)$ is a representative of αz , hence

$$(A.13) \quad \mathfrak{s}_{\mathfrak{b}_J}(az_1 + bz_2, cz_1 + dz_2) = (az_1 + bz_2)\mathbb{Z}_F + (cz_1 + dz_2)\mathfrak{b}_J^{-1} = z_1(a\mathbb{Z}_F + c\mathfrak{b}_J^{-1}) + z_2(b\mathbb{Z}_F + d\mathfrak{b}_J^{-1}).$$

However, since $ad - bc = x$, we see that $adx^{-1} - bcx^{-1} = 1$. From $dx^{-1} \in J^{-1}$ and $b \in \mathfrak{b}^{-1}J$, it follows that $\mathbb{Z}_F = aJ^{-1} + cx^{-1}\mathfrak{b}^{-1}J$, whence

$$(A.14) \quad J = a\mathbb{Z}_F + cx^{-1}\mathfrak{b}^{-1}J^2 = a\mathbb{Z}_F + c(x\mathfrak{b}J^{-2})^{-1} = a\mathbb{Z}_F + c\mathfrak{b}_J^{-1}.$$

Similarly, from $a \in J$ and $cx^{-1} \in \mathfrak{N}\mathfrak{b}J^{-1}$, it follows that $\mathbb{Z}_F = b\mathfrak{N}\mathfrak{b}J^{-1} + dx^{-1}J$, whence

$$\mathfrak{b}^{-1}J = b\mathfrak{N} + dx^{-1}\mathfrak{b}^{-1}J^2 = b\mathfrak{N} + d(x\mathfrak{b}J^{-2})^{-1} = b\mathfrak{N} + d\mathfrak{b}_J^{-1}.$$

Since $b \in \mathfrak{b}^{-1}J$, we also have $b\mathbb{Z}_F + d\mathfrak{b}_J^{-1} = \mathfrak{b}^{-1}J$. Together with (A.14), we plug it in (A.13) to obtain

$$\mathfrak{s}_{\mathfrak{b}_J}(az_1 + bz_2, cz_1 + dz_2) = z_1J + z_2\mathfrak{b}^{-1}J = (z_1\mathbb{Z}_F + z_2\mathfrak{b}^{-1})J = \mathfrak{s}J.$$

By definition of $\langle J \rangle$, $\langle J \rangle z_{\mathfrak{b}} = (\alpha z)_{\mathfrak{b}_J}$ is represented by $(az_1 + bz_2, cz_1 + dz_2)_{\mathfrak{b}_J}$. By [DasguptaKakde], we also have $\mathfrak{M}(\langle J \rangle z_{\mathfrak{b}}) = \mathfrak{M}$.

Since $b \in \mathfrak{b}^{-1}J$ and $z_2 \in \mathfrak{M}\mathfrak{b}\mathfrak{s}$, we have $bz_2 \in \mathfrak{M}\mathfrak{s}J$, showing that $\overline{az_1 + bz_2} = \overline{az_1} = \bar{a} \cdot \bar{z}_1$, where $\bar{a} \in (J/\mathfrak{M}J)^{\times}$ is the image of a . Similarly, since $c \in x\mathfrak{N}\mathfrak{b}J^{-1}$ and $z_1 \in \mathfrak{s}$, we have $cz_1 \in x\mathfrak{N}\mathfrak{b}\mathfrak{s}J^{-1}$, showing that $\overline{cz_1 + dz_2} = \overline{dz_2} = \bar{d} \cdot \bar{z}_2$, where $\bar{d} \in (x\mathfrak{M}\mathfrak{b}\mathfrak{s}J^{-1}/x\mathfrak{N}\mathfrak{b}\mathfrak{s}J^{-1})^{\times}$ is the image of d . Therefore

$$\varphi_{\mathfrak{b}_J}(\langle J \rangle z_{\mathfrak{b}}) = (\mathfrak{s}J, \mathfrak{M}, \bar{a} \cdot \bar{z}_1, \bar{d} \cdot \bar{z}_2) = \langle J \rangle \varphi_{\mathfrak{b}}(z_{\mathfrak{b}}),$$

where the last equality follows from noting that $\bar{a} \equiv 1 \pmod{\mathfrak{N}}$, and $\bar{d} \equiv x \pmod{x(\mathfrak{N}/\mathfrak{M})}$, so that $a_J = \bar{a}$ and $d_{J^{-1}} = \bar{d}$. $\square$

This allows us to refine [DasguptaKakde] in the following way.

Corollary A.15. *There is a bijection $Q_{\mathfrak{M}, \mathfrak{N}} \rightarrow \mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}$, which is $\text{Cl}_F^{\mathfrak{N}}$ -equivariant.*

Proof. Fix $x \in \bigsqcup \mathcal{P}_{\mathfrak{b}}(\mathfrak{M}, \mathfrak{N})/F^{\times}$, and consider the map $Q_{\mathfrak{M}, \mathfrak{N}} \rightarrow \bigsqcup \mathcal{P}_{\mathfrak{b}}(\mathfrak{M}, \mathfrak{N})/F^{\times}$ given by $g \mapsto gx$. By Lemma A.11, this map is a bijection. Compose it with the bijection $\mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}} \leftrightarrow \bigsqcup_{\mathfrak{b}} \mathcal{P}_{\mathfrak{b}}(\mathfrak{M}, \mathfrak{N})/F^{\times}$ to obtain a bijection $\Phi : Q_{\mathfrak{M}, \mathfrak{N}} \rightarrow \mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}$ given by $g \mapsto \varphi(gx)$.

By Lemma A.12, for every $h \in \text{Cl}_F^{\mathfrak{N}}$ we have

$$\Phi(hg) = \varphi((hg)x) = \varphi(h(gx)) = h(\varphi(gx)) = h\Phi(g),$$

establishing the claim. $\square$

The action of $\text{Cl}_F^{\mathfrak{N}}$ on $Q_{\mathfrak{M}, \mathfrak{N}}$ and $\mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}$ induces a linear action on the vector spaces $\mathbb{C}^{Q_{\mathfrak{M}, \mathfrak{N}}}$ and $\mathbb{C}^{\mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}}$ via $\langle J \rangle f(x) = f(\langle J^{-1} \rangle x)$. The above bijection induces a natural pullback map $\Phi^* : \mathbb{C}^{\mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}} \rightarrow \mathbb{C}^{Q_{\mathfrak{M}, \mathfrak{N}}}$, which is an isomorphism of $\text{Cl}_F^{\mathfrak{N}}$ -representations.

In particular, if $\chi : \text{Cl}_F^{\mathfrak{N}} \rightarrow \mathbb{C}$ is a ray class character of modulus $\mathfrak{N}$, Φ^* induces an isomorphism between the corresponding χ -isotypic spaces $\Phi^* : \mathbb{C}^{\mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}}[\chi] \rightarrow \mathbb{C}^{Q_{\mathfrak{M}, \mathfrak{N}}}[\chi]$, where

$$V[\chi] = \{v \in V : Jv = \chi(J)v \quad \forall J \in \text{Cl}_F^{\mathfrak{N}}\}.$$

Let $\text{char}(\mathfrak{M}, \mathfrak{N}, \chi)$ be the set of pairs (η, ψ) where η and ψ are ray class characters, of modulus $\mathfrak{M}$ and $\mathfrak{N}/\mathfrak{M}$ respectively, such that $\eta^{-1}\psi = \chi$.

Recall, as in [DasguptaKakde], that a pair $(\mathfrak{M}, \mathfrak{N})$ is **admissible** if there does not exist a pair of units $\epsilon_1, \epsilon_2 \in \mathbb{Z}_F^\times$ such that $\epsilon_1 \equiv 1 \pmod{\mathfrak{M}}$, $\epsilon_2 \equiv 1 \pmod{\mathfrak{N}/\mathfrak{M}}$ with $\text{Nm}(\epsilon_1) = \text{Nm}(\epsilon_2) = -1$, and ϵ_1/ϵ_2 totally positive.

Lemma A.16. *A basis for $\mathbb{C}^{Q_{\mathfrak{M}, \mathfrak{N}}}[\chi]$ is given by $\text{char}(\mathfrak{M}, \mathfrak{N}, \chi)$, as characters on $Q_{\mathfrak{M}, \mathfrak{N}}$.*

Proof. Assume first that χ is totally even. Dualizing the definition of $Q_{\mathfrak{M}, \mathfrak{N}}$ we have

$$Q_{\mathfrak{M}, \mathfrak{N}}^\vee = \left\{ (\eta, \psi) \in \left(\text{Cl}_F^{\mathfrak{M}} \right)^\vee \times \left(\text{Cl}_F^{\mathfrak{N}/\mathfrak{M}} \right)^\vee : \eta\psi|_{\mathbb{Z}_{F, \mathfrak{N}, 1}} = 1 \right\}.$$

By weak approximation, this is equivalent to the set (η, ψ) of ray class characters of modulus $\mathfrak{M}, \mathfrak{N}/\mathfrak{M}$ respectively, such that $\eta\psi$ is totally even.

It follows that these (η, ψ) form a basis for $\mathbb{C}^{Q_{\mathfrak{M}, \mathfrak{N}}}$. Moreover,

$$\langle J \rangle(\eta, \psi)(I_1, I_2) = (\eta, \psi)(\langle J^{-1} \rangle(I_1, I_2)) = (\eta, \psi)(J^{-1}I_1, JI_2) = (\eta^{-1}\psi)(J)(\eta, \psi)(I_1, I_2),$$

showing that $(\eta, \psi) \in \mathbb{C}^{Q_{\mathfrak{M}, \mathfrak{N}}}[\eta^{-1}\psi]$, hence $\bigoplus_{\eta^{-1}\psi=\chi} \mathbb{C}(\eta, \psi) \subseteq \mathbb{C}^{Q_{\mathfrak{M}, \mathfrak{N}}}[\chi]$, and as we also have $\bigoplus_{\eta, \psi} \mathbb{C}(\eta, \psi) = \mathbb{C}^{Q_{\mathfrak{M}, \mathfrak{N}}}$, there must be an equality.

If χ is totally odd, and $\mathfrak{M} \mid \mathfrak{N}$ is not admissible, then by [DasguptaKakde], $\text{char}(\mathfrak{M}, \mathfrak{N}, \chi) = \emptyset$ and $\mathbb{C}^{Q_{\mathfrak{M}, \mathfrak{N}}}[\chi] = 0$. Therefore, we may assume that $\mathfrak{M} \mid \mathfrak{N}$ is admissible, in which case $\text{char}(\mathfrak{M}, \mathfrak{N}, \chi)$ is a principal homogeneous space for $Q_{\mathfrak{M}, \mathfrak{N}}^\vee$, hence the result. $\square$

The admissible cusps are the cusps in $\bigsqcup_{\mathfrak{M} \mid \mathfrak{N} \text{ admissible}} \mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}$. We say that a cusp $s \in \mathbb{P}^1(F)$ is **singular** for χ if $\text{Stab}(s) \subseteq \ker \chi$. Let $\text{cusp}_1(\mathfrak{M}, \mathfrak{N})$ denote the set of cusps of $\bar{Y}_1(\mathfrak{N})$ corresponding to $\mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}$, and let

$$(A.17) \quad \text{cusp}_k(\mathfrak{M}, \mathfrak{N}, \chi) := \{s \in \text{Cl}_F^{\mathfrak{N}} \setminus \text{cusp}_1(\mathfrak{M}, \mathfrak{N}) : s \text{ is singular for } \chi\}$$

These can be seen as dimensions for the isotypic subspaces of the corresponding vector space based on the cusps.

Lemma A.18. *Assume $k \geq 2$. Then $\dim_{\mathbb{C}} \mathbb{C}^{\mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}}[\chi] = \# \text{cusp}_k(\mathfrak{M}, \mathfrak{N}, \chi)$.*

Proof. Let $T \subseteq \text{cusp}_1(\mathfrak{N})$ be a set of representatives for $\text{cusp}_k(\mathfrak{M}, \mathfrak{N}, \chi)$, and consider the map $\mathbb{C}^T \rightarrow \mathbb{C}^{\mathfrak{Q}_{\mathfrak{M}, \mathfrak{N}}}[\chi]$ given by $t \mapsto \sum_{g \in \text{Cl}_F^{\mathfrak{N}} / \text{Stab}(t)} \chi(g)^{-1} gt$. Note that it is well defined, since by definition $\text{Stab}(t) \subseteq \ker \chi$. It is injective by definition of T , and surjective by definition of the χ -isotypic subspace. $\square$

Aggregating, we also denote

$$(A.19) \quad \text{cusp}_k(\mathfrak{N}, \chi) := \begin{cases} \bigsqcup_{\mathfrak{M} \mid \mathfrak{N}} \text{cusp}_k(\mathfrak{M}, \mathfrak{N}, \chi) & k \geq 2 \text{ even,} \\ \bigsqcup_{\mathfrak{M} \mid \mathfrak{N} \text{ admissible}} \text{cusp}_k(\mathfrak{M}, \mathfrak{N}, \chi) & k \geq 1 \text{ odd.} \end{cases}$$

Note also that there is a natural bijection

$$\text{char}_k(\mathfrak{N}, \chi) \leftrightarrow \begin{cases} \bigsqcup_{\mathfrak{M} \mid \mathfrak{N}} \text{char}(\mathfrak{M}, \mathfrak{N}, \chi) & 2 \mid k, \\ \bigsqcup_{\mathfrak{M} \mid \mathfrak{N} \text{ admissible}} \text{char}(\mathfrak{M}, \mathfrak{N}, \chi) & k \geq 3 \text{ odd,} \\ S_2 \setminus \bigsqcup_{\mathfrak{M} \mid \mathfrak{N} \text{ admissible}} \text{char}(\mathfrak{M}, \mathfrak{N}, \chi) & k = 1. \end{cases}$$

Indeed, a pair $(\psi, \eta)_{\mathfrak{M}}$ on the right hands side corresponds to a triple $(\psi, \eta, \mathfrak{M}/\text{cond}(\psi))$ on the left hand side.

characters}

Corollary A.20. *For all $k \geq 2$ we have*

$$\# \text{ cusp}_k(\mathfrak{N}, \chi) = \# \text{ char}_k(\mathfrak{N}, \chi).$$

Proof. Let $\mathfrak{M} \mid \mathfrak{N}$. Then by Lemma A.16 and Lemma A.18

$$\text{char}(\mathfrak{M}, \mathfrak{N}, \chi) = \dim_{\mathbb{C}} \mathbb{C}^{\mathcal{Q}_{\mathfrak{M}, \mathfrak{N}}}[\chi] = \dim_{\mathbb{C}} \mathbb{C}^{\mathcal{Q}_{\mathfrak{M}, \mathfrak{N}}}[\chi] = \# \text{ cusp}_k(\mathfrak{M}, \mathfrak{N}, \chi).$$

If $k \geq 2$ is even, summing over all $\mathfrak{M} \mid \mathfrak{N}$ yields the result. If $k \geq 3$ is odd, as discussed earlier, $\text{char}(\mathfrak{M}, \mathfrak{N}, \chi) \neq \emptyset$ only if $\mathfrak{M} \mid \mathfrak{N}$ is admissible, hence summing over these yields the result. $\square$

like-basis}

Corollary A.21. *The $E_k(\eta, \psi) \mid \mathfrak{t}$ for $(\eta, \psi, \mathfrak{t}) \in \text{char}_k(\mathfrak{N}, \chi)$ form a basis for $\mathcal{E}_k(\mathfrak{N}, \chi)$.*

Proof. Assume first that $k \geq 2$. Consider the constant term map $\text{CT} : M_k(\mathfrak{N}, \chi) \rightarrow \mathbb{C}^{\text{cusp}_k(\mathfrak{N}, \chi)}$, and note that

$$\mathcal{E}_k(\mathfrak{N}, \chi) \oplus S_k(\mathfrak{N}, \chi) = M_k(\mathfrak{N}, \chi).$$

Since $\ker(\text{CT}) = S_k(\mathfrak{N}, \chi)$, we obtain

$$\begin{aligned} \dim \mathcal{E}_k(\mathfrak{N}, \chi) + \dim S_k(\mathfrak{N}, \chi) &= \dim M_k(\mathfrak{N}, \chi) = \dim \ker(\text{CT}) + \dim \text{CT}(M_k(\mathfrak{N}, \chi)) \\ &\leq \dim S_k(\mathfrak{N}, \chi) + \# \text{ cusp}_k(\mathfrak{N}, \chi) \end{aligned}$$

By Eq. (A.1) and Eq. (A.20), we also have

$$\# \text{ cusp}_k(\mathfrak{N}, \chi) = \# \text{ char}_k(\mathfrak{N}, \chi) \leq \dim \mathcal{E}_k(\mathfrak{N}, \chi),$$

hence equality holds, yielding the result.

For the case $k = 1$, we use an analogue of [Shimizu] for the upper bound. More precisely, with notations as in [Shimizu], let

$$K_1(\mathfrak{N}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(\widehat{\mathbb{Z}}_F) \mid c, a - 1 \in \mathfrak{N}\widehat{\mathbb{Z}}_F, \right\}$$

and let $V(\chi) \subseteq \mathcal{A}(\chi)$ be the space of automorphic forms with central character χ satisfying (A1)-(A3) in [Shimizu]. Denote by $\mathcal{M}_1(\mathfrak{N}, \chi) \subseteq V(\chi)$ the space of forms satisfying $\varphi(gk) = \chi(d(k))\varphi(g)$ for all $k \in K_1(\mathfrak{N})$ as well as (A4) in [Shimizu]. Let $V_0(\chi) = V(\chi) \cap \mathcal{A}_0$ be the cuspidal subspace, and let $V_e(\chi) = V_0(\chi)^\perp$ be the orthogonal complement with respect to the Petersson inner product. The map described in [Shimizu] gives an isomorphism $\mathcal{M}_1(\mathfrak{N}, \chi) \simeq M_1(\mathfrak{N}, \chi)$, and under this identification $V_e(\chi) \cap M_1(\mathfrak{N}, \chi) = E_1(\mathfrak{N}, \chi)$. The restriction of the mapping $V(\chi) \hookrightarrow \bigoplus_{\eta, \psi} \mathcal{B}(\eta, \psi)$ described in [Shimizu] to $V(\chi)$ has kernel $V_0(\chi)$, hence induces an injection on $V_e(\chi)$. Since the $\mathcal{B}(\eta, \psi)$ are irreducible, $V_e(\chi)$ is the direct sum of such spaces. By [Shimizu] and the multiplicity one theorem, the number of these irreducible spaces that intersect $\mathcal{M}_1(\mathfrak{N}, \chi)$ is at most $\# \text{ char}_1(\mathfrak{N}, \chi)$. Therefore $\dim \mathcal{E}_1(\mathfrak{N}, \chi) \leq \# \text{ char}_1(\mathfrak{N}, \chi)$ and Eq. (A.1) finishes the proof. $\square$

Remark A.22. The maps CT defined above involve non-canonical choices. This does not matter for computations, but it is possible to fix that by considering instead the normalized constant terms, and the homology of the space of cusps with respect to the stabilizer, as in [DasguptaKakde].

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